

Chassis and Engine Codes

Vehicle Identification Number

■JHMAAH5330S100001■

Model Name

AF: Coupe CR-X
AL: 1200 2D H/B
AG: 1300 2D H/B
AH: 1500 2D H/B
AM: 1200 4D
AJ: 1300 4D
AK: 1500 4D
AN: Shuttle 4D H/B

Transmission Type

5: 5-speed
4: 4-speed
3: Hondamtic

Number of Doors

3: 2 Door
4: 4 Door
5: 5 Door

The tailgate is counted as a door.

Grade

1: Basic model (Most simple version)
2: Standard model
3: High priced orders-higher than standard version
(3, 4, 5, 6, 7, 8, 9, 0)

Plant

C: Sayama Plant
S: Suzuka Plant

Serial Number

Transmission Number (Hondamatic)

AV-1500001

Transmission Type

Serial Number

Abbreviations:

2 D H/B 2 Door Hatchback
4 D 4 Door Sedan
4 D H/B Wagon (Australia model)
Shuttle (Except Australia model)

Engine Serial Number

EW₂-1200001

Engine Type

ZA₁: 1200 cc
ZA₂: 1200 cc
EV₂: 1300 cc
EW₂: 1500 cc
EW₃: 1500 cc (PGM-FI)

Serial Number

Transmission Number (Manual)

GW-1500001

Transmission Type

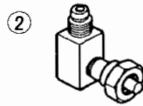
GV: 4-speed
GW: 5-speed

Serial Number

Special Tools

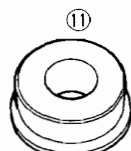
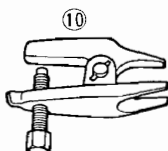
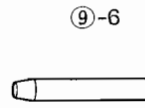
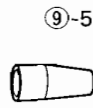
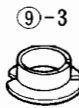
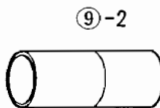
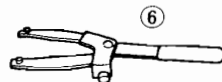
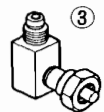
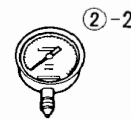
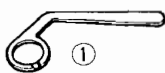
Newly Provided Tool

Newly Provided Tool				
Ref. No.	Tool Number	Description	Q'ty	Remarks
①	07974-SB40100	Thread Adapter	1	
②	07406-0010800	Pump Joint Adapter	1	
③	07406-0010900	Hose Joint Adapter	1	



19. Power Steering

Ref. No.	Tool Number	Description	Q'ty	Remarks
①	07916-6920000 or 07916-6920100	Steering Gear Box Wrench	1	R/H L/H
②	07406-0010200	Power Steering Pressure Gauge	1	
②-1	07406-0010300	Pressure Control Valve	1	
②-2	07406-0010400	Oil Pressure Gauge	1	
③	07406-0010800	Pump Joint Adapter	1	
④	07406-0010900	Hose Joint Adapter	1	
⑤	07953-7190000	Collar Driver	1	
⑥	07725-0030000	Universal Holder	1	07725-0010101 may also be used
⑦	07749-0010000	Driver	1	07949-6110000 may also be used
⑧	07947-6340300	Driver Attachment	1	
⑨	07900-SA50000	Power Steering Tool Kit	1	
⑨-1	07974-SA50100	Piston Seal Ring Guide	1	
⑨-2	07974-SA50200	Sizing Tool	1	
⑨-3	07974-SA50300	Cylinder End Seal Slider	1	
⑨-4	07974-SA50400	End Seal Guide	1	
⑨-5	07974-SA50500	End Cover Seal Guide	1	
⑨-6	07974-SA50600	Pinion Shaft Slider	1	
⑨-7	07974-SA50900	Tool Kit Case	1	
⑩	07941-6920001	Ball Joint Remover	1	
⑪	07746-0010300	Attachment 42x47 mm	1	
⑫	07973-6920001	Adjustment Guide	1	
⑬	07974-SB40100	Thread Adapter	1	



Standards and Service Limits

*1300 and 1200

Unit: mm (in.)

Cylinder Head/Valve Train — Section 6

	MEASUREMENT	STANDARD (NEW)	SERVICE LIMIT
Compression	350 min ⁻¹ (rpm) and wide-open throttle	Nominal PGM-FI 1,274 kPa (13.0 kg/cm ² , 185 psi) Carburetor 1,225 kPa (12.5 kg/cm ² , 176 psi) Minimum PGM-FI 1,078 kPa (11.0 kg/cm ² , 156 psi) Carburetor 1,029 kPa (10.5 kg/cm ² , 149 psi) Maximum variation 196 kPa (2 kg/cm ² , 28 psi)	
Cylinder head	Warpage Height	— 90 (3.54)	0.05 (0.002) 89.8 (3.53)
Camshaft	End play Oil clearance Runout Cam lobe height	0.05–0.15 (0.002–0.006) 0.050–0.098 (0.002–0.004) 0.03 (0.001) max. PGM-FI IN 40.865 (1.6089) EX 40.884 (1.6096) 1500 IN 40.370 (1.5894) EX 40.391 (1.5902) 1300 IN 40.056 (1.5770) EX 40.078 (1.5779) 1200 IN 39.095 (1.5392) EX 39.120 (1.5402)	0.5 (0.02) 0.15 (0.006) 0.06 (0.002) — — — — — — —
Valve	Valve clearance Valve stem O.D. Stem-to-guide clearance Stem installed height	IN 0.17–0.22 (0.007–0.009) EX 0.22–0.27 (0.009–0.011) IN 6.58–6.59 (0.2591–0.2594) EX 6.55–6.56 (0.2579–0.2583) IN 0.02–0.05 (0.001–0.002) EX 0.05–0.08 (0.002–0.003) IN 48.16 (1.896) EX 48.16 (1.896)	— — 6.55 (0.258) 6.52 (0.257) 0.08 (0.003) 0.11 (0.004) 48.95 (1.927) 48.95 (1.927)
Valve seat	Width	IN and EX 1.25–1.55 (0.049–0.061)	2.0 (0.08)
Valve spring	Free length Squareness Inner and Outer	IN and EX 47.6 (1.87) —	46.6 (1.83) 1.75 (0.068)
Valve guide	I.D.	IN 6.61–6.63 (0.260–0.261) EX 6.61–6.63 (0.260–0.261)	6.65 (0.262) 6.65 (0.262)
Rocker arm	Arm-to-shaft clearance	0.018–0.054 (0.0007–0.0021)	0.08 (0.003)

Engine Block — Section 7

	MEASUREMENT	STANDARD (NEW)	SERVICE LIMIT
Cylinder block	Warpage of deck surface Bore diameter Bore taper Reboring limit	0.07 (0.003) max. 74.00–74.02 (2.9133–2.9142) 0.07–0.012 (0.0003–0.0005) —	0.10 (0.004) 74.10 (2.9173) 0.05 (0.002) 0.5 (0.02)
Piston	Skirt O.D. At 16 mm (0.63 in) from bottom of skirt Clearance in cylinder Piston-to-ring clearance (Top) (Second)	73.97–73.99 (2.9122–2.9133) 0.01–0.05 (0.0004–0.0020) 0.03–0.06 (0.0012–0.0024) 0.030–0.055 (0.0012–0.0022)	73.96 (2.912) 0.07 (0.003) 0.13 (0.005) 0.13 (0.005)
Piston ring	Ring end gap (Top and second) Ring end gap (Oil)	0.15–0.35 (0.006–0.014) 0.30–0.90 (0.012–0.035)	0.6 (0.024) 1.1 (0.043)
Connecting rod	Pin-to-rod interference Large end bore diameter End play installed on crankshaft	0.02–0.04 (0.0008–0.0016) Nominal 45 (1.77) *43 (1.69) 0.15–0.30 (0.006–0.012)	0.02 (0.0008) — 0.40 (0.016)
Crankshaft	Main journal diameter Taper/out-of-round, main journal Rod journal diameter Taper/out-of-round, rod journal End play Runout	49.976–50.000 (1.9676–1.9685) 0.005 (0.0002) max. 41.976–42.000 (1.6526–1.6535) *39.976–40.000 (1.5739–1.5748) 0.005 (0.0002) max. 0.10–0.35 (0.004–0.014) 0.03 (0.0012) max.	— 0.010 (0.0004) — — 0.010 (0.0004) 0.45 (0.018) 0.06 (0.0024)
Bearings	Main bearing-to-journal oil clearance Rod bearing-to-journal oil clearance	0.024–0.042 (0.0009–0.0017) 0.020–0.038 (0.0008–0.0015)	0.07 (0.003) 0.07 (0.003)

Engine Lubrication — Section 8

	MEASUREMENT	STANDARD (NEW)	SERVICE LIMIT
Engine oil	Capacity ℓ (US. qt., Imp. qt.)	4.0 (4.2, 3.4) After engine disassembly 3.5 (3.7, 3.1) After oil change, including oil filter 3.0 (3.2, 2.6) After oil change, without oil filter	
Oil pump	Displacement Inner-to-outer rotor radial clearance Pump body-to-rotor radial clearance Pump body-to-rotor side clearance	35ℓ (9.2 US gal., 7.7 Imp gal.) 3,000 min ⁻¹ (rpm) 0.14 (0.006) max. 0.10–0.175 (0.004–0.007) 0.03–0.08 (0.001–0.003)	0.2 (0.008) 0.2 (0.008) 0.15 (0.006)
Relief valve	Pressure setting	333–340 kPa (3.4–4.2 kg/cm ² , 48–60 psi)	

Cooling — Section 10

	MEASUREMENT	STANDARD (NEW)
Radiator	Capacity (incl. heater) ℓ (US. Gal., Imp. Gal.) Includes reservoir tank 0.4 (0.11, 0.09)	Coupe (PGM-FI) 5.1 (1.3, 1.1) 1200 4MT 4.4 (1.4, 1.2) KG only 4MT 5.2 (1.4, 1.1) HM 4.9 (1.3, 1.1) 1300 4MT 4.9 (1.3, 1.1) KC only 4MT 5.2 (1.4, 1.1) Other models 5MT 4.9 (1.3, 1.1) KT only 5MT 4.4 (1.2, 1.0) Other models HM 4.9 (1.4, 1.1) KC and EC models HM 5.2 (1.3, 1.1) Other models 1500 5.2 (1.4, 1.1) KC and EC models 5.5 (1.5, 1.2) Other models Subtract 1ℓ (0.26 US Gal., 0.22 Imp. Gal.) from capacities at replacement
	Pressure cap opening pressure	74–103 kPa (0.75–1.05 kg/cm ² , 11–15 psi)
Thermostat	Starts to open Full open Valve lift at full open	76–78°C (169–173°F) 91°C (196°F) 8 (0.31) max.
Cooling fan	Fan-to-core clearance Thermoswitch "ON" temperature Thermoswitch "OFF" temperature	ND 22 mm (0.87 in.) TOYO 17.5 mm (0.69 in.) 88.5–91.5°C (191–197°F) 85.5–86.5°C (186–188°F)

Carburetor — Section 11

	MEASUREMENT	STANDARD (NEW)
Carburetor	Choke fast idle	1,500–2,500 min ⁻¹ (rpm)
	Idle speed with headlights and cooling fan off (On Swedish model: on)	Manual 700–800 min ⁻¹ (rpm) Hondamatic 650–750 min ⁻¹ (rpm)
	Idle CO	KS and KQ below 2.0% KX 0.5–2.0% Other models below 3.0%
	Float level	35.4–37.4 (13.9–14.7 in.)
PGM-FI	Choke fast idle	1,200–2,000 min ⁻¹ (rpm)
	Idle speed with headlights and cooling fan off (on Swedish model: on)	700–800 min ⁻¹ (rpm)
	Idle CO	KS below 1.5% KX 0.5–2%

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Standards and Service Limits(cont'd)

Fuel — Section 12

	MEASUREMENT		STANDARD (NEW)
Fuel pump	Delivery pressure Displacement		17.7–26.5 kPa (0.18–0.27 kg/cm ² , 27–38 psi) 170 cc/min at camshaft rpm 300 min ⁻¹ (rpm)
Fuel pump (PGM-FI)	Delivery pressure Displacement		230–270 kPa (2.35–2.75 kg/cm ² , 33–39 psi) 230 cc/min in 10 seconds
Fuel tank	Capacity	Coupe 2D H/B 4D, 4D H/B	41 ℓ (10.8 US. Gal., 9.0 Imp. Gal.) 45 ℓ (11.9 US. Gal., 9.9 Imp. Gal.) 46 ℓ (12.1 US. Gal., 10.1 Imp. Gal.)

Clutch — Section 13

	MEASUREMENT		STANDARD (NEW)	SERVICE LIMIT
Clutch pedal	Pedal height	Coupe	175 (6.89) to floor	—
		Except Coupe	179 (7.05) to floor	—
	Stroke		135–140 (5.3–5.5)	—
	Pedal play	Coupe	10–30 (0.39–1.18)	—
		Except Coup	16–21 (0.63–0.83)	—
Disengagement height		Coupe	61 (2.4) min. to floor 31 (1.2) min. to carpet	
		2D H/B, 4D	83 (3.3) min. to floor 53 (2.1) min. to carpet	
		4D H/B	78 (3.1) min. to floor 48 (1.9) min. to carpet	
Clutch arm	Release arm adjustment		4.0–5.0 (0.16–0.20)	—
Flywheel	Clutch surface runout		0.05 (0.002) max.	0.15 (0.006)
Clutch plate	Rivet head depth		1.3 (0.05) min.	0.2 (0.008)
	Surface runout		0.8 (0.03) max.	1.0 (0.04)
	Radial play in splines		0.036–0.112 (0.0014–0.0044)	0.5 (0.02)
	Thickness		8.1–8.8 (0.32–0.35)	5.7 (0.22)
Clutch release bearing holder	I.D.		29.000–29.059 (1.142–1.144)	29.20 (1.150)
	Holder-to-guide sleeve clearance		0.040–0.132 (0.0016–0.0052)	0.2 (0.008)
Clutch cover	Unevenness of diaphragm spring		0.8 (0.03) max.	1.0 (0.04)

Manual Transmission — Section 14 and 15

	MEASUREMENT		STANDARD (NEW)	SERVICE LIMIT
Transmission oil	Capacity & (US. qt., Imp. qt)		2.5 (2.6, 2.2) at assembly 2.3 (2.4, 2.0) at oil change	
Mainshaft	End play Diameter of needle bearing contact area Diameter of fifth gear contact area Diameter of 62/22 ball bearing contact area Diameter of 6304 ball bearing contact area Runout		0.11–0.25 (0.004–0.010) 27.997–28.010 (1.1022–1.1028) 24.987–25.000 (0.9837–0.9843) 21.987–22.000 (0.8656–0.8661) 19.983–19.996 (0.7867–0.7872) 0.02 (0.0008) max.	— 27.94 (1.100) 24.93 (0.981) 21.93 (0.863) 19.93 (0.7846) 0.05 (0.0019)
Mainshaft fifth gear	I.D. End play		30.007–30.020 (1.1814–1.1819) 0.05–0.35 (0.0020–0.0138)	30.07 (1.184) —
Countershaft	End play Diameter of needle bearing contact area Diameter of ball bearing contact area Diameter of low gear contact area Runout		0.35 (0.0138) 30.004–30.017 (1.1813–1.1818) 24.9935–25.0065 (0.9840–0.9845) 31.984–32.000 (1.2592–1.2598) 0.04 (0.0016)	0.65 (0.026) 29.94 (1.179) 24.94 (0.982) 31.93 (1.257) 0.10 (0.004)
Countershaft low gear	I.D. End play		37.009–37.025 (1.4570–1.4577) 0.03–0.08 (0.0012–0.0031)	37.08 (1.460) 0.18 (0.007)
Countershaft second,, third/fourth gear	I.D. End play		37.009–37.025 (1.4570–1.4577) 0.05–0.12 (0.0020–0.0047)	37.08 (1.460) 0.18 (0.007)
Spacer collar	Second, Third	I.D.	25.980–25.991 (1.0228–1.0233)	26.04 (1.025)
		O.D. Length	31.989–32.000 (1.2594–1.2598) 28.01–28.13 (1.1028–1.1074)	31.93 (1.257) —
Fourth		I.D.	25.007–25.037 (0.9845–0.9857)	25.08 (0.987)
		O.D. Length	31.989–32.000 (1.2594–1.2598) 28.01–28.13 (1.1028–1.1074)	31.93 (1.257)
Reverse idler gear	I.D. Gear-to-reverse gear shaft clearance		15.016–15.043 (0.5912–0.5922) 0.032–0.077 (0.0013–0.0030)	15.08 (0.594) 0.14 (0.006)
Synchronizer ring	Ring-to-gear clearance (ring pushed against gear)		0.85–1.10 (0.033–0.043)	0.4 (0.016)
Shift fork	Synchronizer sleeve gear		6.95–7.05 (0.2736–0.2776)	—
	Fork-to-synchronizer sleeve clearance		0.45–0.65 (0.018–0.026)	1.0 (0.039)
Reverse shift fork	End gap		6.9–7.0 (0.27–0.28)	—
	Fork-to-reverse idler gear clearance		0.1–0.3 (0.004–0.012)	0.7 (0.028)
	Groove width		7.05–7.25 (0.278–0.285)	—
	Fork-to-fifth/reverse shift shaft clearance		0.05–0.35 (0.002–0.014)	0.5 (0.020)
Shift arm B	I.D.		14.016–14.043 (0.5518–0.5529)	—
	Shift arm-to-shift guide clearance		0.022–0.067 (0.0009–0.0026)	0.15 (0.006)

(cont'd)

Standards and Service Limits(cont'd)

Unit: mm (in.)

Hondamatic Transmission — Section 16

	MEASUREMENT	STANDARD (NEW)	SERVICE LIMIT
Transmission oil	Capacity & (US. qt., Imp. qt)	2.8 (3.0, 2.5) at oil change 5.4 (5.7, 4.8) at assembly	
Hydraulic pressure	Line pressure at 2,000 min ⁻¹ (rpm) 1500 1200, 1300	735–784 kPa (7.5–8.0 kg/cm ² , 107–114 psi) 637–686 kPa (6.5–7.0 kg/cm ² , 92–100 psi)	686 kPa (7.0 kg/cm ² , 100 psi) 588 kPa (6.0 kg/cm ² , 85 psi)
	OD clutch pressure at 2,000 min ⁻¹ (rpm) 1500 1200, 1300	686–784 kPa (7.0–8.0 kg/cm ² , 100–114 psi) 588–686 kPa (6.0–7.0 kg/cm ² , 85–100 psi)	637 kPa (6.5 kg/cm ² , 92 psi) 539 kPa (5.5 kg/cm ² , 78 psi)
	☆clutch pressure at 2,000 min ⁻¹ (rpm) 1500 1200, 1300	686–784 kPa (7.0–8.0 kg/cm ² , 100–114 psi) 588–686 kPa (6.0–7.0 kg/cm ² , 85–100 psi)	637 kPa (6.5 kg/cm ² , 92 psi) 539 kPa (5.5 kg/cm ² , 78 psi)
	1st clutch pressure at 2,000 min ⁻¹ (rpm) 1500 1200, 1300	686–784 kPa (7.0–8.0 kg/cm ² , 100–114 psi) 588–686 kPa (6.0–7.0 kg/cm ² , 85–100 psi)	637 kPa (6.5 kg/cm ² , 92 psi) 539 kPa (5.5 kg/cm ² , 78 psi)
	Governor pressure at 60 km/h	221–230 kPa (2.25–2.35 kg/cm ² , 32–33 psi)	216 kPa (2.2 kg/cm ² , 31 psi)
	Throttle pressure 1500 1200, 1300	735–784 kPa (7.5–8.0 kg/cm ² , 107–114 psi) 637–686 kPa (6.5–7.0 kg/cm ² , 92–100 psi)	686 kPa (7.0 kg/cm ² , 100 psi) 588 kPa (6.0 kg/cm ² , 85 psi)
Stall speed	Check with car on level ground	2,700 min ⁻¹ (rpm)	2,300–2,900 min ⁻¹ (rpm)
Clutch	Clutch initial clearance 1st	0.4–0.7 (0.016–0.028)	—
	2nd	0.65–0.80 (0.026–0.031)	—
	3rd	0.4–0.6 (0.016–0.024)	—
	Clutch return spring free length	30.5 (1.20)	28.5 (1.12)
	Clutch disc thickness	1.88–2.0 (0.074–0.079)	Until grooves worn out
	Clutch plate thickness	1.95–2.05 (0.077–0.079)	Discoloration
	Clutch end plate thickness Mark 1	2.3–2.4 (0.091–0.094)	
	Mark 2	2.4–2.5 (0.094–0.098)	
	Mark 3	2.5–2.6 (0.098–0.102)	
	Mark 4	2.6–2.7 (0.102–0.106)	
	Mark 5	2.7–2.8 (0.106–0.110)	
	Mark 6	2.8–2.9 (0.110–0.114)	
	Mark 7	2.9–3.0 (0.114–0.118)	
	Mark 8	3.0–3.1 (0.118–0.122)	
	Mark 9	3.1–3.2 (0.122–0.126)	
	Mark 10	3.2–3.3 (0.126–0.130)	
Transmission	Diameter of needle bearing contact area on main and stator shaft	19.980–19.993 (0.7866–0.7871)	
	Diameter of needle bearing contact area on main 2nd gear collar	31.975–31.991 (1.2588–1.2594)	
	Diameter of needle bearing contact area on mainshaft 1st gear collar	30.975–30.991 (1.2195–1.2201)	
	Diameter of needle bearing contact area on countershaft (L side)	32.984–33.000 (1.2986–1.2993)	
	Diameter of needle bearing contact area on countershaft 3rd gear	31.975–31.991 (1.2589–1.2595)	
	Diameter of needle bearing contact area on countershaft 2nd gear	27.980–27.993 (1.1016–1.1021)	
	Diameter of needle bearing contact area on countershaft reverse gear collar	29.980–29.993 (1.1803–1.1808)	
	Diameter of needle bearing contact area on reverse idle gear	13.994–14.000 (0.5509–0.5512)	
	Reverse idler shaft holder diameter	14.016–14.034 (0.5518–0.5525)	
	Mainshaft 2nd gear I.D.	38.000–38.016 (1.4961–1.4967)	
	Mainshaft 1st gear I.D.	36.000–36.016 (1.4173–1.4179)	
	Countershaft 3rd gear I.D.	38.000–38.016 (1.4966–1.4966)	
	Countershaft 2nd gear I.D.	33.000–33.016 (1.4173–1.4179)	
	Countershaft 1st gear I.D.	35.000–35.016 (1.3779–1.3785)	
	Countershaft reverse gear I.D.	36.000–36.016 (1.4173–1.4179)	
	Reverse idler gear I.D.	18.007–18.020 (0.7086–0.7094)	

	MEASUREMENT	STANDARD (NEW)	SERVICE LIMIT
Transmission (cont'd)	Mainshaft 2nd gear end play	0.07–0.15 (0.003–0.006)	—
	Mainshaft 1st gear end play	0.08–0.20 (0.003–0.008)	—
	Countershaft 3rd gear end play	0.07–0.15 (0.003–0.006)	—
	Countershaft 2nd gear end play	0.07–0.15 (0.003–0.006)	—
	Reverse idle gear end play	0.05–0.18 (0.0020–0.0071)	—
	Countershaft reverse gear end play	0.10–0.20 (0.004–0.008)	—
	Reverse gear hub O.D.	51.87–51.90 (2.0421–2.0433)	Wear or damage
	Thrust washer thickness		
	Mainshaft 2nd gear, Countershaft 3rd gear	A 2.97–3.00 (0.1169–0.1181)	—
		B 3.02–3.05 (0.1189–0.1201)	—
		C 3.07–3.10 (0.1209–0.1220)	—
		D 3.12–3.15 (0.1228–0.1240)	—
		E 3.17–3.20 (0.1248–0.1260)	—
		F 3.22–3.25 (0.1268–0.1280)	—
		G 3.27–3.30 (0.1287–0.1299)	—
		H 3.32–3.35 (0.1307–0.1319)	—
		I 3.37–3.40 (0.1327–0.1339)	—
	Mainshaft R side bearing	3.95–4.05 (0.1555–0.1594)	Wear or damage
	Mainshaft 1st gear	2.43–2.50 (0.0957–0.0984)	Wear or damage
	Countershaft 2nd gear thickness	A 2.27–2.30 (0.0894–0.0906)	—
		B 2.32–2.35 (0.0913–0.0925)	—
		C 2.37–2.40 (0.0933–0.0945)	—
		D 2.42–2.45 (0.0953–0.0965)	—
		E 2.47–2.50 (0.0972–0.0984)	—
		F 2.50–2.55 (0.0972–0.1004)	—
		G 2.52–2.60 (0.0992–0.1024)	—
Regulator valve body	Mainshaft 1st gear collar length	22.50–22.55 (0.8858–0.8878)	—
	Mainshaft 1st gear collar flange thickness	2.5–2.6 (0.098–0.102)	Wear or damage
	Countershaft reverse gear collar length	14.0–14.1 (0.551–0.555)	—
	Countershaft reverse gear collar flange thickness	2.45–2.50 (0.096–0.098)	Wear or damage
	Mainshaft and countershaft feed pipe O.D. (at 20 mm from end)	7.97–7.98 (0.3138–0.3142)	7.95 (0.31)
	Mainshaft sealing ring 32 mm thickness	1.980–1.995 (0.0780–0.0785)	—
	Mainshaft bushing I.D.	8.000–8.015 (0.3150–0.3156)	8.03 (0.316)
	Countershaft bushing I.D.	8.000–8.015 (0.3150–0.3156)	8.03 (0.316)
	Mainshaft sealing ring groove width	2.025–2.060 (0.0797–0.0811)	2.08 (0.082)
Shifting device and parking brake control	Sealing ring contact area diameter	32.000–32.025 (1.2598–1.2608)	32.05 (1.26)
	Reverse shift fork thickness	5.9–6.0 (0.232–0.236)	5.4 (0.21)
	Parking brake ratchet pawl	—	Wear or other defect
Servo body	Parking gear	—	Wear or other defect
	Throttle cam stopper	18.5–18.6 (0.7283–0.7323)	—
Valve body	Shift fork shaft bore I.D.	A 14.000–14.005 (0.5512–0.5514)	—
		B 14.006–14.010 (0.5514–0.5516)	—
		C 14.011–14.015 (0.5516–0.5518)	—
Valve body	Shift fork shaft valve bore I.D.	37.000–37.039 (1.4567–1.4582)	37.045 (1.4583)
	Oil pump gear side clearance	0.03–0.05 (0.0012–0.0020)	0.07 (0.003)
	Oil pump gear-to-body clearance	Drive: 0.21–0.27 (0.0083–0.0106)	—
		Driven: 0.05–0.09 (0.0020–0.0035)	—
	Stator camshaft needle bearing bore I.D.	24.000–24.021 (0.9449–0.9457)	Wear or damage
	Stator camshaft needle bearing contact and O.D.	26.000–26.013 (1.0236–1.0241)	Wear or damage
	Oil pump driven gear I.D.	14.016–14.034 (0.5518–0.5525)	Wear or damage
	Oil pump shaft O.D.	13.980–13.990 (0.5503–0.5507)	Wear or damage

(cont'd)

Standards and Service Limits(cont'd)

Unit: mm (in.)

Differential – Section 17

Differential – Section 17			
	MEASUREMENT	STANDARD (NEW)	SERVICE LIMIT
Ring gear	Backlash	0.073–0.132 (0.0029–0.0052)	0.25 (0.010)
Differential carrier	Pinion shaft bore diameter	18.000–18.018 (0.7087–0.7094)	18.1 (0.71)
	Carrier-to-pinion shaft clearance	0.016–0.052 (0.0006–0.0020)	0.1 (0.004)
	Driveshaft bore diameter	26.005–26.025 (1.0238–1.0246)	—
	Manual. Hondamatic	28.000–28.021 (1.1024–1.1032)	—
		Carrier-to-driveshaft clearance	0.025–0.066 (0.0010–0.0026)
	Side clearance	0.10–0.20 (0.004–0.008)	0.15 (0.006)
Differential pinion gear	Backlash	0.05–0.15 (0.002–0.006)	—
	Pinion gear bore diameter	18.041–18.061 (0.7103–0.7111)	—
	Pinion gear-to-pinion shaft clearance	0.057–0.095 (0.0022–0.0037)	0.15 (0.006)

Driveshaft – Section 18

	MEASUREMENT		STANDARD (NEW)	SERVICE LIMIT
Driveshaft	Right boot	As installed	471–476 (18.5–18.7)	—
	Left boot	As installed	771–776 (30.4–30.6)	—

Steering – Section 19

	MEASUREMENT		STANDARD (NEW)	SERVICE LIMIT
Steering wheel	Play		10.0 (0.39) Max.	—
	Pinion-starting torque N-m (kg-m, lb-ft)		0.5–1.3 (0.05–0.13, 0.36–0.94)	—

Suspension – Section 20

	MEASUREMENT			STANDARD (NEW)	SERVICE LIMIT
Wheel alignment	Front camber	Coupe		0°00'±1°	
		2D H/B		0°00'±1° KY: 0°20'±1°	
		4D		0°00'±1° KY: 0°30'±1°	
		4D H/B		0°30'±1° KC: 0°20'±1°	
				EC: 0°26'±1°	
	Rear camber	Coupe		–0°45'±15'	
		2D H/B		2°25'±1°	
		4D		2°20'±1° KY: 2°10'±1°	
		4D H/B		2°20'±1° KY: 2°15'±1°	
				2°00'±1° KC: 2°05'±1°	
	Caster	Coupe		EC: 1°47'±1°	
		2D H/B		0±3 mm (0±0.118 in.)	
		4D		IN 2±2 mm (0.079±0.079 in.)	
		4D H/B		12°55'±30'	
				12°50'±30' KY: 12°30'±30'	
	Front toe	Coupe		12°50'±30' KY: 12°15'±30'	
		2D H/B		11°45'±30' KC: 12°00'±30'	
		4D		EC: 11°54'±30'	
		4D H/B			
	Rear toe	Coupe			
		2D H/B			
		4D			
		4D H/B			
	Kingpin inclination	Coupe			
		2D H/B			
		4D			
		4D H/B			
Steering angle	R/L	Inside		41°30' ± 2°	
		Outside		34°30' ± 2°	
Wheel	Rim runout	Steel	Axial	0–1.0 (0–0.039)	—
			Radial	0–1.0 (0–0.039)	—
		Aluminum	Axial	0–0.7 (0–0.028)	—
			Radial	0–0.7 (0–0.028)	—
					—

Brake — Section 21

	MEASUREMENT	STANDARD (NEW)	SERVICE LIMIT
Parking brake lever	Play in stroke 200N (20 kg, 44 lbs)	To be locked when pulled 4–8 notches	
Foot brake pedal	Pedal height Free play	174 (6.8) 4D H/B:168 (6.6) to floor 1–5 (0.04–0.20)	— 5 (0.20)
Master cylinder	Piston-to-push rod clearance	0–0.4 (0–0.016)	—
Brake drum	I.D. 4D H/B	180 (7.09) 200 (7.87)	181 (7.13) 201 (7.91)
Lining	Thickness	4.5 (0.18)	2.0 (0.08)
Disc brake	Disc thickness Disc runout Disc parallelism Pad thickness	Ventilated Solid — 0.007 (0.0003) 10.0 (0.39) 9.5 (0.37)	15.0 (0.59) 10.0 (0.39) 0.10 (0.004) 0.015 (0.0006) 3.0 (0.12) 3.0 (0.12)
		4D H/B EC and KX Other models	

Brake Booster	Characteristic	Vacuum (mmHg)	Pedal Pressure kg (lbs)	Line Pressure kg/cm ² (psi)
		0 300 500	20 (44) 20 (44) 20 (44)	16 (227) min 46 (654) min 66 (939) min

(cont'd)

Standards and Service Limits(cont'd)

Engine Electrical — Section 25, 26 and 27

Unit: mm (in.)

	MEASUREMENT		STANDARD (NEW)						
Ignition coil	Rated voltage		12 Volts						
	Insulation resistance		10,000 ohms min.						
	Performance: Make sure strong sparks jump across electrodes (3-point tester)								
	Voltage	Camshaft	Secondary Voltage	3-point gap	Condition				
	12V	3,000 min ⁻¹ (rpm)	17 ± 4 kV	13–19 (0.51–0.75)	At 80°C (176°F)				
Ignition wire	Resistance		25,000 ohms max.						
Spark plug	Type	Standard	KC, EC						
			Other models						
			NGK: BPR6EY-11, ND: W20EXR-U11						
			NGK: PP6EY-11, ND: W20EX-U11						
	Gap		1.0–1.1 (0.039–0.043)						
Ignition timing	At idling	1200		17±2° BTDC					
		1300		12±2° BTDC					
		1500	Canadian model MT	10±2° BTDC					
			Canadian model HM	14±2° BTDC					
			European model	16±2° BTDC					
			PGM-F1 model	16±2° BTDC					
	General export model	16±2° BTDC							
Battery	Lighting capacity (20-hour ratio)		47 Ampere Hours						
	Starting capacity (5-second ratio)		8.4V minimum at 300 Ampere draw						
Alternator		ND		MITSUBISHI					
		STANDARD (NEW)		SERVICE LIMIT	STANDARD (NEW)	SERVICE LIMIT			
		Output at no-load		14V at 1,090 min ⁻¹ (rpm)	14V at 1,400 min ⁻¹ (rpm)	14V at 1,100 min ⁻¹ (rpm)	14V at 1,400 min ⁻¹ (rpm)		
	Output		14V/55A at 6,000 min ⁻¹ (rpm)	60A at 5,000 min ⁻¹ (rpm)	14V/55A at 6,000 min ⁻¹ (rpm)	50A at 5,000 min ⁻¹ (rpm)			
	Coil resistance (rotor)		2.9 ohm	2.8–3.0 ohms	3 ohm				
	Slip ring O.D.		14.4 (0.57)	13.5 (0.53)	23.0 (0.91)	22.5 (0.89)			
	Brush length		13.5 (0.53)	5.0 (0.20)	18.0 (0.71)	8.0 (0.31)			
Brush spring tension		330g (11.6 oz)	200g (7.05 oz)	370g (13.05 oz)	210g (7.41 oz)				
Alternator belt	Deflection midway between pulleys/load		7–10 (0.28–0.39)/98 (10 kg, 22 lb) for used belt						
			4–6.5 (0.16–0.26)/98 (10 kg, 22 lb) after replacement of belt						
Starting motor	MEASUREMENT	ND 0.8kW		HITACHI 0.8kW		ND 1.0kW, 1.4kW		MITSUBA 1.0kW, 1.4kW	
		STANDARD (NEW)	SERVICE LIMIT	STANDARD (NEW)	SERVICE LIMIT	STANDARD (NEW)	SERVICE LIMIT	STANDARD (NEW)	SERVICE LIMIT
	Mica depth	0.5–0.8 (0.020–0.031)	0.2 (0.008)	0.5–0.8 (0.020–0.031)	0.2 (0.008)	0.5–0.8 (0.020–0.031)	0.2 (0.008)	0.4–0.5 (0.016–0.020)	0.15 (0.006)
	Commutator runout	0–0.5 (0.020)	0.3 (0.012)	0–0.1 (0.004)	0.4 (0.016)	0–0.02 (0.0008)	0.05 (0.020)	0–0.02 (0.008)	0.05 (0.020)
	Commutator O.D.	28.0 (1.10)	27.0 (1.06)		39.0 (1.54)	30.0 (1.18)	29.0 (1.14)	28.0 (1.10)	27.5 (1.08)
	Brush length	15.5–16.5 (0.61–0.65)	10.0 (0.39)	14.5–15.5 (0.61–0.65)	12.0 (0.47)	12.5–13.5 (0.49–0.53)	8.5 (0.33)	14.3–14.7 (0.56–0.58)	9.3 (0.37)
	Spring pressure (new)	1.2 kg (2.6 lb)	—	1.6 kg (3.5 lb)	—	1.75 kg (3.8 lb)	—	2.1 kg (4.6 lb)	—

Design Specifications

Coupe

	ITEMS	METRIC	ENGLISH	NOTES
DIMENSIONS	Overall Length	3,675 mm	144.7 in.	
	Overall Width	1,625 mm	63.9 in.	
	Overall Height	1,290 mm	50.8 in.	
	Wheelbase	2,200 mm	86.6 in.	
	Tread Front/Rear	1,400/1,415 mm	55.1/55.7 in.	
	Ground Clearance	165 mm	6.5 in.	
	Seating Capacity		4 2 (KS,KW)	
	Overhang Front/Rear	770/705 mm	30.3/27.8 in.	Includes bumper
WEIGHTS	Engine Weight	107 kg	236 lb.	Excludes transmission
	Curb Weight	820 kg (KE) 825 kg (KF,KG,KB,KW,KX) 830 kg (KS, Finland)	1,808 lb. 1,819 lb. 1,830 lb.	
	Weight Distribution (F/R)	500/320 kg (KE) 505/320 kg (KF,KG,KB,KW,KX)	1,102/705 lb. 1,113/705 lb.	
	Max. Permissible Weight (EC)	510/320 kg (KS, Finland) 1,220 kg (KF,KG,KB,KE,KX,KW) 1,150 kg (KS)	1,124/705 lb. 2,690 lb. 2,535 lb.	Curb weight + A/C—21.3 kg (47 lb.) + Cargo—45 kg (100 lb.) + Passengers—68 kg x 2 (150 lb. x 2) + Tolerance = G.V.W.R.
	Gross Vehicle Weight Rating (MVSS)	1,075 kg	2,370 lb.	
	Carrying (loading) Weight	45 kg	100 lb.	
GAPACITIES	Engine Oil: drain and refill (with filter) initial fill	3.5 ℓ 4.0 ℓ	3.7 US qt, 3.1 Imp qt 4.2 US qt, 3.4 Imp qt	
	Transmission Oil: drain and refill initial fill	2.3 ℓ 2.5 ℓ	2.4 US qt, 2.0 Imp qt 2.6 US qt, 2.2 Imp qt	
	Fuel Tank	41 ℓ	10.8 US gal, 9.0 Imp gal	
	Cooling System/Drain and Refill Radiator	5.1/4.1 ℓ	5.4/4.3 US qt, 4.5/3.6 Imp qt	
ENGINE	Type	Water cooled, 4-cycle O.H.C.		
	Cylinder Arrangement	4-cylinder in-line, transverse		
	Bore and Stroke	74 x 86.5 mm	2.91 x 34.1 in.	
	Compression Ratio	8.7 : 1		
	Displacement	1,488 cm ³	91 cu in.	
	Valve Train	Timing belt driven, single overhead camshaft		
	Lubrication System	Trochoid pump		
	Fuel Required	Super or premium grade gasoline with 97 Research Octane Number or higher.		
	Valve Timing	OPEN	10° ATDC	OPEN measurement begins CLOSE measurement ends at point where valve reaches 1 mm lift. Measurement begins/ends at point where valve reaches 0.6 mm lift.
	Intake	CLOSE	40° ABDC	
		OPEN	40° BBDC	
	Exhaust	CLOSE	10° BTDC	
TRANS MISSION	Clutch	Single plate dry, diaphragm spring		
	Transmission	Synchronized 5 forward 1 reverse		
	Primary Reduction	Direct 1 : 1		
	Gear Ratio 1st	2.916		
	2nd	1.764		
	3rd	1.181		
	4th	0.846		
	5th	0.714		
	Reverse	2.916		
	Final Reduction	Single helical gear, 4.266		
	Clutch Lining Area	147 cm ²	22.9 sq. in.	
STEERING	Gear Type	Rack and Pinion		
	Overall Ratio	19.5 : 1		
	Turns, lock-to-lock	4.1		
	Steering Wheel Diameter	370 x 360 mm	14.6 x 14.2 in.	
SUSPENSION	Front	Independent, Strut, Torsion bar spring		
	Rear	Rigid axle, Trailing arm, Coil spring		

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Design Specifications (cont'd)

Coupe

	ITEMS	METRIC	ENGLISH	NOTES
WHEEL ALIGNMENT	Camber Front and Rear		0°/-0°45'	
	Caster		2°25'	
	Toe Front	0 mm	0 in.	
	Rear	IN 2 mm	IN 0.08 in.	
	Steering Axis Inclination		12°55'	
BRAKES	Type, Front	Power assisted ventilated disc		
	Type, Rear	Power assisted drum		
	Lining Surface Area Front/Rear	36/50.2 cm ²	5.6/7.8 sq. in.	
	Effective Disc Diameter	190 mm	7.5 in.	
	Brake Drum I.D.	180 mm	7.1 in.	
	Parking Brake Type	Mechanical expanding, rear two wheel brakes		
TIRES	Size	175/70 HR13		
ELECTRICAL	Battery	12 V - 45 AH		KE and KF
	Starter	12 V - 47 AH		
	Alternator	12 V - 1.0 kW		
	Fuses	12 V - 55 amps		
	Main Fuse	20 A x 3, 15 A x 5, 10 A x 9		
	Headlights	55 A x 1, 45 A x 2 30A x 1 (Sunroof)		
	Speedometer/Gauge Illumination Lights	12 V - 60/55 W		
	Front Turn Signal Lights	12 V - 3.4 W		
	Position Lights	12 V - 21 W		
	Front Side Marker Lights	12 V - 5 W		
	Warning Indicator Lights	12 V - 5 W		
	Interior Light	12 V - 1.4 W		
	Rear Turn Signal Lights	12 V - 5 W (Sunroof 12V-8W)		
	Stop/Taillights	12 V - 21 W		
	Turn Signal Indicator Lights	12 V - 21/5 W		
	Tail Gate Light	12 V - 1.4 W		
	Back-up Lights	12 V - 3.4 W		
	License Lights	12 V - 21 W		
	Rear Fog Lights	12 V - 5 W		
	Heater Panel Light	12 V - 21 W (EC)		
		12 V - 1.4 W, 12 V - 1.2 W		

2D H/B

	ITEMS		METRIC	ENGLISH	NOTES
DIMENSIONS	Overall Length		3,810 mm	150 in.	
	Overall Width		1,635 mm	64.4 in.	
			1,625 mm (KC)	63.9 in.	
	Overall Height		1,340 mm	52.8 in.	
			1,335 mm (KC)	52.6 in.	
	Wheelbase		2,380 mm	93.7 in.	
	Tread Front/Rear		1,400/1,415 mm	55.1/55.7 in.	
	Ground Clearance		165 mm	6.5 in.	
	Seating Capacity		175 mm (KY)	6.9 in.	
	Overhang	Front/Rear	735/695 mm	28.9/27.4 mm	Includes bumper
WEIGHTS	Engine Weight	1500	107 kg	236 lb.	Excludes transmission
		1300	98.6 kg	217.4 lb.	
	Curb Weight				
		1500	5-speed	825 kg (KF,KG,KB,KX,KQ)	1,819 lb.
				830 kg (KW,KS,KE)	1,830 lb.
				866 kg (KC)	1,909 lb.
			Hondamatic	840 kg (KQ)	1,852 lb.
		1300	5-speed	795 kg (KF,KG,KB,KW,KS)	1,753 lb.
				805 kg (KE,KX)	1,775 lb.
				811 kg (KC)	1,788 lb.
			4-speed	814 kg (KC)	1,795 lb.
			Hondamatic	810 kg (KF,KG,KB,KW,KS)	1,786 lb.
				820 kg (KE,KX)	1,808 lb.
				828 kg (KC)	1,825 lb.
		1200	4-speed	775 kg (KP,KT,KU)	1,709 lb.
				785 kg (KG)	1,731 lb.
			Hondamatic	790 kg (KP,KT,KU)	1,742 lb.
	Weight Distribution (F/R)	1500	5-speed	495/330 kg (KF,KG,KB,KX)	1,091/728 lb.
				500/330 kg (KW,KS,KE)	1,102/728 lb.
				520/346 kg (KC)	1,146/763 lb.
				500/325 kg (KQ)	1,102/716 lb.
		1300	5-speed	475/320 kg (KF,KG,KB,KW,KS)	1,047/705 lb.
				475/330 kg (KE,KX)	1,047/728 lb.
				486/325 kg (KC)	1,071/716 lb.
			4-speed	488/326 kg (KC)	1,076/719 lb.
			Hondamatic	490/320 kg (KF,KG,KB,KW,KS)	1,080/705 lb.
				490/330 kg (KE,KX)	1,080/728 lb.
				505/323 kg (KC)	1,113/712 lb.
		1200	4-speed	465/310 kg (KP,KT,KU)	1,025/683 lb.
				470/315 kg (KG)	1,036/694 lb.
			Hondamatic	480/310 kg (KP,KT,KU)	1,058/683 lb.
	Max. Permissible Weight (EC)		1,270 kg	2,800 lb.	Curb weight + A/C—22 kg (49 lb.) + Cargo—45 kg (100 lb.) + Passengers—68 kg x 5 (150 lb. x 5) + Tolerance = G.V.W.R.
			1,200 kg (KS)	2,646 lb.	
	Gross Vehicle Weight Rating (MVSS)		1,290 kg (1500 engine)	2,850 lb.	
			1,240 kg (1300 engine)	2,740 lb.	
	Carring (loading) Weight Capacity		25 kg (KC)	55 lb. (KC)	
CAPACITIES	Engine Oil: drain and refill (with filter)		3.5 ℓ	3.7 US qt, 3.1 Imp qt	
	initial fill		4.0 ℓ	4.2 US qt, 3.4 Imp qt	
	Transmission Oil: drain and refill		2.3 ℓ	2.4 US qt, 2.0 Imp qt	
	initial fill		2.5 ℓ	2.6 US qt, 2.2 Imp qt	
	Hondamatic Fluid: transmission system		2.8 ℓ	3.0 US qt, 2.5 Imp qt	
			5.4 ℓ	5.7 US qt, 4.8 Imp qt	

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Design Specifications (cont'd)

2D H/B

	ITEMS		METRIC		ENGLISH			NOTES
CAPACITIES	Fuel Tank		45 ℓ		11.9 US gal, 9.9 Imp gal			
	Cooling System/ Drain and Refill Radiator	1500	5.2/4.2 ℓ (KC,KF,KG,KB,KW,KS,KE,KX) 5.5/4.5 ℓ (KQ,KY,KP,KD,KT,KU)		5.5/4.4 US qt, 4.6/3.7 Imp qt 5.8/4.8 US qt, 4.8/4.0 Imp qt			
		1300						
		5-speed	4.4/3.4 ℓ (KC,KF,KG,KB,KW,KS,KE,KX) 4.9/3.9 ℓ (KC)		4.7/3.6 US qt, 3.9/3.0 Imp qt			
		4-speed	5.2/4.2 ℓ (KP,KD,KT,KU) 4.9/3.9 ℓ (KC,KF,KG,KB,KW,KS,KE,KX) 5.2/4.2 ℓ (KP,KD,KT,KU)		5.2/4.1 US qt, 4.3/3.4 Imp qt 5.5/4.4 US qt, 4.6/3.7 Imp qt 5.2/4.1 US qt, 4.3/3.4 Imp qt			
		Hondamatic						
		1200						
		4-speed	4.4/3.4 ℓ (KG) 5.2/4.2 ℓ (KP,KT,KU) 4.9/3.9 ℓ (KP,KT,KU)		4.7/3.6 US qt, 3.9/3.0 Imp qt 5.5/4.4 US qt, 4.6/3.7 Imp qt 5.2/4.1 US qt, 4.3/3.4 Imp qt			
		Hondamatic						
ENGINE	Type		Water cooled, 4-cycle O.H.C. 4-cylinder in-line, transverse					OPEN measurement begins CLOSE measurement ends at point where valve reaches 1 mm lift.
	Cylinder Arrangement							
	Bore and Stroke	1500	74 x 86.5 mm		2.91 x 3.41 in.			
		1300	74 x 78 mm		2.91 x 3.07 in.			
		1200	74 x 69 mm		2.91 x 2.72 in.			
	Compression Ratio	1500			8.7 : 1			
		1300			8.7 : 1			
		1200			8.2 : 1			
	Displacement	1500	1,488 cm ³		91 cu in.			
		1300	1,342 cm ³		82 cu in.			
		1200	1,187 cm ³		72 cu in.			
	Valve Train		Timing belt driven, single overhead camshaft					
	Lubrication System		Trochoid pump					
	Fuel Required		Low-lead or regular grade gasoline with 91 Research Octane number or higher.					
	Valve Timing		1500	1300	1200			
Intake	OPEN	10° ATDC * ¹	10° ATDC	15° ATDC * ¹	10° ATDC	15° ATDC		
	CLOSE	20° ABDC	20° ABDC	15° ABDC	15° ABDC	15° ABDC		
Exhaust	OPEN	25° BBDC	25° BBDC	20° BBDC	20° BBDC	15° BBDC		
	CLOSE	10° BTDC	5° BTDC	15° BTDC	10° BTDC	15° BTDC		
TRANS- MISSION	Clutch	4/5-speed	Single plate dry, diaphragm spring					
		Hondamatic	Torque converter					
	Transmission	5-speed	Synchronized 5 forward 1 reverse					
		4-speed	Synchronized 4 forward 1 reverse					
		Hondamatic	3 forward speeds 1 reverse with torque converter					
	Primary Reduction		Direct 1 : 1					
	Gear Ratio	1500	5-speed		Hondamatic			
		1st	2.916		1.782			
		2nd	1.764		1.133			
		3rd	1.181		0.777			
		4th	0.846		—			
		5th	0.714		—			
		Reverse	2.916		1.954			
		Final Reduction	4.266		3.588			
		1300	5-speed		4-speed Hondamatic			
			KS, KX, KC	Others				
		1st	2.916	3.272	2.916		1.782	
		2nd	1.764	1.666	1.764		1.206	
		3rd	1.181	1.041	1.181		0.828	
		4th	0.846	0.777	0.846		—	
		5th	0.714	0.655	—		—	
		Reverse	2.916	2.916	2.916		1.954	
		Final Reduction	4.266	4.066	4.266		3.588	
		1200	4-speed		Hondamatic			
		1st	2.916		1.782			
		2nd	1.764		1.206			
		3rd	1.181		0.828			
	4th	0.846		—				
	5th	—		—				
	Reverse	2.916		1.954				
	Final Reduction	4.428		3.588				
	Clutch Lining Area	147 cm ²		22.9 sq. in.				

*¹ : KS, KX model with Hondamatic

	ITEMS	METRIC	ENGLISH	NOTES
STEERING	Gear Type	Rack and Pinion		1500 Civic S Others
	Overall Ratio	19.5 : 1		
	Turns, lock-to-lock	4.1		
	Steering Wheel Diameter	370 x 360 mm 377 mm	14.6 x 14.2 in. 14.8 in.	
SUSPENSION	Type, Front Type, Rear	Independent, Strut, Torsion bar spring Rigid axle, Trailing arm, Coil spring		
WHEEL ALIGNMENT	Camber Front and Rear	0°/-0°45'		
	Caster	0°20'/-0°45' (KY) 2°20' 2°10' (KY)		
	Toe Front Rear	0 mm IN 2 mm	0 in. IN 0.08 in.	
	Steering Axis Inclination	12°50' 12°30'		
BRAKES	Type, Front	Self-adjusting power-disc brake		
	Type, Rear	Power assisted ventilated disc (1500, KC, KX) Power assisted drum		
	Lining Surface Area Front/Rear	36/50.2 cm ²	5.6/7.9 sq. in.	
	Effective Disc Diameter	190 mm	7.5 in.	
	Brake Drum I.D.	180 mm	7.1 in.	
	Parking Brake Type	Mechanical expanding, rear two wheel brakes		
TIRES	Size	155-13/6.15-13, P165/70R13, 165/70SR13 (KC) 155SR13 (EC-DX, STD) 165/70SR13 (EC-S, EXCEPT KW, KS) 165SR13 (KY) 165/70SR13 (KQ) 155SR13, 6.15-13-4PR (KP, KD, KT, KU)		
ELECTRICAL	Battery	12 V - 47 AH 12 V - 45 AH (KF, KE) 12 V - 40 AH (KQ, KT, KU, KP, KD, KY) 12 V - 1.0 kW, 12 V - 0.8 kW 12 V - 55 amps 20 A x 2, 15 A x 4, 10 A x 9 55 A x 1, 45 A x 2 12 V - 60/55 W 12 V - 65/55 W (KC) 12 V - 45/40 W (KT, KU, KP, KD, KY) 12 V - 3.4 W, 14 V - 1.4 W 12 V - 21 W 12 V - 32 CP (KC) 12 V - 5 W 12 V - 5 W 12 V - 2 CP (KC) 12 V - 1.4 W 12 V - 5 W 12 V - 21 W 12 V - 32 CP (KC) 12 V - 21/5 W 12 V - 32/3 CP (KC) 12 V - 1.4 W 12 V - 3.4 W 12 V - 21 W 12 V - 32 CP (KC) 12 V - 5 W 12 V - 4 CP (KC) 12 V - 21 W (EC) 12 V - 1.4 W		
	Starter			
	Alternator			
	Fuses			
	Main Fuse			
	Headlights			
	Speedometer/Gauge Illumination Lights			
	Front Turn Signal Lights			
	Position Lights			
	Front Side Marker Lights			
	Rear Side Marker Lights			
	Warning Indicator Lights			
	Interior Light			
	Rear Turn Signal Lights			
	Stop/Taillights			
	Turn Signal Indicator Lights			
	Tail Gate Light			
	Back-up Lights			
	License Lights			
	Rear Fog Lights			
	Heater Panel Light			

(cont'd)

Design Specifications (cont'd)

4 D

	ITEMS		METRIC	ENGLISH	NOTES
DIMENSIONS	Overall Length	1500	4,145 mm	163.2 in.	
	Overall Width		1,630 mm	64.2 in.	
		1200/1300	1,625 mm (KC)	63.9 in.	
			1,625 mm	64.0 in.	
	Overall Height		1,385 mm	54.5 in.	
	Wheelbase		2,450 mm	96.5 in.	
	Tread F/R		1,400/1,415 mm	55.1/55.7 in.	
	Ground Clearance		165 mm	6.3 in.	
	Seating Capacity				5
	Overhang F/R		795/900 mm	31.3/35.4 in.	Includes bumper
WEIGHTS	Engine Weight	1500	107 kg	236 lb.	Excludes transmission
	(Wet)	1300	98.6 kg	217.4 lb.	
	Curb Weight	1500			
		5-speed	845 kg (KX)	1,863 lb.	
			850 kg (KW,KS)	1,874 lb.	
			865 kg (KQ)	1,907 lb.	
			874 kg (KC)	1,927 lb.	
		Hondamatic	860 kg (KX)	1,896 lb.	
			865 kg (KW,KS)	1,907 lb.	
			880 kg (KQ)	1,940 lb.	
			889 kg (KC)	1,960 lb.	
		1300			
		5-speed	815 kg (KX)	1,797 lb.	
			820 kg (KW,KS)	1,808 lb.	
		Hondamatic	830 kg (KX)	1,830 lb.	
			835 kg (KW,KS)	1,841 lb.	
		1200			
		4-speed	810 kg (KP,KT,KU)	1,786 lb.	
		Hondamatic	825 kg (KP,KT,KU)	1,819 lb.	
	Weight Distribution (F/R)	1500			
		5-speed	500/345 kg (KX)	1,102/761 lb.	
			505/345 kg (KW,KS)	1,113/761 lb.	
			517/357 kg (KC)	1,140/787 lb.	
		Hondamatic	515/345 kg (KX)	1,135/761 lb.	
			520/345 kg (KW,KS)	1,146/761 lb.	
			534/355 kg (KC)	1,177/783 lb.	
		1300			
		5-speed	480/335 kg (KX)	1,058/739 lb.	
			485/335 kg (KW,KS)	1,069/739 lb.	
		Hondamatic	495/335 kg (KX)	1,091/739 lb.	
			500/335 kg (KW,KS)	1,102/739 lb.	
		1200			
		4-speed	480/330 kg (KP,KT,KU)	1,058/728 lb.	
		Hondamatic	495/330 kg (KP,KT,KU)	1,091/728 lb.	
	Max. permissible weight (EC)		1,300 kg	2,866 lb.	
			1,240 kg (KS)	2,734 lb.	
	Gross Vehicle Weight Rating (MVSS)		1,320 kg	2,910 lb.	
	Carrying (loading) Weight Capacity		25 kg	55 lb.	
CAPACITIES	Engine Oil:				
	drain and refill (with filter)		3.5 ℓ	3.7 US qt, 3.1 Imp qt	
	initial fill		4.0 ℓ	4.2 US qt, 3.4 Imp qt	
	Transmission Oil:				
	drain and refill		2.3 ℓ	2.4 US qt, 2.0 Imp qt	
	initial fill		2.5 ℓ	2.6 US qt, 2.2 Imp qt	
	Hondamatic Fluid:				
	transmission		2.8 ℓ	3.0 US qt, 2.5 Imp qt	
	system		5.4 ℓ	5.7 US qt, 4.8 Imp qt	
	Fuel Tank		46 ℓ	12.1 US gal, 10.1 Imp gal	
	Cooling System/	1500	5.2/4.2 ℓ (KC,KW,KS,KX)	5.5/4.4 US qt, 4.6/3.7 Imp qt	
	Drain and Refill		5.5/4.5 ℓ	5.8/4.8 US qt, 4.8/4.0 Imp qt	
	Radiator		(KO,KY,KP,KD,KT,KU)		
		1300			
	5-speed		4.4/3.4 ℓ (KF,KW,KS,KE,KX)	4.7/3.6 US qt, 3.9/3.0 Imp qt	
			4.9/3.9 ℓ (KT,KU)	5.2/4.1 US qt, 4.3/3.4 Imp qt	
	4-speed		5.2/4.2 ℓ (KP,KD,KT,KU)	5.5/4.4 US qt, 4.6/3.7 Imp qt	
	Hondamatic		4.9/3.9 ℓ (KF,KW,KS,KE,KX)	5.2/4.1 US qt, 4.3/3.4 Imp qt	
			5.2/4.2 ℓ (KP,KD,KT,KU)	5.5/4.4 US qt, 4.6/3.7 Imp qt	
		1200			
	4-speed		5.2/4.2 ℓ (KP,KT,KU)	5.5/4.4 US qt, 4.6/3.7 Imp qt	
	Hondamatic		4.9/3.9 ℓ (KP,KT,KU)	5.2/4.1 US qt, 4.3/3.4 Imp qt	

Curb weight
+ A/C—22 kg (49 lb.)
+ Cargo—25 kg (55 lb.)
+ Passengers—70 kg x 5
(154 lb. x 5)
+ Tolerance
= G.V.W.R.

	ITEMS		METRIC		ENGLISH		NOTES
ENGINE	Type		Water cooled, 4-cycle O.H.C.				OPEN measurement begins CLOSE measurement ends at point where valve reaches 1 mm lift.
	Cylinder Arrangement		4-cylinder in-line, transverse				
	Bore and Stroke	1500	74 x 86.5 mm		2.91 x 3.41 in.		
		1300	74 x 78 mm		2.91 x 3.07 in.		
		1200	74 x 69 mm		2.91 x 2.72 in.		
	Compression Ratio	1500	8.7 : 1		9.2 : 1 (KC)		
		1300	8.7 : 1				
		1200	8.2 : 1				
	Displacement	1500	1,488 cm ³		91 cu in.		
		1300	1,342 cm ³		82 cu in.		
	1200	1,187 cm ³		72 cu in.			
Valve Train		Timing belt driven, single overhead camshaft					
Lubrication System		Trochoid pump					
Fuel Required		Low-lead or regular grade gasoline with 91 Research Octane number or higher.					
Valve Timing		1500	1300		1200		
Intake	OPEN	10° ATDC *1	10° ATDC	15° ATDC *1	10° ATDC	15° ATDC	
	CLOSE	20° ABDC	20° ABDC	15° ABDC	15° ABDC	15° ABDC	
Exhaust	OPEN	25° BBDC	25° BBDC	20° BBDC	20° BBDC	15° BBDC	
	CLOSE	10° BTDC	5° BTDC	15° BTDC	10° BTDC	15° BTDC	
TRANS-MISSION	Clutch	4/5-speed Hondamatic	Single plate dry, diaphragm spring Torque converter				
	Transmission	5-speed 4-speed Hondamatic	Synchronized 5 forward 1 reverse Synchronized 4 forward 1 reverse 3 forward speeds 1 reverse with torque converter Direct 1 : 1				
	Primary Reduction						
	Gear Ratio	1500	5-speed		Hondamatic		
		1st	2.916		1.782		
		2nd	1.764		1.133		
		3rd	1.181		0.777		
		4th	0.846		—		
		5th	0.714		—		
		Reverse	2.916		1.954		
		Final Reduction	4.266		3.588		
		1300	5-speed		4-speed Hondamatic		
		1st	2.916		2.916 1.782		
		2nd	1.764		1.764 1.206		
		3rd	1.181		1.181 0.828		
		4th	0.846		0.846 —		
		5th	0.714		— —		
		Reverse	2.916		2.916 1.954		
		Final Reduction	4.428		4.428 3.588		
		1200	4-speed		Hondamatic		
	1st	2.916		1.782			
	2nd	1.764		1.206			
	3rd	1.181		0.828			
	4th	0.846		—			
	5th	0.714		—			
	Reverse	2.916		1.954			
	Final Reduction	4.642		3.588			
	Clutch Lining Area	147 cm ²		22.9 sq. in.			
STEERING	Gear Type	Rack and Pinion					
	Overall Ratio	19.5 : 1					
	Turns, lock-to-lock	4.1					
	Steering Wheel Diameter	377 mm	14.8 in.				
SUSPENSION	Type, Front	Independent, Strut, Torsion bar spring					
	Type, Rear	Rigid axle, Trailing arm, Coil spring					
WHEEL ALIGNMENT	Camber	Front and Rear	0° / -0°45'				
	Caster		0°30' / -0°45' (KY)				
			2°20'				
	Toe		2°15' (KY)				
			0 mm 0 in.				
	Steering Axis Inclination		IN 2 mm IN 0.08 in.				
			12°50'				
			12°15' (KY)				
BRAKES	Type, Front	Self-adjusting power-disc brake					
	Type, Rear	Power assisted ventilated disc (1500 KC, KX)					
	Lining Surface Area Front/Rear	Power assisted drum					
	Effective Disc Diameter	36/50.2 cm ²	5.6/7.9 sq. in.				
	Brake Drum I.D.	190 mm	7.5 in.				
		180 mm	7.1 in.				
	Parking Brake Type	Mechanical expanding, rear two wheel brakes					

*1 : KS, KX model with Hondamatic

(cont'd)

Design Specifications (cont'd)

4 D

	ITEMS	METRIC	ENGLISH	NOTES
TIRES	Size	165/70SR13 (KC, 1500 KX) 155SR13 (KF, KS, KW, KX, KE) 165SR13 (KY) 165/70SR13 (KQ) 155SR13, 6.15-13-4PR (KP, KD, KT, KU)		
ELECTRICAL	Battery Starter Alternator Fuses Main Fuse Headlights Speedometer/Gauge Illumination Lights Front Turn Signal Lights Position Lights Front Side Marker Lights Rear Side Marker Lights Warning Indicator Lights Interior Light Rear Turn Signal Lights Stop/Taillights Turn Signal Indicator Lights Tail Gate Light Back-up Lights License Lights Rear Fog Lights Heater Panel Light	12 V - 47 AH 12 V - 45 AH (KE) 12 V - 40 AH (KQ, KT, KU, KP, KD, KY) 12 V - 1.4 kW, 12 V - 1.0 kW, 12 V - 0.8 kW 12 V - 55 amps 20 A x 2, 15 A x 4, 10 A x 9 55 A x 1, 45 A x 2 12 V - 60/55 W 12 V - 65/55 W (KC) 12 V - 45/40 W (KT, KU, KP, KD, KY) 12 V - 3.4 W, 14 V - 1.4 W 12 V - 21 W 12 V - 32 CP (KC) 12 V - 5 W 12 V - 5 W 12 V - 2 CP (KC) 12 V - 1.4 W 12 V - 5 W 12 V - 21 W 12 V - 32 CP (KC) 12 V - 21/5 W 12 V - 32/3 CP (KC) 12 V - 1.4 W 12 V - 3.4 W 12 V - 21 W 12 V - 32 CP (KC) 12 V - 5 W 12 V - 4 CP (KC) 12 V - 21 W (EC) 12 V - 1.4 W		

4D H/B

	ITEMS	METRIC	ENGLISH	NOTES
DIMENSIONS	Overall Length	3,990 mm	157.1 in.	
	Overall Width	1,650 mm	65.0 in.	
		1,625 mm (KQ)	63.9 in.	
		1,624 mm (KC)	63.9 in.	
	Overall Height	1,490 mm	58.7 in.	
		1,480 mm (KC)	58.3 in.	
	Wheelbase	2,450 mm	96.5 in.	
	Tread F/R	1,400/1,415 mm	55.1/55.7 in.	
	Ground Clearance	165 mm	6.5 in.	
		175 mm (KY)	6.9 in.	
WEIGHTS	Seating Capacity			
	Overhang F/R	745/795 mm	29.3/31.3 in.	Includes bumper
	Engine Weight (Wet)	107 kg	236 lb.	
	Curb Weight	5-speed	875 kg (KP,KD,KT,KU)	1,929 lb.
			885 kg (KF,KG,KB,KX)	1,951 lb.
			890 kg (KW,KS,KE)	1,962 lb.
			900 kg (KY)	1,984 lb.
			905 kg (KQ)	1,995 lb.
			911 kg (KC)	2,008 lb.
		Hondamatic	890 kg (KP,KD,KT,KU)	1,962 lb.
WEIGHTS			900 kg (KF,KG,KB,KX)	1,984 lb.
			905 kg (KW,KS,KE)	1,995 lb.
			915 kg (KY)	2,017 lb.
			920 kg (KQ)	2,028 lb.
			927 kg (KC)	2,044 lb.
	Weight Distribution	5-speed	505/370 kg (KP,KD,KT,KU)	1,113/816 lb.
	(F/R)		505/380 kg (KF,KG,KB,KX)	1,113/838 lb.
			510/380 kg (KW,KS)	1,124/838 lb.
			505/385 kg (KE)	
			515/385 kg (KY)	1,135/849 lb.
WEIGHTS			520/385 kg (KQ)	1,146/849 lb.
			530/381 kg (KC)	1,168/840 lb.
		Hondamatic	520/370 kg (KP,KD,KT,KU)	1,146/816 lb.
			520/380 kg (KF,KG,KB,KX)	1,146/838 lb.
			525/380 kg (KW,KS)	1,157/838 lb.
			520/385 kg (KE)	
			530/385 kg (KY)	1,168/849 lb.
			535/385 kg (KQ)	1,179/849 lb.
			548/379 kg (KC)	1,208/836 lb.
	Max. permissible weight (EC)	1,365 kg	3,009 lb.	
WEIGHTS	Gross Vehicle Weight Rating (MVSS)	1,300 kg (KS)	2,866 lb.	
	Carrying (loading) Weight Capacity	1,365 kg	3,010 lb.	
		45 kg	100 lb.	
				Curb weight +A/C—22 kg (49 lb.)
				+Cargo—45 kg (100 lb.)
				+Passengers—70 kg x 5 (154 lb.) x 5
				+Sun roof—20 kg (44 lb.)
				+Tolerance =G.V.W.R.
CAPACITIES	Engine Oil:			
	drain and refill (with filter)	3.5 ℓ	3.7 US qt, 3.1 Imp qt	
	initial fill	4.0 ℓ	4.2 US qt, 3.4 Imp qt	
	Transmission Oil:			
	drain and refill	2.3 ℓ	2.4 US qt, 2.0 Imp qt	
	initial fill	2.5 ℓ	2.6 US qt, 2.2 Imp qt	
	Hondamatic Fluid:			
	transmission	2.8 ℓ	3.0 US qt, 2.5 Imp qt	
	system	5.4 ℓ	5.7 US qt, 4.8 Imp qt	
	Fuel Tank	46 ℓ	12.1 US gal, 10.1 Imp gal	
CAPACITIES	Cooling System/Drain and Refill Radiator	5.1/4.1 ℓ (KC,KF,KG,KB,KW,KS,KE,KX)	5.5/4.4 US qt, 4.6/3.7 Imp qt	
		5.5/4.5 ℓ (KQ,KY,KP,KD,KT,KU)	5.8/4.8 US qt, 4.8/4.0 Imp qt	
ENGINE	Type	Water cooled, 4-cycle O.H.C.		
	Cylinder Arrangement	4-cylinder in-line, transverse		
	Bore and Stroke	74 x 86.5 mm	2.91 x 3.4 in.	
	Compression Ratio	8.7 : 1		
		9.2 : 1 (KC)		
	Displacement	1,488 cm ³	91 cu in.	
	Valve Train	Timing belt driven, single overhead camshaft		
	Lubrication System	Trochoid pump		
	Fuel Required	Low-lead or regular grade gasoline with 91 Research Octane number or higher.		
	Valve Timing			
ENGINE	Intake	OPEN	10° ATDC	
		CLOSE	20° ABDC	
	Exhaust	OPEN	25° BBDC	
		CLOSE	5° BTDC	
ENGINE				OPEN measure-ment beings
				CLOSE measure-ment ends at point where valve reaches 1 mm lift.

*1: KS, KX model with Hondamatic

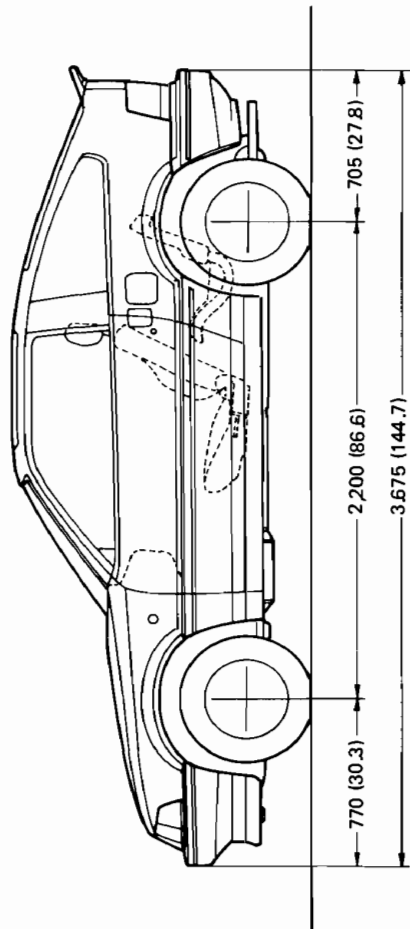
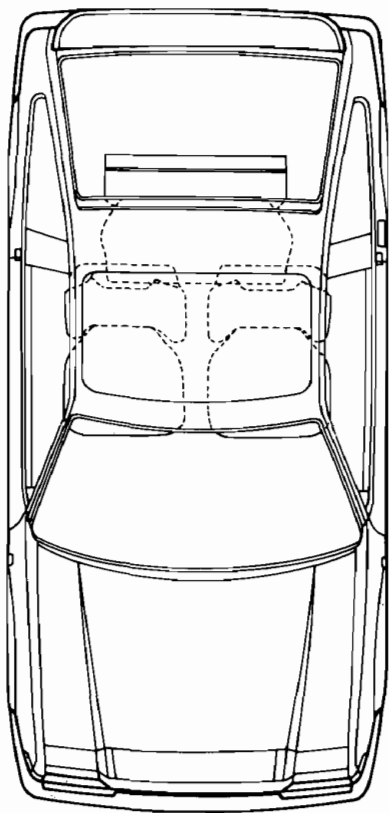
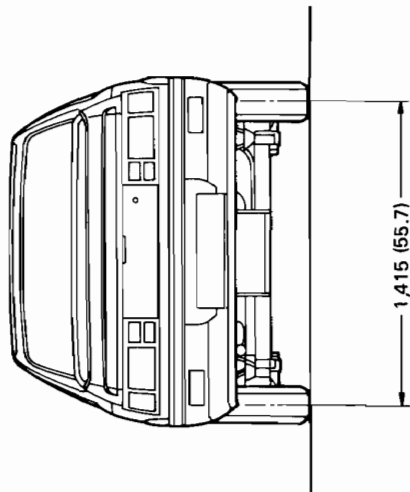
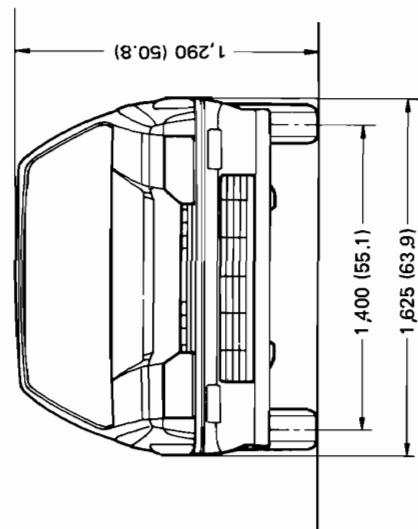
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4D H/B

3-20

Body Specifications

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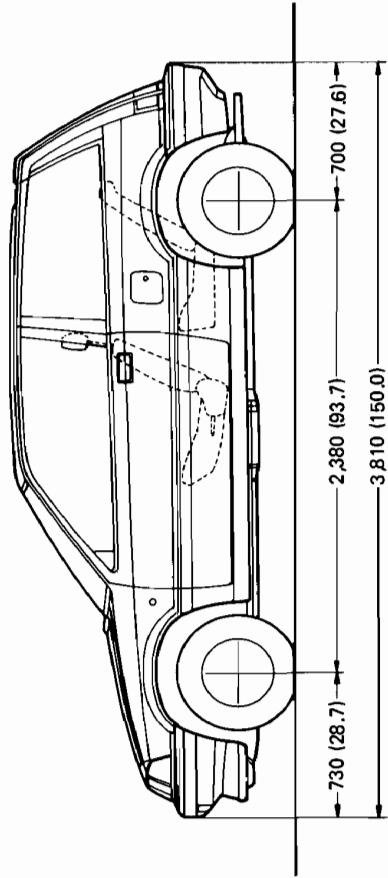
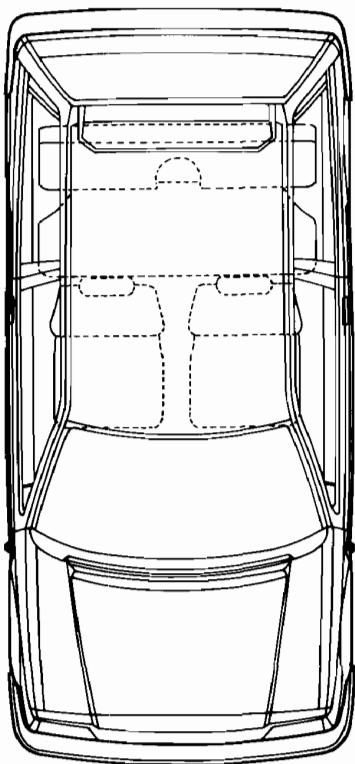
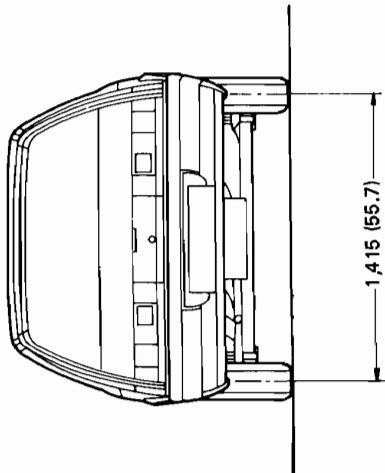
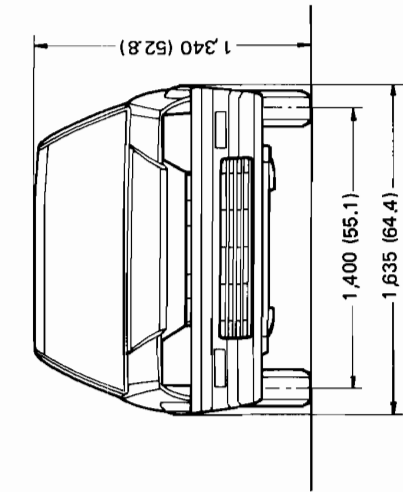


Unit: mm (in.)

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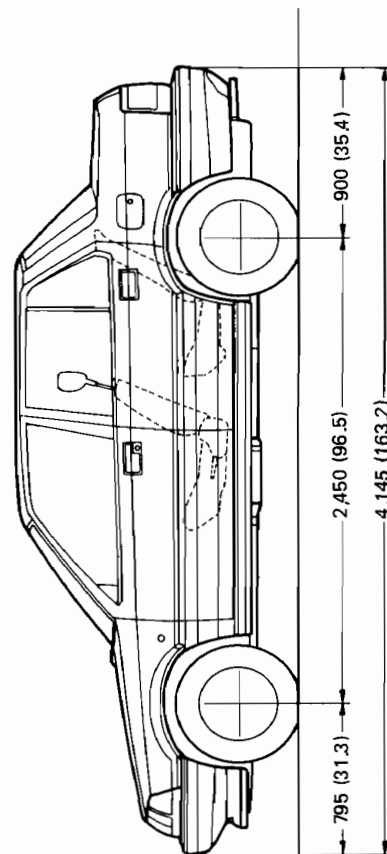
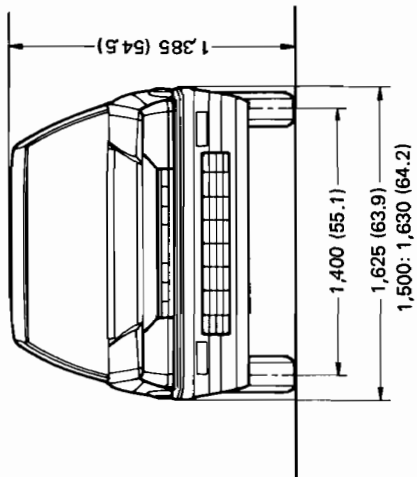
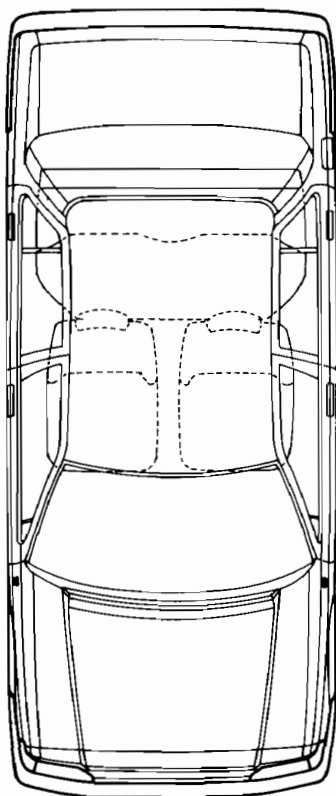
Body Specifications (cont'd)

2D H/B



Unit: mm (in.)

4D

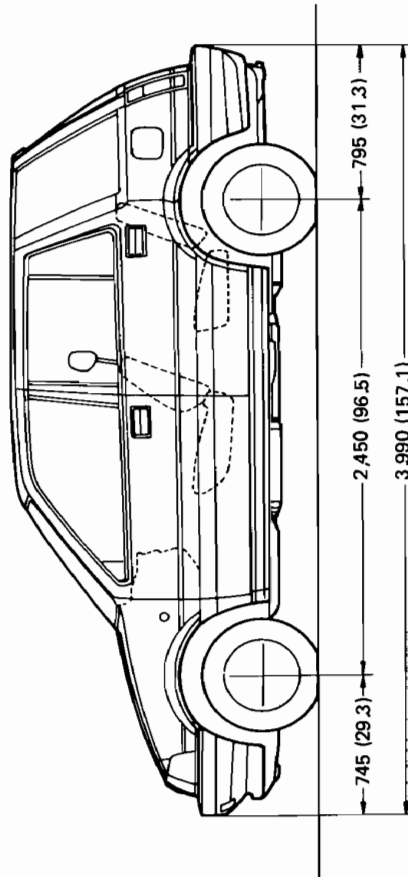
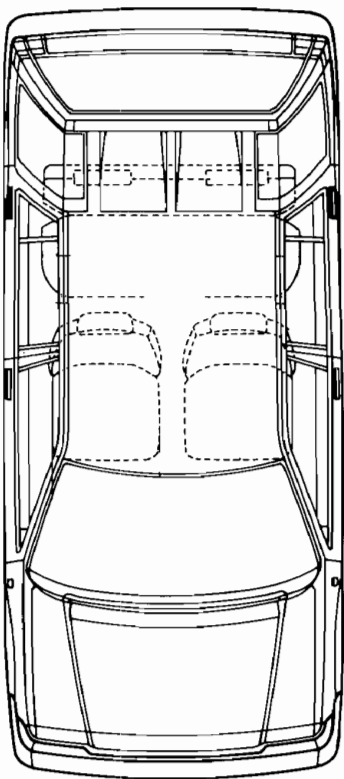
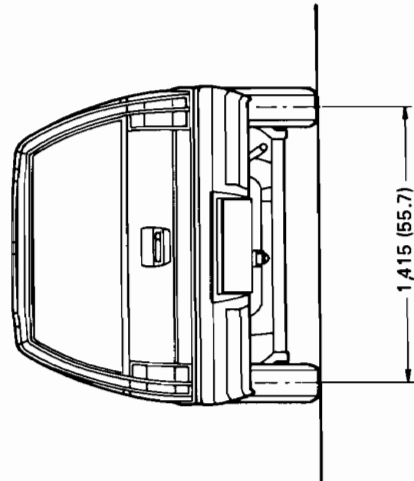
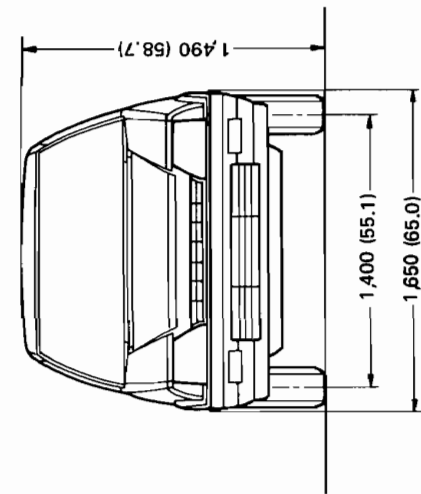


Unit: mm (in.)

(cont'd)

Body Specifications (cont'd)

4D H/B

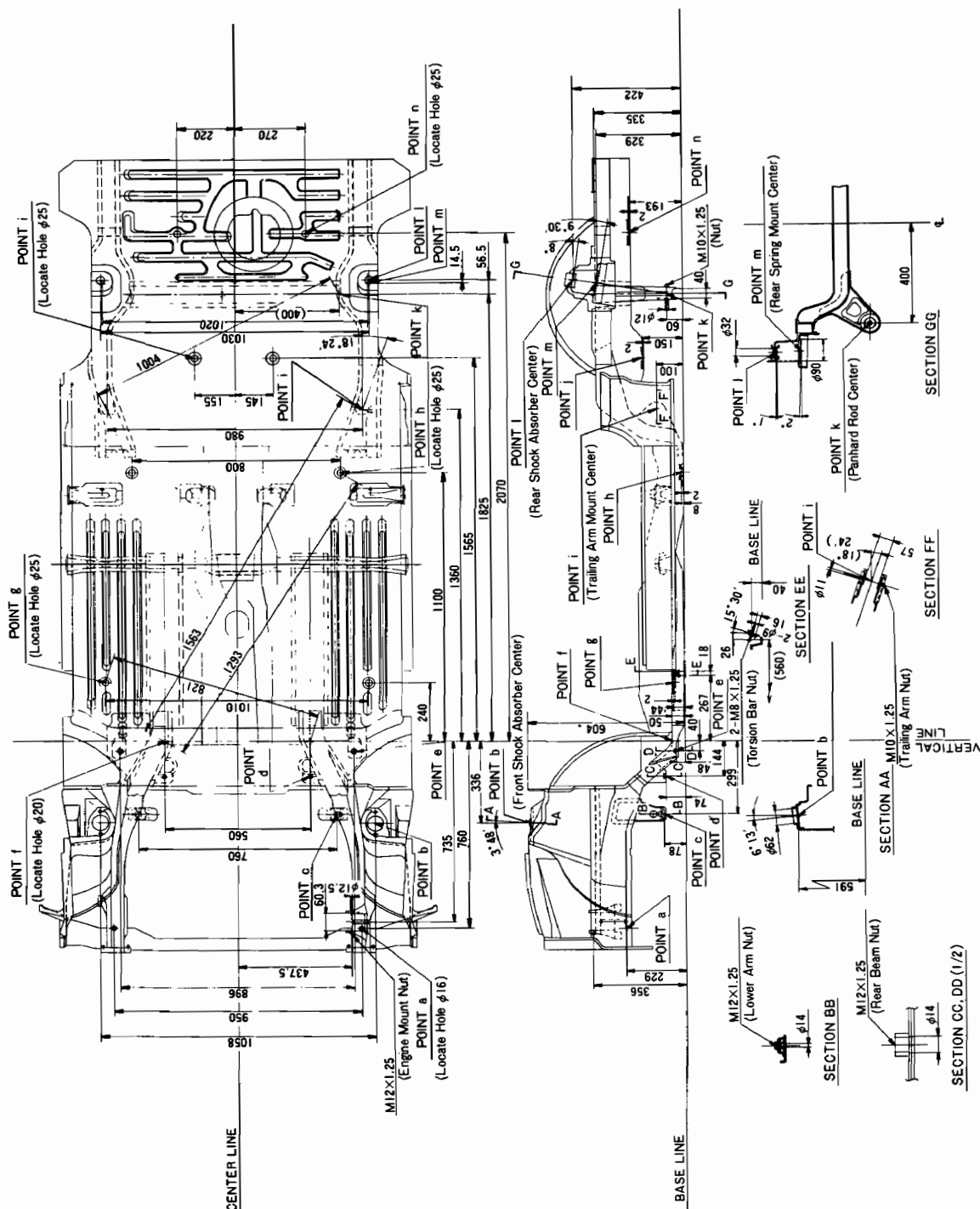


Unit: mm (in.)

Frame Repair Chart

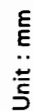
Coupe

specs



Unit : mm

(cont'd)



(cont'd)

4D H/B

Unit : mm



Required Maintenance Schedule

CRX

SERVICE AT THE INTERVAL OF LISTED KM (MILES) OR MONTHS, WHICHEVER OCCURS FIRST.						
ITEMS	x 1,000 km	20	40	60	80	100
	x 1,000 miles	12	24	36	48	60
	months	12	24	36	48	60
IDLE SPEED AND IDLE CO		I	I	I	I	I
VALVE CLEARANCE		I	I	I	I	I
ALTERNATOR DRIVE BELT			I		I	
■ ENGINE OIL AND OIL FILTER		Replace every 10,000 km (6,000 miles) or 6 months				
■ TRANSMISSION OIL			R		R	
■ RADIATOR COOLANT					R*	
COOLING SYSTEM, HOSES AND CONNECTIONS			I		I	
AIR CLEANER ELEMENT		R	R	R	R	R
FUEL FILTER			R		R	
TANK, FUEL LINE AND CONNECTIONS			I		I	
IGNITION TIMING AND CONTROL SYSTEM			I		I	
SPARK PLUGS		R	R	R	R	R
DISTRIBUTOR CAP AND ROTOR			I		I	
IGNITION WIRING			I		I	
PCV VALVE			R		R	
BRAKE HOSES, LINES		I	I	I	I	I
BRAKE FLUID			R		R	
REAR BRAKES			I		I	
FRONT BRAKE DISCS AND CALIPERS		I	I	I	I	I
FRONT BRAKE PADS		Inspect every 10,000 km (6,000 miles) or 6 months				
PARKING BRAKE		I	I		I	
CLUTCH RELEASE ARM TRAVEL		I	I	I	I	I
ENGINE EXHAUST SILENCER, SUSPENSION MOUNTING BOLTS		I	I	I	I	I
FRONT WHEEL ALIGNMENT		I	I	I	I	I
STEERING OPERATION, TIE ROD ENDS, STEERING GEAR BOX AND BOOTS		I	I		I	

R—Replace I—Inspect. After inspection, clean, adjust, repair or replace if necessary

■ REMARK: Day to day care (such as oil, coolant check and replenishment) should be done practically according to the Owner's Manual.

* Thereafter, replace every 2 years or 48,000 km (30,000 miles), whichever comes first.

CAUTION: The following items must be serviced more frequently on cars normally used under severe driving conditions. Refer to the chart below for the appropriate maintenance intervals.

"Severe driving conditions" include:

- A: Repeated short-distance driving
- B: Driving in dusty conditions
- C: Driving in severe, cold weather
- D: Driving in areas using road salt or other corrosive materials
- E: Driving on rough and/or muddy roads
- F: Towing a trailer

R—Replace

I—Inspect. After inspection, clean, adjust, repair or replace if necessary.

Condition	Maintenance item	Maintenance operation	Interval
A, B, F	Engine oil and oil filter	R	Every 5,000 km (3,000 miles) or 3 months
A, B, D, E, F	Front brake discs and calipers	I	Every 10,000 km (6,000 miles) or 6 months
A, B, C, E, F	Clutch release arm travel	I	Every 10,000 km (6,000 miles) or 6 months



2D H/B and 4D

SERVICE AT THE INTERVAL OF LISTED KM (MILES) OR MONTHS, WHICHEVER OCCURS FIRST.						
ITEMS	x 1,000 km	20	40	60	80	100
	x 1,000 miles	12	24	36	48	60
	months	12	24	36	48	60
IDLE SPEED AND IDLE CO		I	I	I	I	I
VALVE CLEARANCE		I	I	I	I	I
ALTERNATOR DRIVE BELT			I		I	
ENGINE OIL AND OIL FILTER		Replace every 10,000 km (6,000 miles) or 6 months				
TRANSMISSION OIL	MANUAL		R		R	
TRANSMISSION OIL**	HONDAMATIC	R		R		R
RADIATOR COOLANT					R*	
COOLING SYSTEM, HOSES AND CONNECTIONS			I		I	
AIR CLEANER ELEMENT		R	R	R	R	R
FUEL FILTER			R		R	
INTAKE AIR TEMP. CONTROL SYSTEM**						I
TANK, FUEL LINE AND CONNECTIONS			I		I	
THROTTLE CONTROL SYSTEM**			I		I	
CHOKE MECHANISM*			I		I	
CHARCOAL CANISTER**						I
TWO-WAY VALVE**						I
IGNITION TIMING AND CONTROL SYSTEM			I		I	
SPARK PLUGS		R	R	R	R	R
DISTRIBUTOR CAP AND ROTOR			I		I	
IGNITION WIRING			I		I	
PCV VALVE			R		R	
BLOW-BY FILTER*			R		R	
BRAKE HOSES, LINES		I	I	I	I	I
BRAKE FLUID			R		R	
REAR BRAKES			I		I	
FRONT BRAKE DISCS AND CALIPERS		I	I	I	I	I
FRONT BRAKE PADS		Inspect every 10,000 km (6,000 miles) or 6 months				
PARKING BRAKE		I	I		I	
CLUTCH RELEASE ARM TRAVEL		I	I	I	I	I
ENGINE EXHAUST SILENCER, SUSPENSION MOUNTING BOLTS		I	I	I	I	I
FRONT WHEEL ALIGNMENT		I	I	I	I	I
STEERING OPERATION, TIE ROD ENDS, STEERING GEAR BOX AND BOOTS		I	I		I	
POWER STEERING PUMP BELT*			I		I	
POWER STEERING SYSTEM*		I	I	I	I	I

R—Replace I—Inspect. After inspection, clean, adjust, repair or replace if necessary

■ **REMARK:** Day to day care (such as oil, coolant check and replenishment) should be done practically according to the Owner's Manual.

*1 Thereafter, replace every 2 years or 48,000 km (30,000 miles), whichever comes first.

*2 Only for manual transmission on KW, KS and KQ types

*3 Only for KQ and KY types

*4 Except for cars equipped with Fuel Injection Engine.

*5 Cars equipped with power steering.

CAUTION: The following items must be serviced more frequently on cars normally used under severe driving conditions. Refer to the chart below for the appropriate maintenance intervals.

"Severe driving conditions" include:

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- B: Driving in dusty conditions
- C: Driving in severe, cold weather
- D: Driving in areas using road salt or other corrosive materials
- E: Driving on rough and/or muddy roads
- F: Towing a trailer

R—Replace

I—Inspect. After inspection, clean, adjust, repair or replace if necessary.

Condition	Maintenance item	Maintenance operation	Interval
A, B, F	Engine oil and oil filter	R	Every 5,000 km (3,000 miles) or 3 months
A, B, D, E, F	Front brake discs and calipers	I	Every 10,000 km (6,000 miles) or 6 months
A, B, C, E, F	Clutch release arm travel	I	Every 10,000 km (6,000 miles) or 6 months
B, C, E	Power steering system	I	Every 10,000 km (6,000 miles) or 6 months

Required Maintenance Schedule

4D H/B

SERVICE AT THE INTERVAL OF LISTED KM (MILES) OR MONTHS, WHICHEVER OCCURS FIRST.						
ITEMS	x 1,000 km	20	40	60	80	100
	x 1,000 miles	12	24	36	48	60
	months	12	24	36	48	60
IDLE SPEED AND IDLE CO		I	I	I	I	I
VALVE CLEARANCE		I	I	I	I	I
ALTERNATOR DRIVE BELT			I		I	
■ ENGINE OIL AND OIL FILTER		Replace every 10,000 km (6,000 miles) or 6 months				
■ TRANSMISSION OIL	MANUAL		R		R	
■ TRANSMISSION OIL	HONDAMATIC	R		R		R
■ RADIATOR COOLANT					R*1	
COOLING SYSTEM, HOSES AND CONNECTIONS			I		I	
AIR CLEANER ELEMENT		R	R	R	R	R
FUEL FILTER			R		R	
INTAKE AIR TEMP. CONTROL SYSTEM						I
TANK, FUEL LINE AND CONNECTIONS			I		I	
THROTTLE CONTROL SYSTEM*2			I		I	
CHOKE MECHANISM			I		I	
CHARCOAL CANISTER*3						I
TWO-WAY VALVE*3						I
IGNITION TIMING AND CONTROL SYSTEM			I		I	
SPARK PLUGS		R	R	R	R	R
DISTRIBUTOR CAP AND ROTOR			I		I	
IGNITION WIRING			I		I	
PCV VALVE			R		R	
BLOW-BY FILTER			R		R	
BRAKE HOSES, LINES		I	I	I	I	I
BRAKE FLUID			R		R	
REAR BRAKES			I		I	
FRONT BRAKE DISCS AND CALIPERS		I	I	I	I	I
FRONT BRAKE PADS		Inspect every 10,000 km (6,000 miles) or 6 months				
PARKING BRAKE		I	I		I	
CLUTCH RELEASE ARM TRAVEL		I	I	I	I	I
ENGINE EXHAUST SILENCER, SUSPENSION MOUNTING BOLTS		I	I	I	I	I
FRONT WHEEL ALIGNMENT		I	I	I	I	I
STEERING OPERATION, TIE ROD ENDS, STEERING GEAR BOX AND BOOTS		I	I		I	
POWER STEERING PUMP BELT			I		I	
POWER STEERING SYSTEM		I	I	I	I	I

R—Replace

I—Inspect. After inspection, clean, adjust, repair or replace if necessary

■ **REMARK:** Day to day care (such as oil, coolant check and replenishment) should be done practically according to the Owner's Manual.

*1 Thereafter, replace every 2 years or 48,000 km (30,000 miles), whichever comes first.

*2 Only for manual transmission on KW, KS and KQ types

*3 Only for KQ and KY types

CAUTION: The following items must be serviced more frequently on cars normally used under severe driving conditions. Refer to the chart below for the appropriate maintenance intervals.

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- E: Driving on rough and/or muddy roads
- F: Towing a trailer

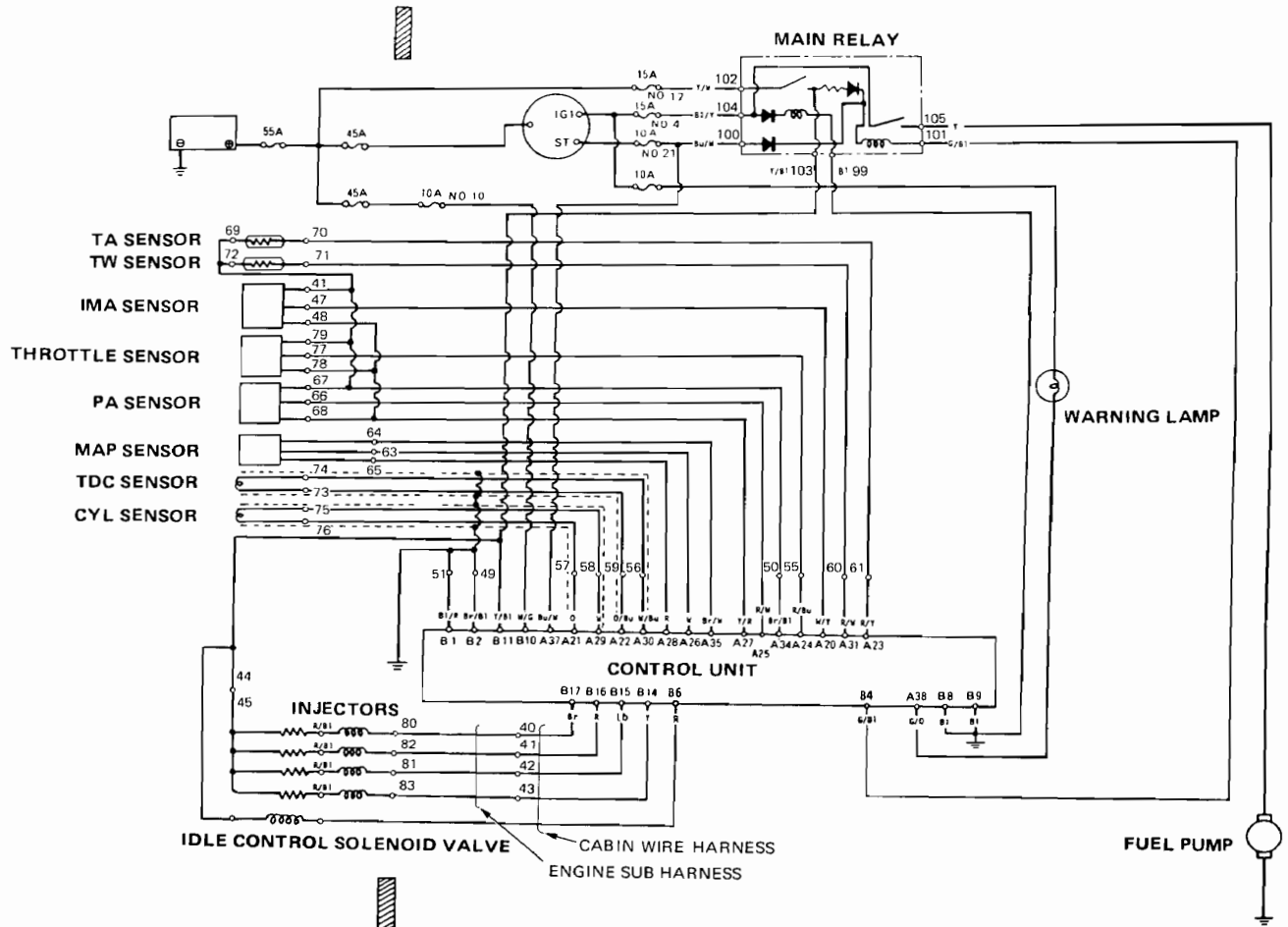
R—Replace

I—Inspect. After inspection, clean, adjust, repair or replace if necessary.

Condition	Maintenance item	Maintenance operation	Interval
A, B, F	Engine oil and oil filter	R	Every 5,000 km (3,000 miles) or 3 months
A, B, D, E, F	Front brake discs and calipers	I	Every 10,000 km (6,000 miles) or 6 months
A, B, C, E, F	Clutch release arm travel	I	Every 10,000 km (6,000 miles) or 6 months
B, C, E	Power steering system	I	Every 10,000 km (6,000 miles) or 6 months

PGM-FI System

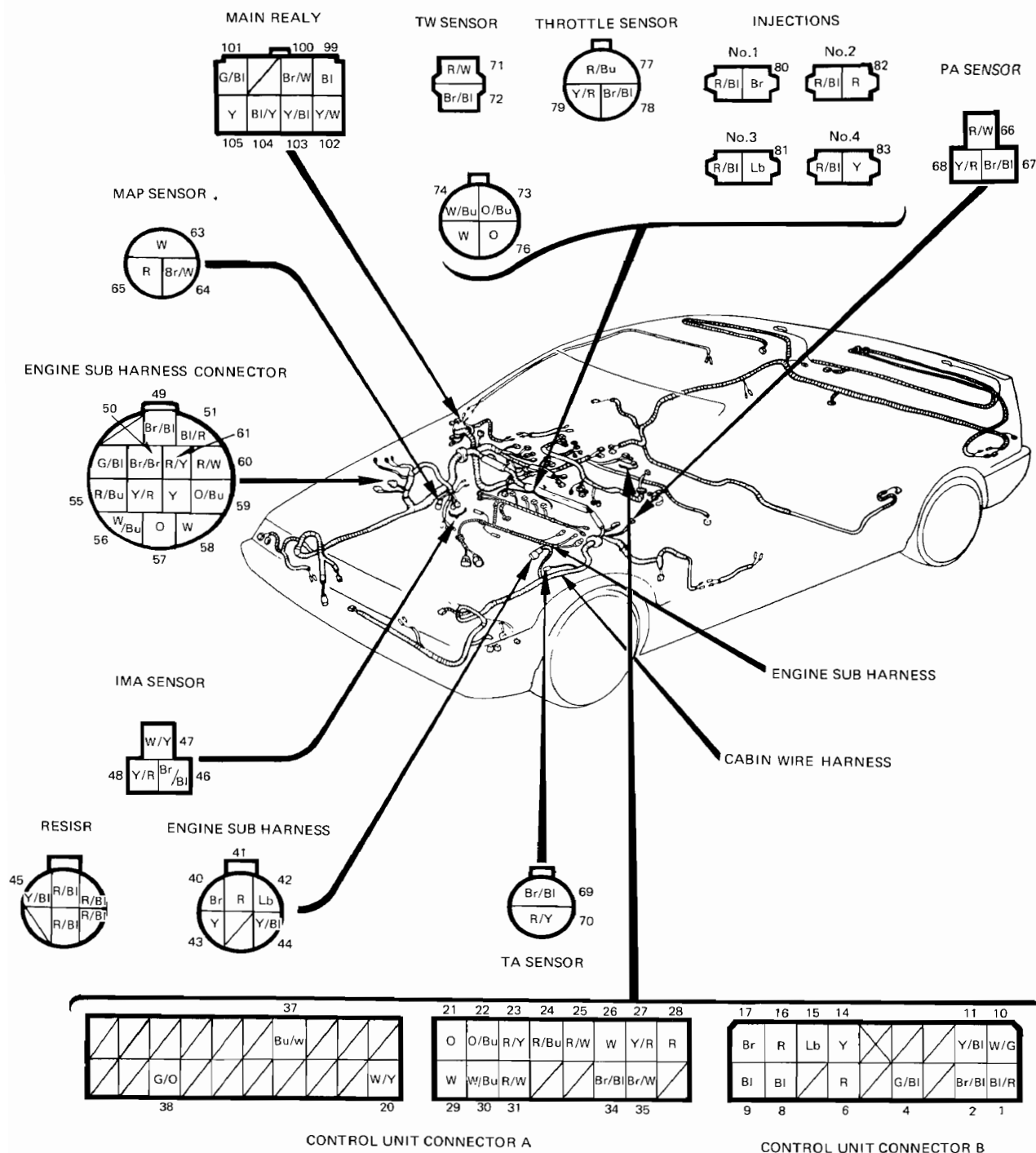
Wiring Diagram





Wire Arrangement in Connectors

NOTE: THE ILLUSTRATED WIRE ARRANGEMENT IS THE VIEW FROM THE WIRE SIDE OF THE CONNECTOR.

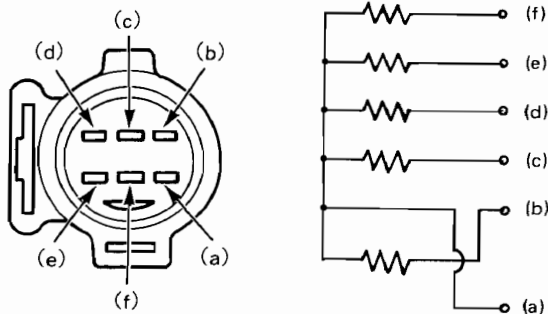


Resistor

Testing

1. Disconnect the connectors of the resistor.
2. Measure resistance between a terminal and each

Resistance should be 5–7 Ω .



- If resistance is outside above ranges, replace the resistor.

Fuel Pump

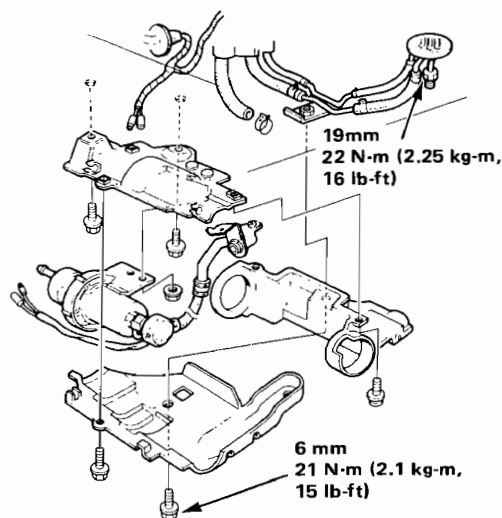
Replacement

(2D H/B Only)

WARNING

- Do not smoke while working on fuel system. Keep open flame away from work area.
- Block front wheels before jacking up rear of car.

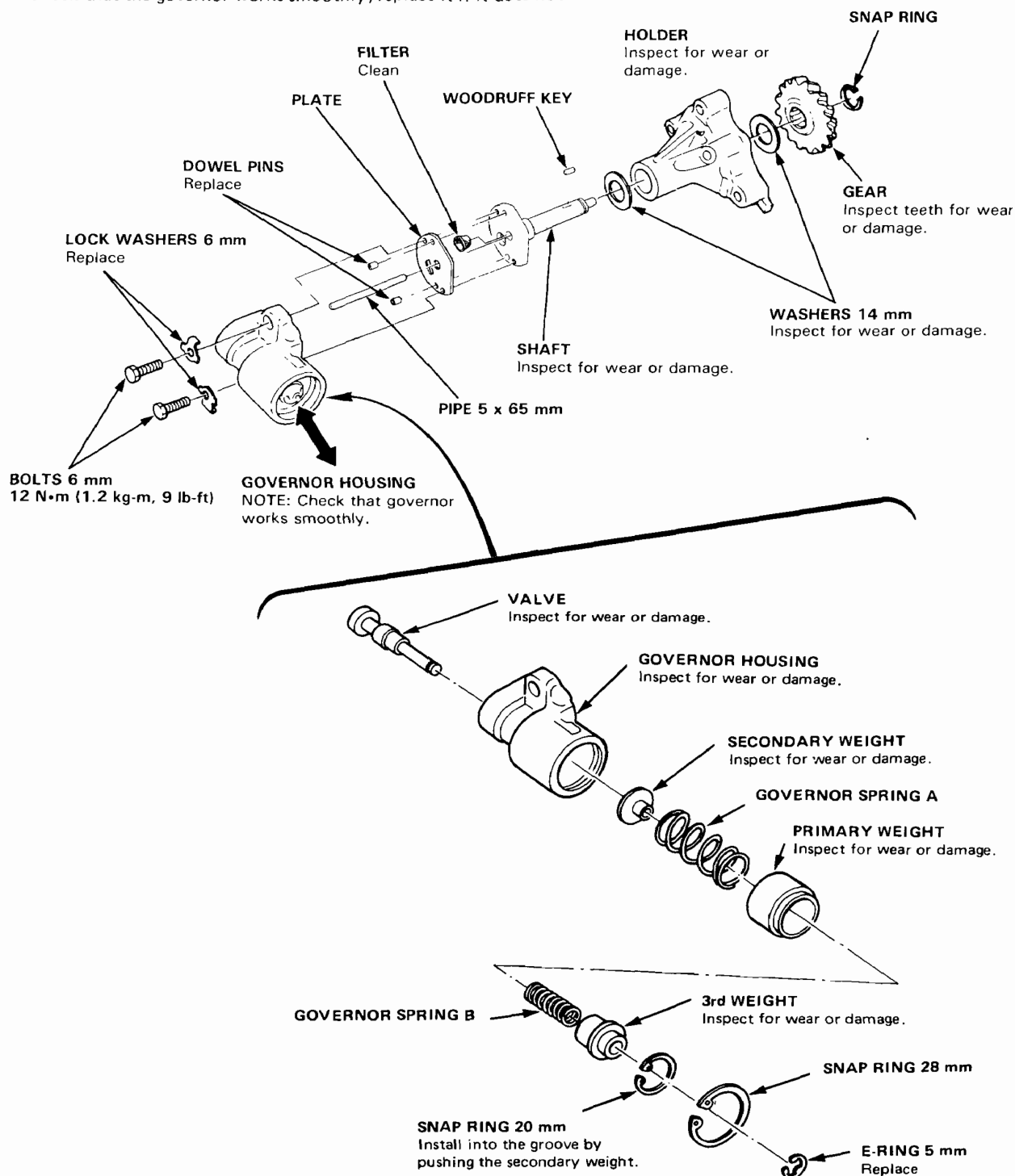
1. Jack up rear of car and place jack stands in proper locations.
2. Remove left rear wheel.
3. Remove the fuel pump cover.
4. Disconnect fuel lines and electrical leads at the fuel pump.
5. Removing three bolts, remove fuel pump with mount.



Hondamatic

Governor Disassembly/Inspection

- Clean all parts thoroughly in solvent or carburetor cleaner, and dry with compressed air. Blow out all passages.
- Check that the governor works smoothly; replace it if it does not.





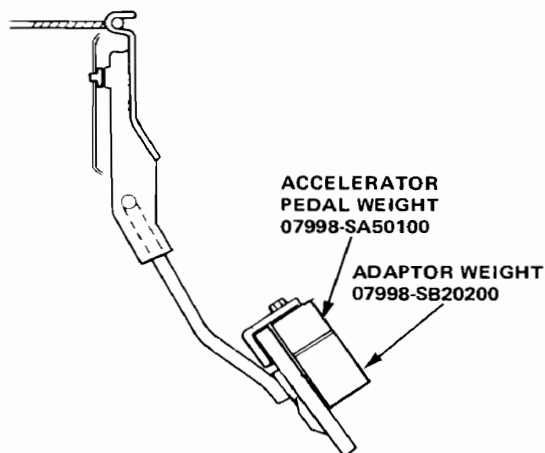
Throttle Control Cable Adjustment/Inspection

NOTE: Perform the following inspections before adjusting the throttle control cable.

- The carburetor throttle cable play is correct
- The engine is warmed-up to operating temperature.

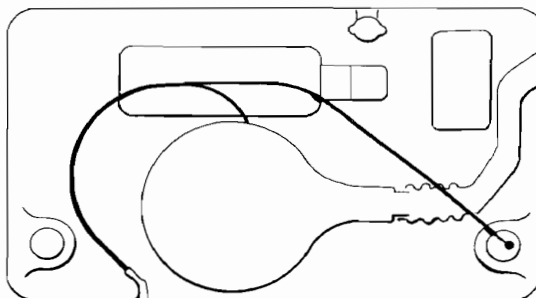
NOTE: The cooling fan should come on twice or more.

- The idle speed is correct.
 - The distance between the throttle control lever and the throttle control bracket is correct as shown in left column.
1. With the engine off, disconnect the throttle control cable from the throttle control lever.
 2. Attach a weight of about 1.3 kg (2.6 lbs) to the accelerator pedal. Raise the pedal, then release it, this will allow the weight to remove the normal free play from the throttle cable.

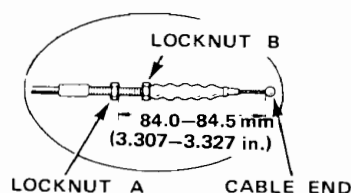
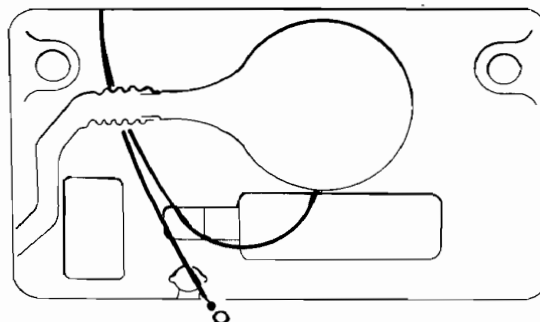


3. Secure the throttle control cable with clamps as shown.

L/H



R/H

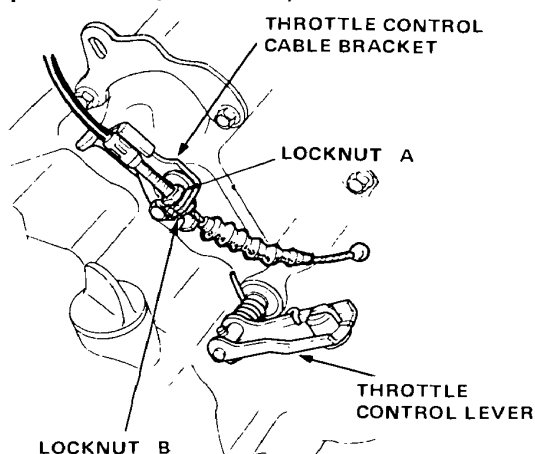


4. L/H: Lay the end of the throttle control cable on the R damper shock tower.
R/H: Lay the end of the throttle control cable on the engine hood latch.
5. Adjust the distance between the throttle control cable end and nut (A) to 84.0-84.5 mm (3.308-3.327 in.).
6. Insert the end of throttle control cable in the groove of the throttle control lever.

(cont'd)

Hondamatic

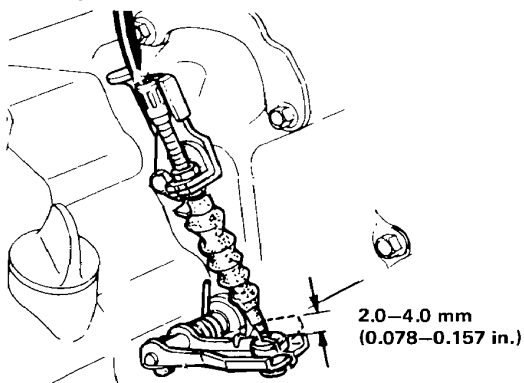
Throttle Control Cable Adjustment/ Inspection (Cont'd)



7. Insert the throttle control cable in the bracket and secure with lock nut (B).

NOTE: Make sure the cable is not kinked or twisted.

8. Check that the cable moves freely by depressing the accelerator.
9. Remove the weight on the accelerator pedal and push the pedal to make sure that there is the specified play at the throttle control lever.



10. Start the engine and check the synchronization between the carburetor and the throttle control cable.

NOTE: The throttle control lever should start to move as engine speed increases.

- If the throttle control lever moves before engine speed increases, turn the cable lock nut A counter clockwise and re-tighten lock nut B.
- If the throttle control lever moves after engine speed increases, turn lock nut A clockwise and re-tighten the lock nut B.

Driveshafts

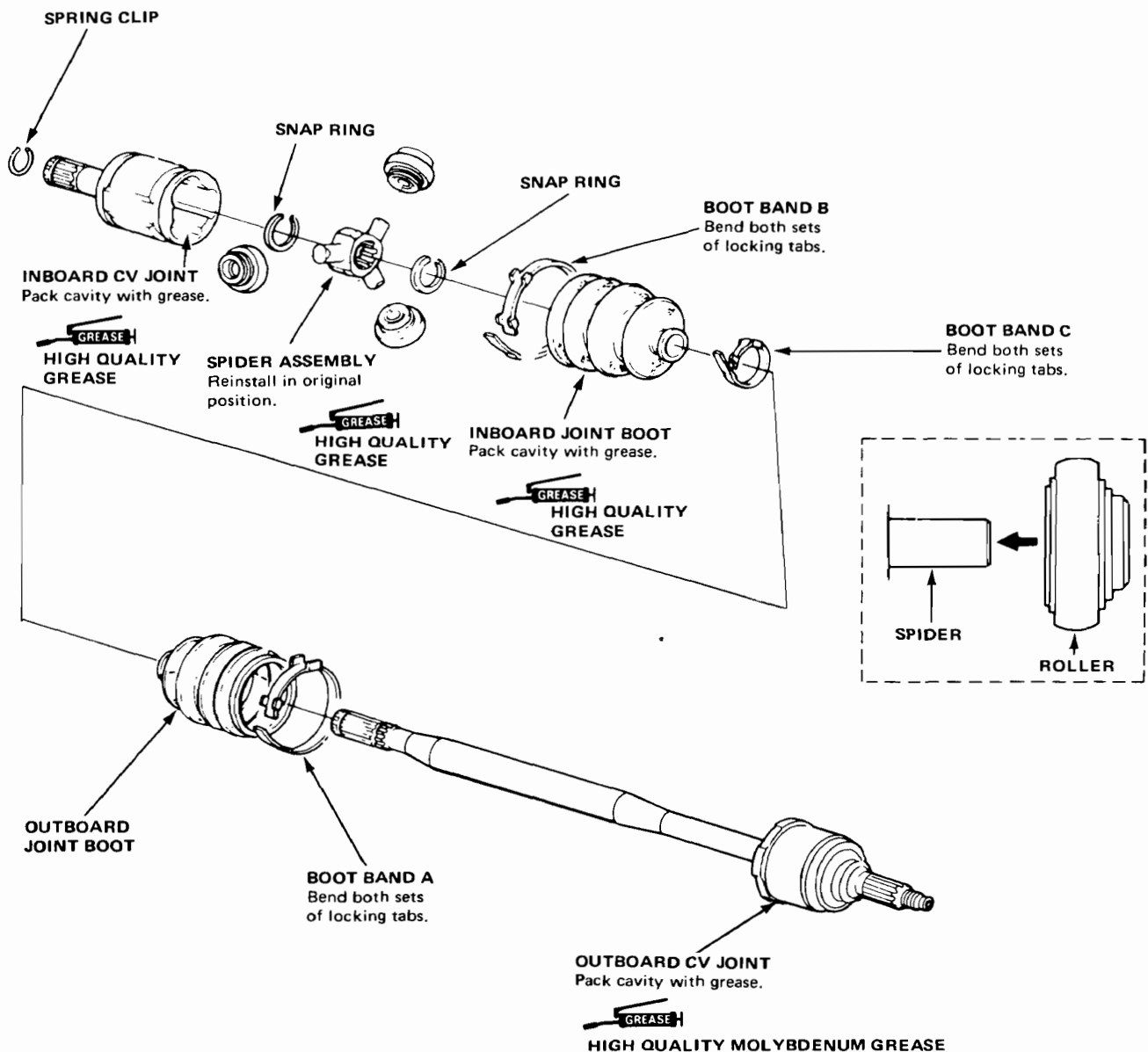
Reassembly : Roller type joint

1. Reassemble the driveshafts in reverse order of disassembly.

GREASE : Thouroughly, pack the inboard CV joint with high quality grease and the outboard CV joint and bearings with high quality molybdenum grease when reassembling the driveshaft.

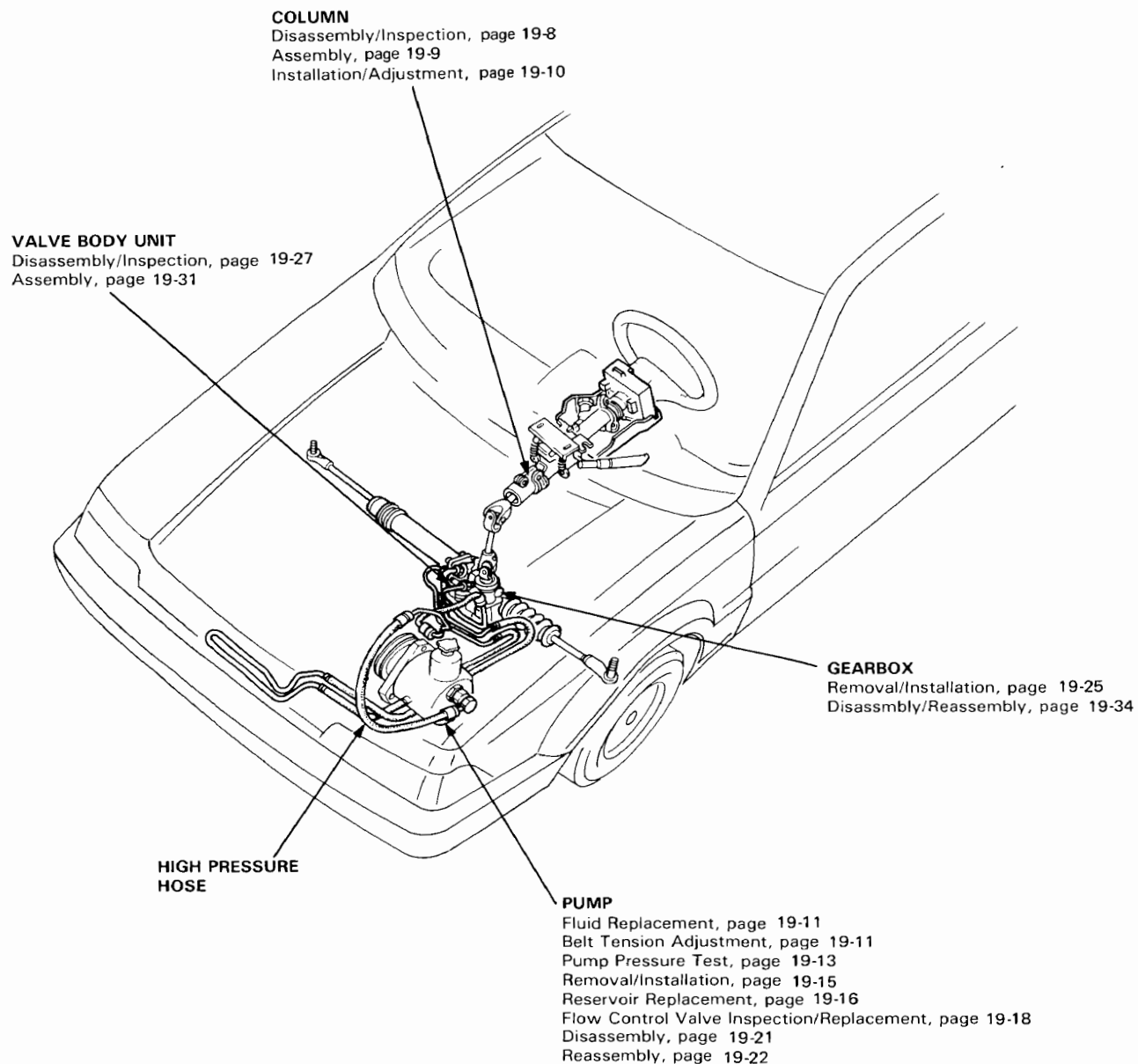
2. Install the rollers and bearing races on the spider shafts, then slide the spider assembly into the inboard shaft joint.
CAUTION: Avoid getting oil or grease on the rubber parts.

3. Slide the boots into place and install new boot bands C on the small ends.
Position the bands so they are centered between the locating humps at each end of the driveshaft. Expand and compress the boots until they return to their normal shape and length.



Power Steering

Index



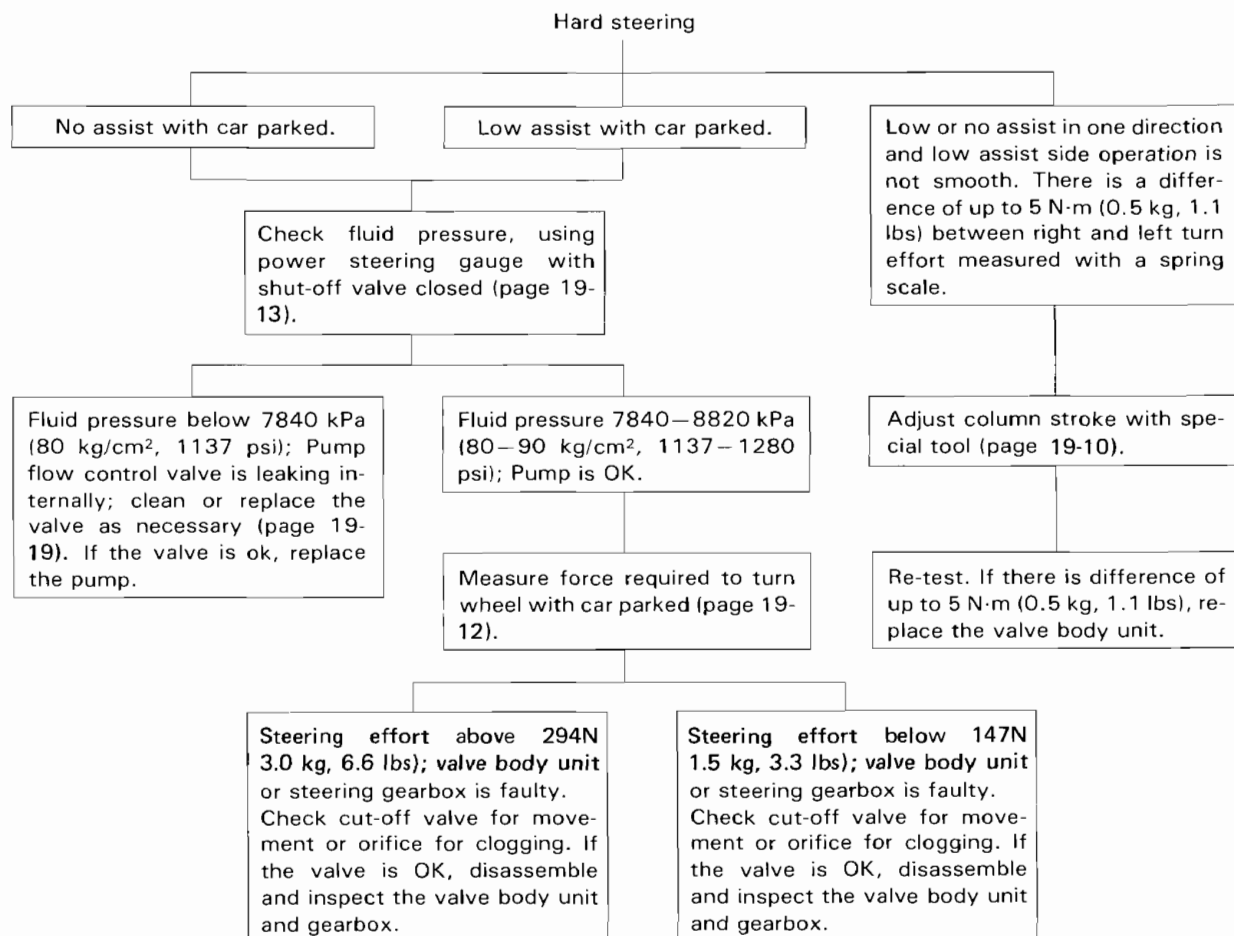


Troubleshooting

General

Check the following before you begin:

- Has the suspension been modified in away that might affect steering?
- Are tire sizes and air pressures correct.
- Is the steering wheel original equipment or equivalent?
- Is the power steering pump belt properly adjusted?
- Is steering fluid reservoir filled to proper level?
- Does the steering fluid contain bubbles?
- Is the engine idle speed correct and steady?

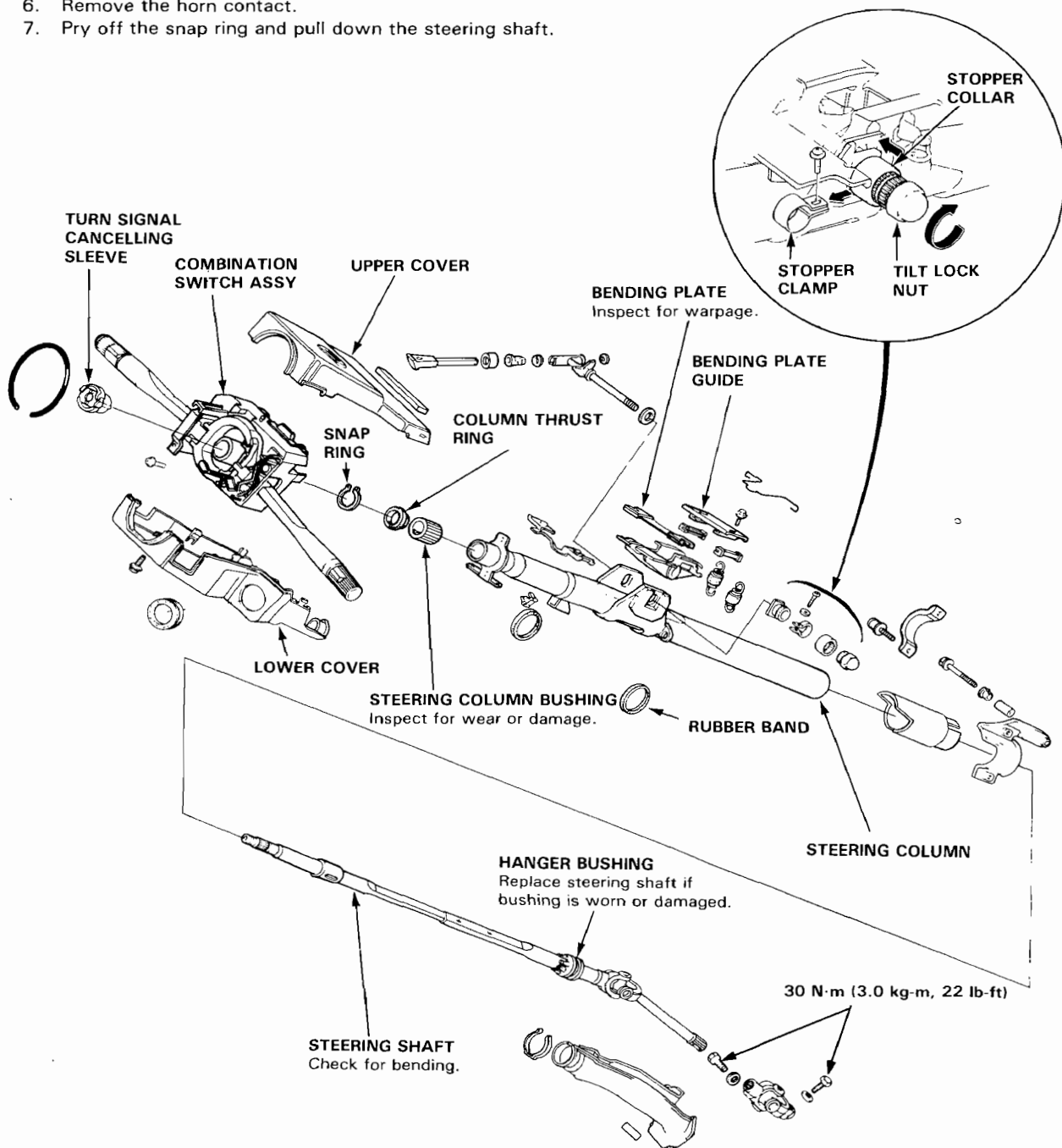


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Column

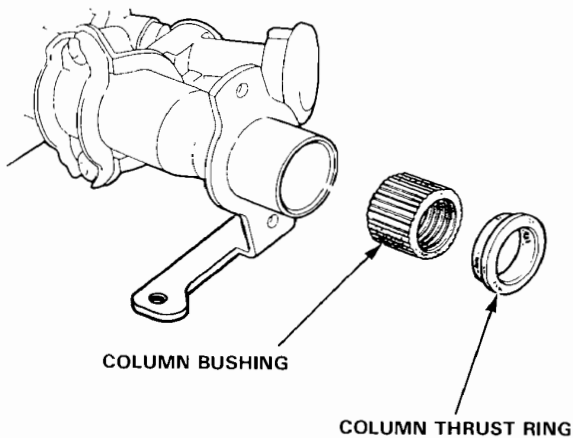
Disassembly/Inspection

1. Remove the turn signal cancelling sleeve.
2. Remove the rubber bands, then remove the upper column holder and bending plate.
3. Remove the bending plate guide.
4. Remove the combination switch.
5. Turn the ignition switch to the "I" position.
6. Remove the horn contact.
7. Pry off the snap ring and pull down the steering shaft.

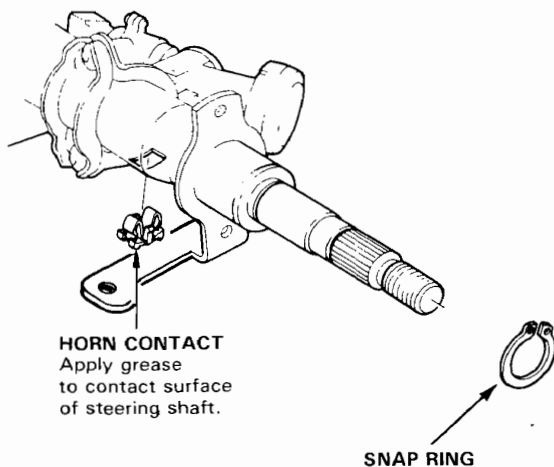


Assembly

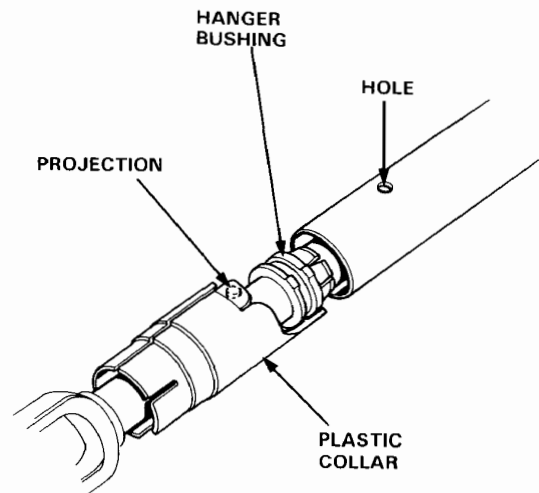
1. Insert the column bushing.
2. Set the column thrust ring in position.



3. Insert the steering shaft in the steering column from the bottom.
4. Install the snap ring on the steering shaft.
5. Insert the horn contact in the hole in the steering column.



6. Apply grease to the lower end of the steering shaft and inside the column.
7. Push the hanger bushing into the bottom end of the column as far as it will go.
8. Install the plastic collar over the bottom end of the steering column by aligning the round projection inside the collar with the hole in the column.



9. To install other parts, refer to the base manual.

Adjustment

NOTE: A special tool (adjustment guide) is required to position the steering column during installation.

1. Install the adjustment guide on the top end of the steering shaft and turn it as far as it will go.
2. Loosely install the upper bracket nuts and bending plate guide attaching bolts and pull the column down to be sure the bending plate is seated snugly against the hook.
3. Loosely install the lower bracket and pull the column down so that there is no clearance between the bending plate and hook.
4. Tighten the upper bracket nuts and bending plate guide bolts to the specified torque.

TORQUE: 14 N·m (1.4 kg-m, 10 lb-ft)

5. Tighten the lower bracket bolts to the specified torque.

6. Pull the column joint down fully and tighten the top bolt.

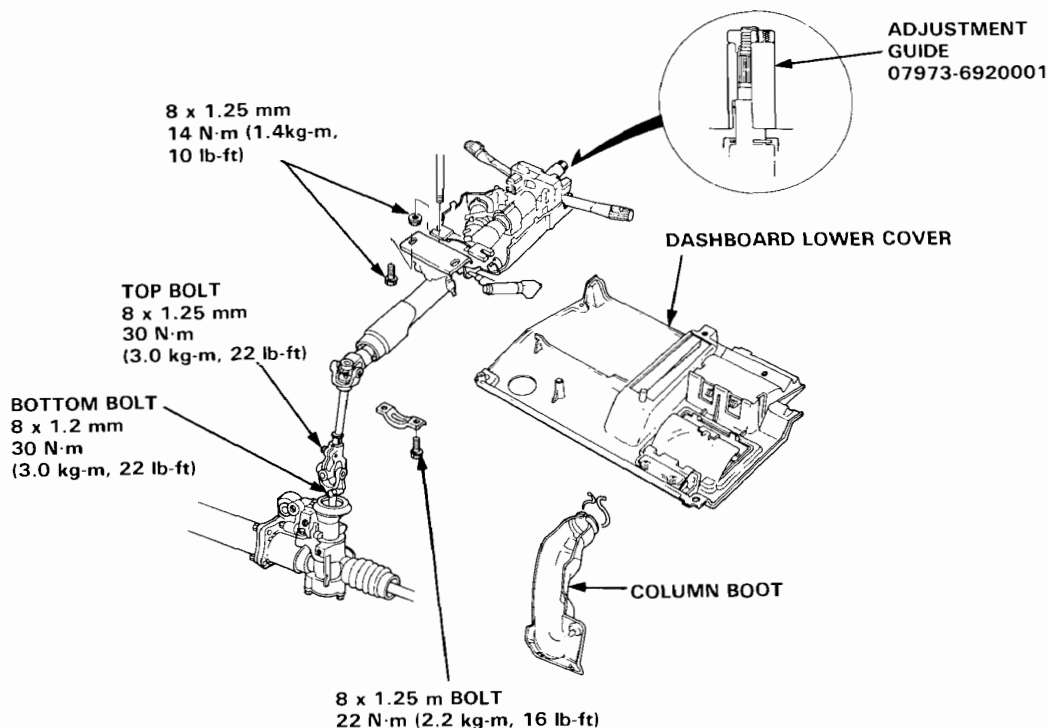
TORQUE: 30 N·m (3.0 kg-m, 22 lb-ft)

7. Tighten the bottom bolt.

TORQUE: 30 N·m (3.0 kg-m, 22 lb-ft)

NOTE: Make sure the end of the adjustment guide bottoms against the turn signal switch as shown.

8. Install the column boot and the dashboard lower cover.



Troubleshooting

General (cont'd)

Uneven or rough steering

Belt slipping on pulley.

Adjust belt tension (page 19-11). Replace if necessary.

Cut-off valve sticking or leaking.

Check cut-off valve. Replace if necessary.

Engine idle speed low or erratic. If engine stalls when wheel is turned while car is stopped or moving at low speeds, adjust engine idle speed.

Check power steering pump preload (page 19-15).

Improperly adjusted steering shaft stroke.

Adjust shaft.

Air bubbles visible in reservoir fluid. Check fluid level. If low, check for leaks. Add fluid to proper level.

If fluid level is OK, check oil strainer for clogging or O-ring for deterioration. Replace if necessary.

Improperly adjusted rack guide. Adjust rack guide (page 19-12).

If rack guide adjustment is OK, check pinion bearings for wear. If worn, replace them (19-37).

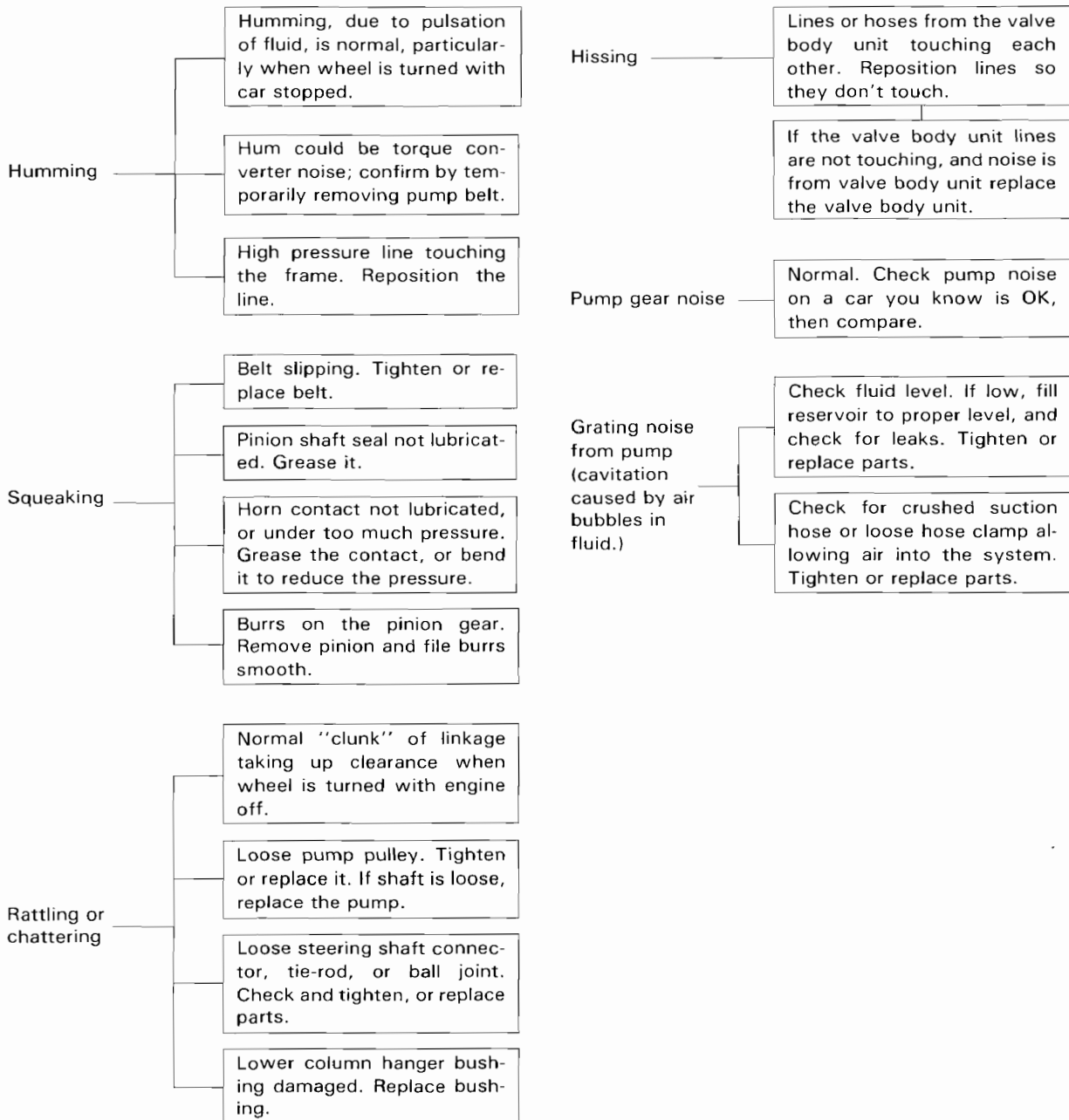


Shock or vibration when steering wheel is turned fully.	Belt slipping on pulley (pump stops momentarily).	Adjust belt tension or replace belt if necessary.
	Sticking relief valves; pulsation exceeds 980 kPa (10 kg/cm ² , 145 psi) during pump pressure test (page 19-13).	Check flow control valve. If OK, replace pump.
Excessively light steering at high speed.	Measure force, required to turn wheel with car parked on dry, paved surface.	If below 147N (1.5 kg, 3 lbs), check cut-off valve and valve body unit.
	Improperly adjusted tire air pressure.	Adjust.
	Improperly adjusted wheel alignment.	Adjust.
	Check rack guide adjustment.	Adjust if necessary.
Vehicle steers to one side.	Pump belt slipping on pulley.	Adjust belt tension or replace belt if necessary.
	Check cut-off valve for movement and valve body unit.	Replace if necessary.
	Rack guide adjusted too loose.	Adjust.
	Improperly adjusted wheel alignment.	Adjust.
	Improperly adjusted tire air pressure.	Adjust.
Wheel will not return smoothly.	Tire air pressure too low.	Inflate to correct pressure.
	Improperly adjusted wheel alignment.	Adjust.
	Improperly adjusted rack guide.	Adjust.
	Improperly adjusted column stroke.	Adjust.

Troubleshooting

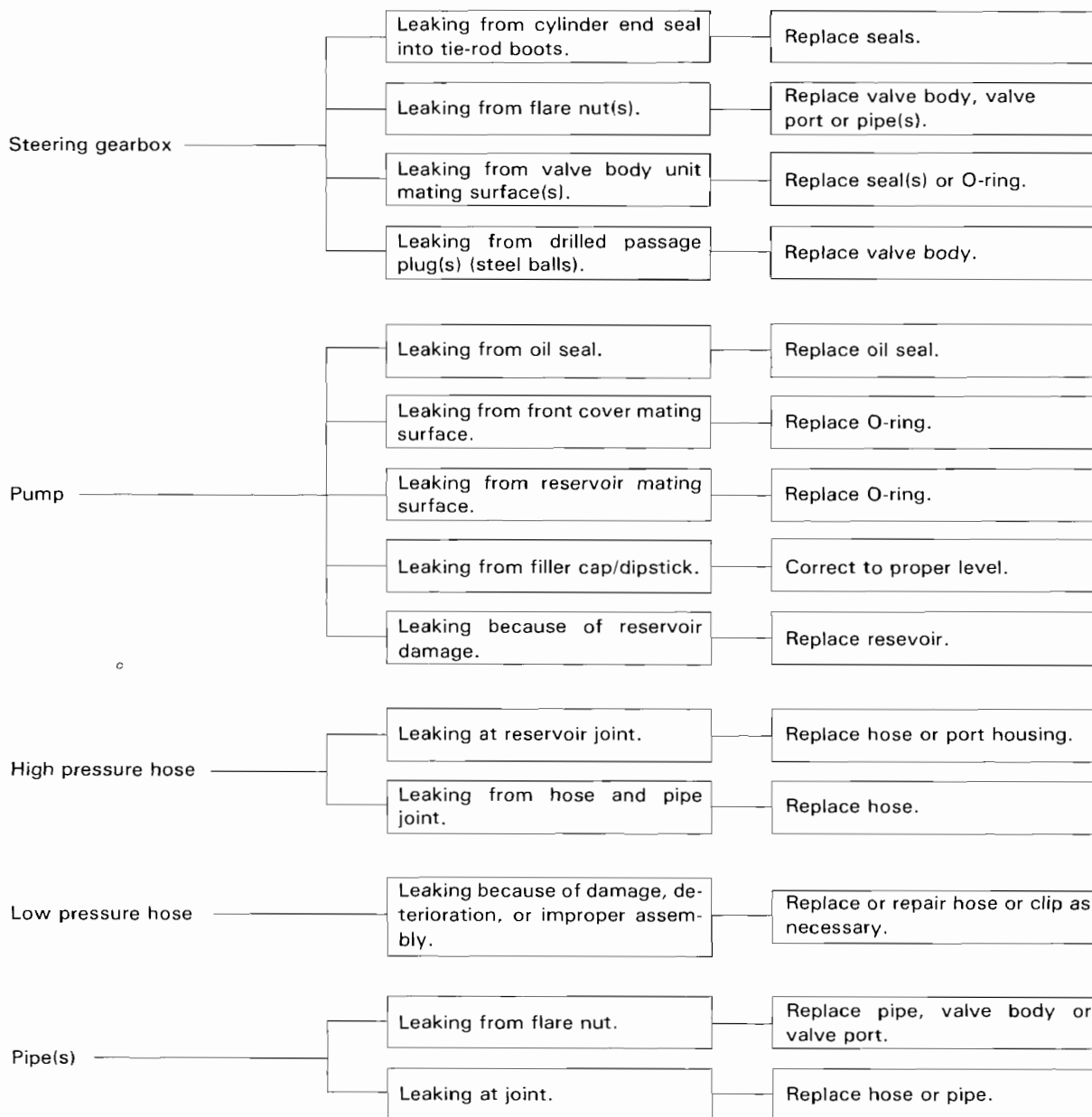
Noise and Vibration

NOTE: Pump noise within 2–3 minutes after starting in cold weather (–20°F or colder) is normal.





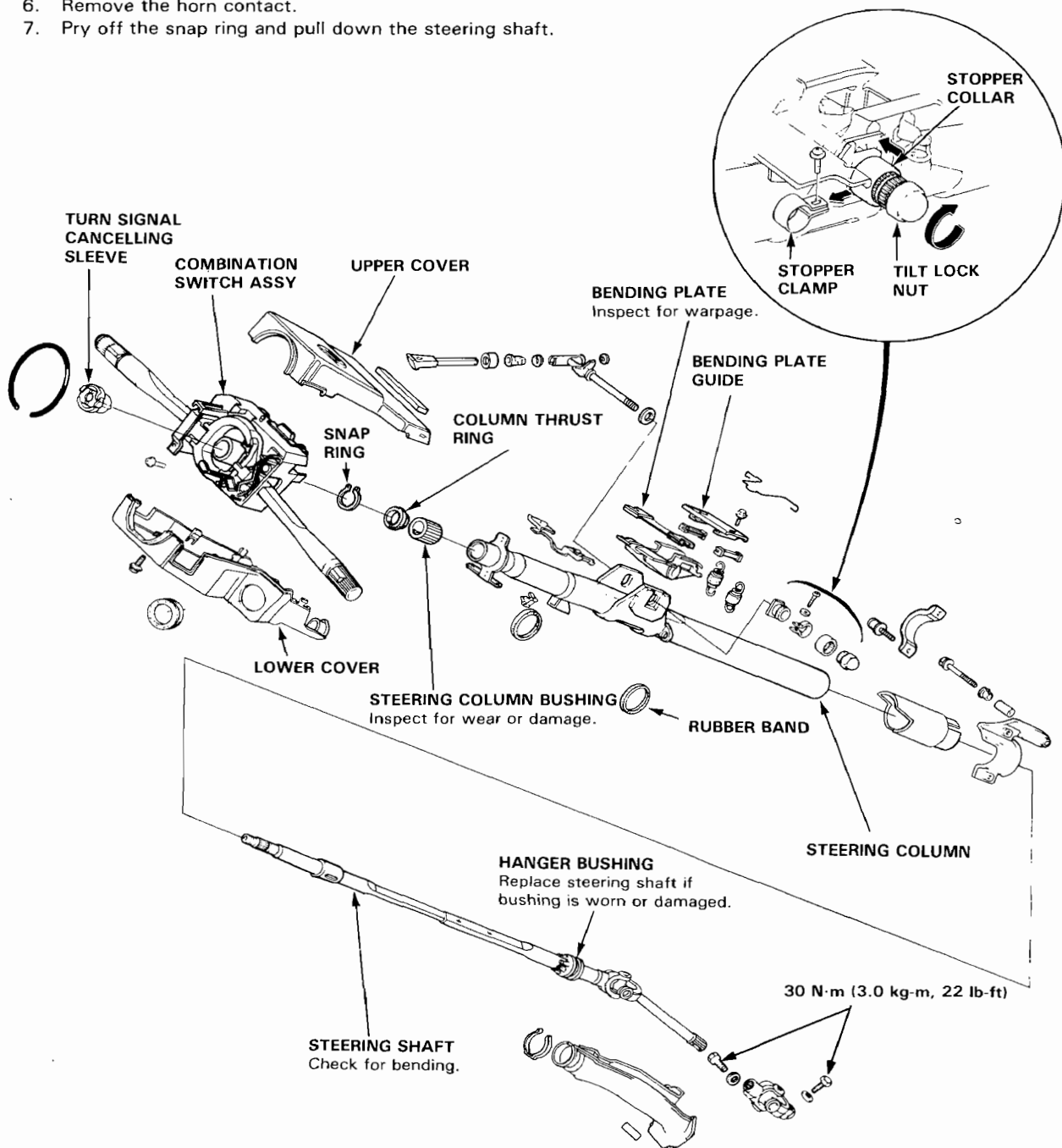
Fluid Leaks



Column

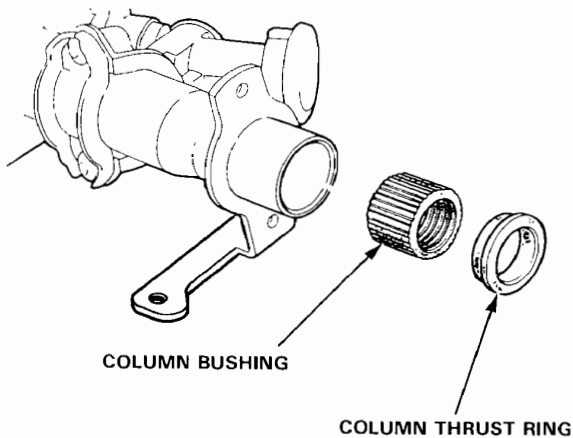
Disassembly/Inspection

1. Remove the turn signal cancelling sleeve.
2. Remove the rubber bands, then remove the upper column holder and bending plate.
3. Remove the bending plate guide.
4. Remove the combination switch.
5. Turn the ignition switch to the "I" position.
6. Remove the horn contact.
7. Pry off the snap ring and pull down the steering shaft.

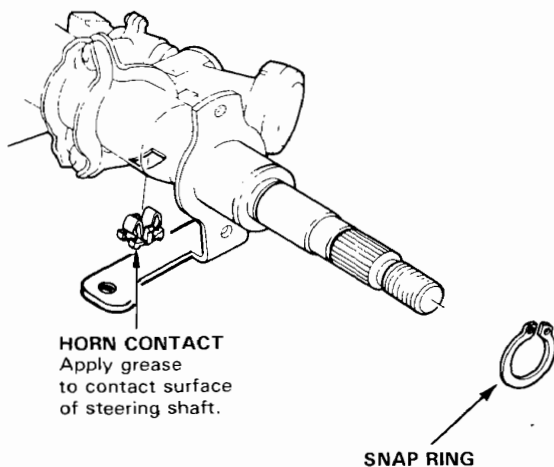


Assembly

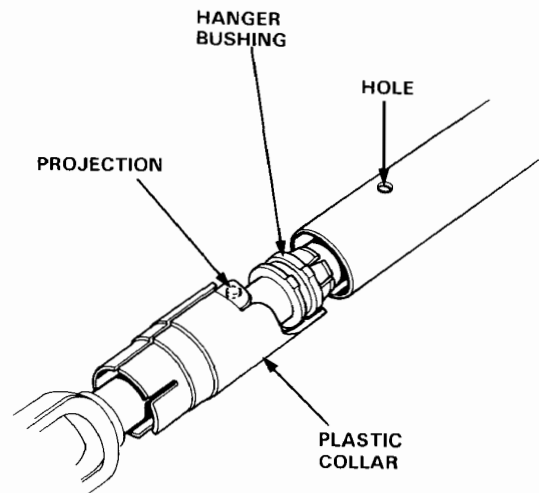
1. Insert the column bushing.
2. Set the column thrust ring in position.



3. Insert the steering shaft in the steering column from the bottom.
4. Install the snap ring on the steering shaft.
5. Insert the horn contact in the hole in the steering column.



6. Apply grease to the lower end of the steering shaft and inside the column.
7. Push the hanger bushing into the bottom end of the column as far as it will go.
8. Install the plastic collar over the bottom end of the steering column by aligning the round projection inside the collar with the hole in the column.



9. To install other parts, refer to the base manual.

Adjustment

NOTE: A special tool (adjustment guide) is required to position the steering column during installation.

1. Install the adjustment guide on the top end of the steering shaft and turn it as far as it will go.
2. Loosely install the upper bracket nuts and bending plate guide attaching bolts and pull the column down to be sure the bending plate is seated snugly against the hook.
3. Loosely install the lower bracket and pull the column down so that there is no clearance between the bending plate and hook.
4. Tighten the upper bracket nuts and bending plate guide bolts to the specified torque.

TORQUE: 14 N·m (1.4 kg-m, 10 lb-ft)

5. Tighten the lower bracket bolts to the specified torque.

6. Pull the column joint down fully and tighten the top bolt.

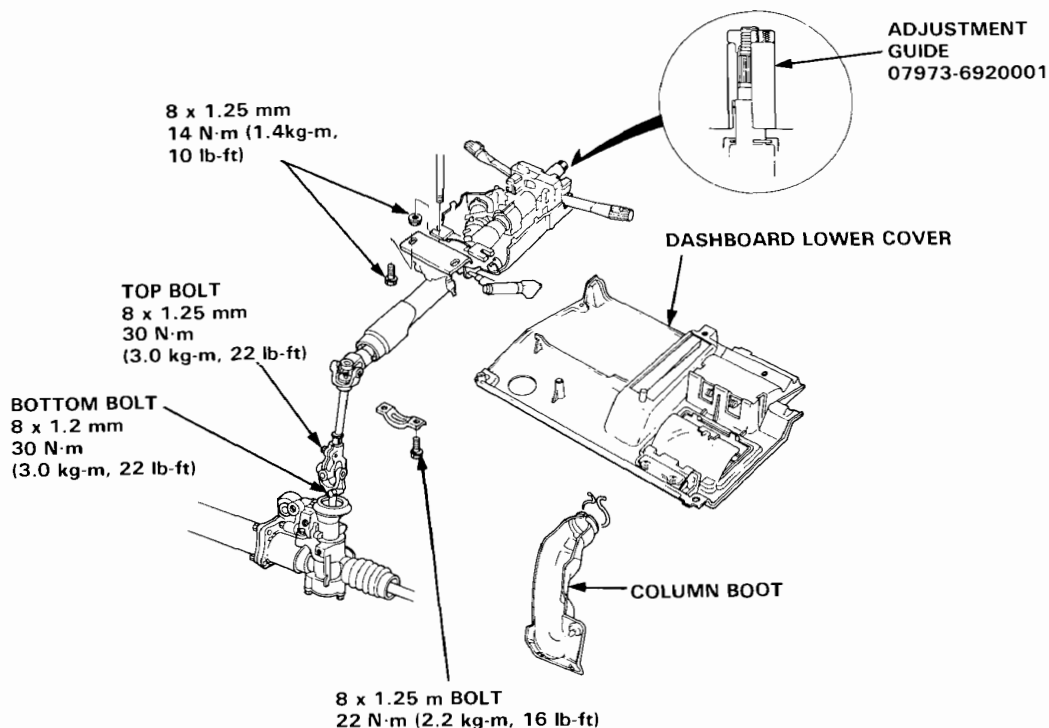
TORQUE: 30 N·m (3.0 kg-m, 22 lb-ft)

7. Tighten the bottom bolt.

TORQUE: 30 N·m (3.0 kg-m, 22 lb-ft)

NOTE: Make sure the end of the adjustment guide bottoms against the turn signal switch as shown.

8. Install the column boot and the dashboard lower cover.





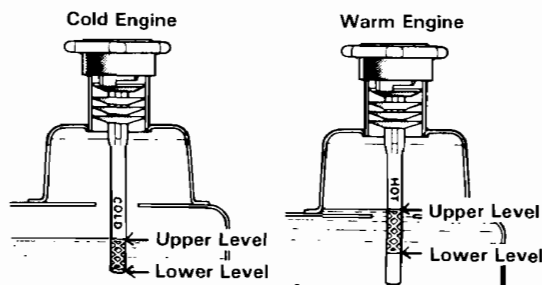
Power Steering Fluid

Replacement

Check the reservoir level at regular intervals, and add fluid as necessary.

CAUTION: Use only **GENUINE HONDA** Power Steering Fluid. Use of fluids such as ATF or other manufacturers' power steering fluid will damage the system.

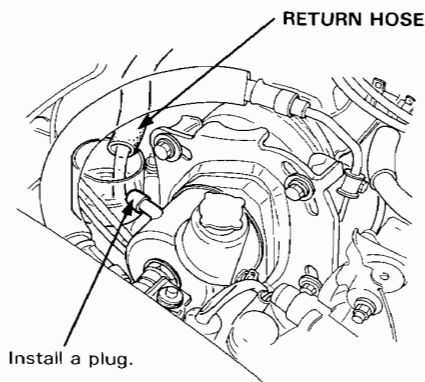
Fluid Level



Fluid Replacement

CAPACITY: 1.1 U.S. qt (1.0ℓ) at change

1. Raise the front end of the car and place safety stands in the proper locations.
2. Disconnect the return hose from the gearbox at the reservoir, and put the end in a suitable container.
3. Start the engine and let it run at idle, and turn the steering wheel from lock-to-lock several times. When fluid stops running out of the hose, shut off the engine. Discard the fluid.



3. Refit the return hose to the reservoir.
4. Fill the reservoir to the upper level mark.
5. Start the engine and run it at idle, then turn the steering wheel from lock-to-lock several times to bleed the air from the system.
6. Recheck the fluid level and add fluid if necessary.

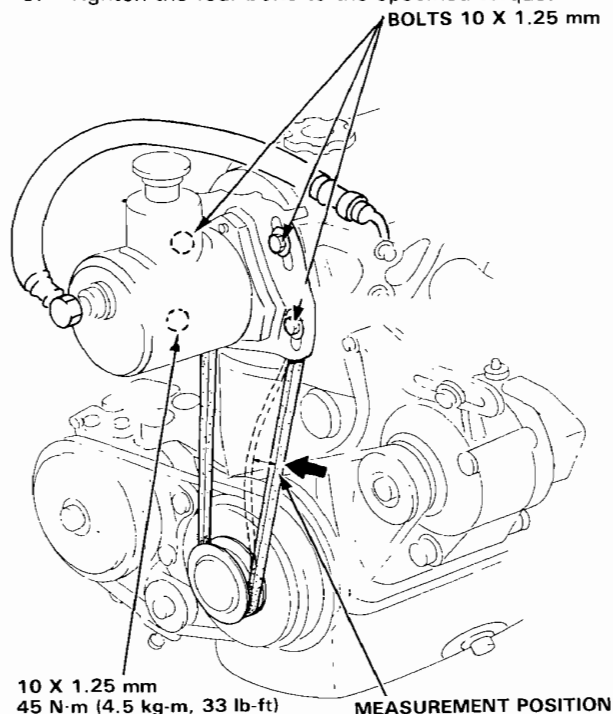
CAUTION: Do not fill the reservoir beyond the upper level mark.

On-Car Checks

Belt tension Adjustment

A properly adjusted belt should deflect about 18–22 mm (3/4–7/8 in.) when you push on it mid-way between the pulleys with a force of about 100N (10kg, 22 lbs).

1. Loosen the four bolts.
2. Slide the pump body.
3. Tighten the four bolts to the specified torque.



2. Start the engine and let it idle, and turn the steering wheel from lock-to-lock several times.
3. Stop the engine. Check and readjust belt deflection if necessary.



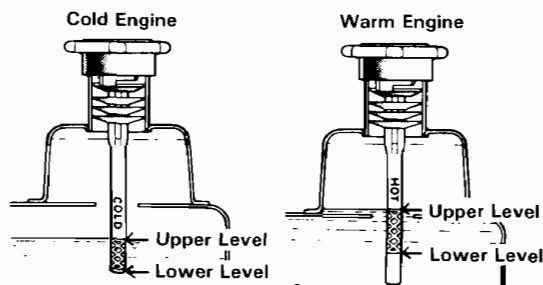
Power Steering Fluid

Replacement

Check the reservoir level at regular intervals, and add fluid as necessary.

CAUTION: Use only **GENUINE HONDA** Power Steering Fluid. Use of fluids such as ATF or other manufacturers' power steering fluid will damage the system.

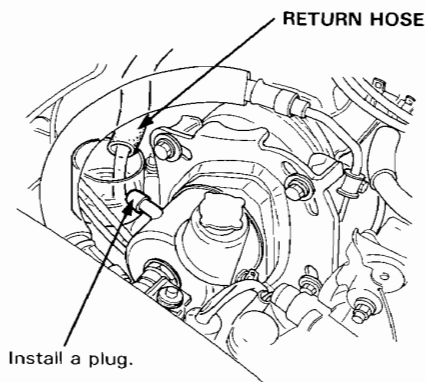
Fluid Level



Fluid Replacement

CAPACITY: 1.1 U.S. qt (1.0ℓ) at change

1. Raise the front end of the car and place safety stands in the proper locations.
2. Disconnect the return hose from the gearbox at the reservoir, and put the end in a suitable container.
3. Start the engine and let it run at idle, and turn the steering wheel from lock-to-lock several times. When fluid stops running out of the hose, shut off the engine. Discard the fluid.



3. Refit the return hose to the reservoir.
4. Fill the reservoir to the upper level mark.
5. Start the engine and run it at idle, then turn the steering wheel from lock-to-lock several times to bleed the air from the system.
6. Recheck the fluid level and add fluid if necessary.

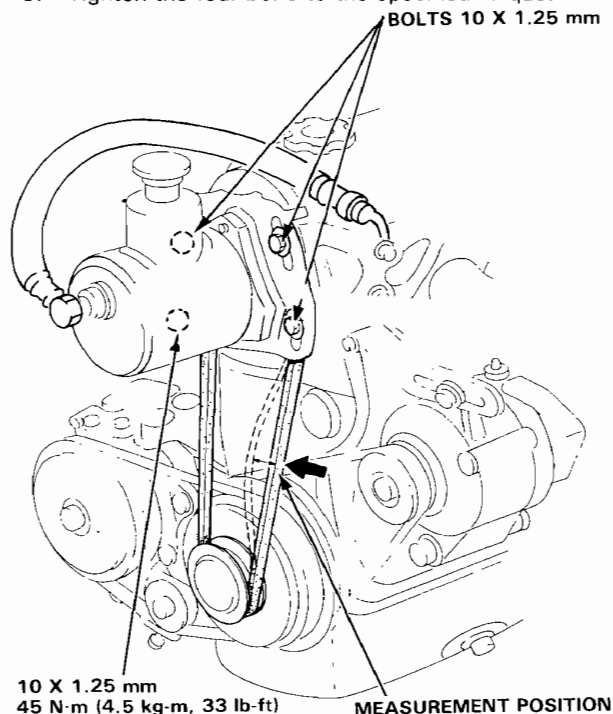
CAUTION: Do not fill the reservoir beyond the upper level mark.

On-Car Checks

Belt tension Adjustment

A properly adjusted belt should deflect about 18–22 mm (3/4–7/8 in.) when you push on it mid-way between the pulleys with a force of about 100N (10kg, 22 lbs).

1. Loosen the four bolts.
2. Slide the pump body.
3. Tighten the four bolts to the specified torque.



2. Start the engine and let it idle, and turn the steering wheel from lock-to-lock several times.
3. Stop the engine. Check and readjust belt deflection if necessary.

On-Car Checks

Assist Check With Car Parked

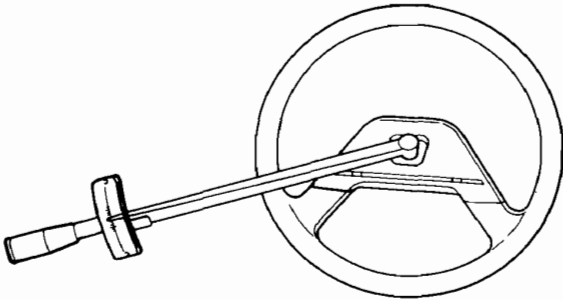
1. Check the power steering fluid level and pump belt tension.
2. Start the engine, allow to idle, and turn the steering wheel from lock-to-lock several times to warm up the fluid.

Measuring Fluid Temperature: 40–50°C
(104–122°F)

Check With Torque Wrench

Attach a torque wrench to the steering wheel nut. With the engine idling and the car on a clean, dry floor. Turn the wrench as shown and read the torque as soon as the tires begin to turn.

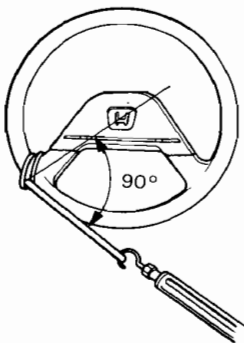
TORQUE: 5.6 N·m (56.5 kg-cm, 4.0 lb-ft) MAX.



Check With Spring Scale

Attach a spring scale to the steering wheel. With the engine idling and the car on a clean, dry floor, pull the scale as shown and read it as soon as the tires begin to turn.

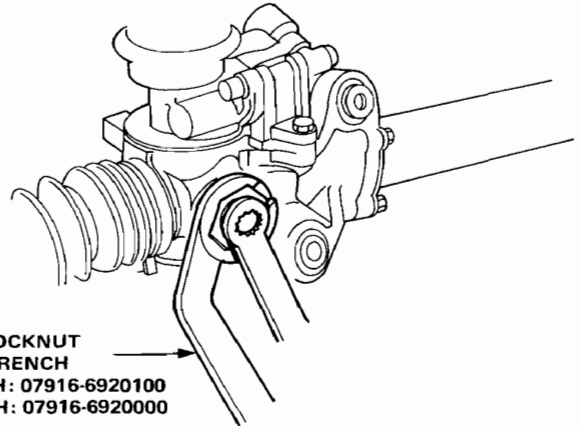
FORCE: 294N (3.0 kg, 6.6 lbs) MAX.



If values are not within the specification, inspect the valve body unit (see page 19-27).

Rack Guide Adjustment

1. Loosen the locknut on the rack guide screw with the special tool as shown.



2. Tighten the guide screw until it compresses the spring against the guide; then loosen it, and tighten it to about 4 N·m (0.4 kg-m, 3 lb-ft) and back it off about 25° (about 1/12 of a turn). Tighten the locknut to about 25 N·m (2.5 kg-m, 18 lb-ft) while preventing the guide screw from moving.



Pump Pressure Test

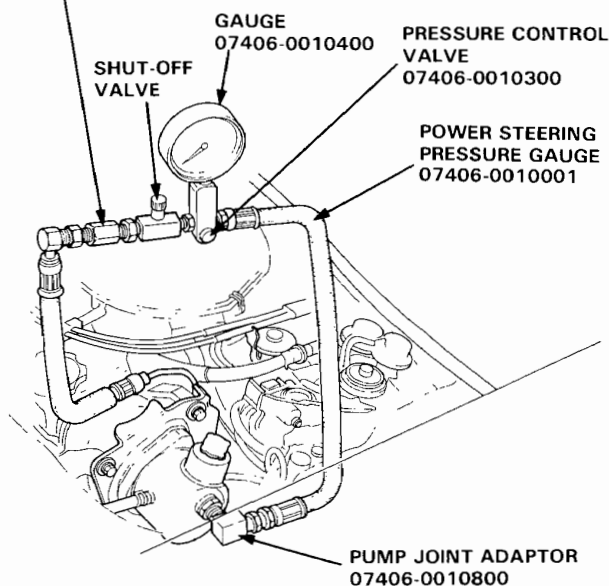
Check fluid pressure as follows to determine whether the trouble is in the pump or gearbox:

NOTE: First check the power steering fluid level and pump belt tension.

1. Disconnect the outlet hose from the pump outlet fitting, and install the pressure gauge and the adaptors between the hose and pump as shown.

HOSE JOINT ADAPTOR
07406-0010900

NOTE: If Power Steering Gauge (07406-0010000) is used, the pressure valve (07406-0010300) and gauge (07406-0010400) must be installed as shown.



2. Open the shut-off valve fully.
3. Open the pressure control valve fully.
4. Start the engine and let it idle.
5. Turn the steering wheel from lock-to-lock several times to get the fluid up to operating temperature.

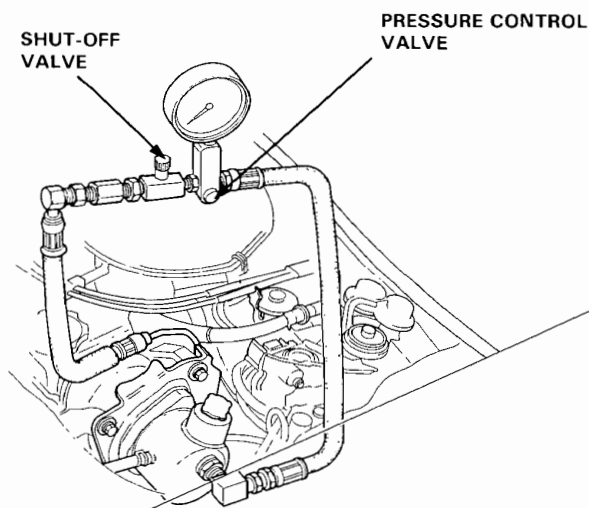
Measuring Fluid Temperature: 40–50°C
(104–120°F)

6. Close the shut-off valve, then, close the pressure control valve gradually until the pressure gauge needle is stable, then read pressure.

7. Open the shut-off valve fully.

CAUTION: Do not keep the shut-off valve closed more than 5 seconds or the pump could be damaged by over-heating.

If the pump is OK, the gauge should read at least 7845 kPa (80 kg/cm² 1135 psi). A low reading means pump output is too low for full assist. Repair or replace the pump.



Pump

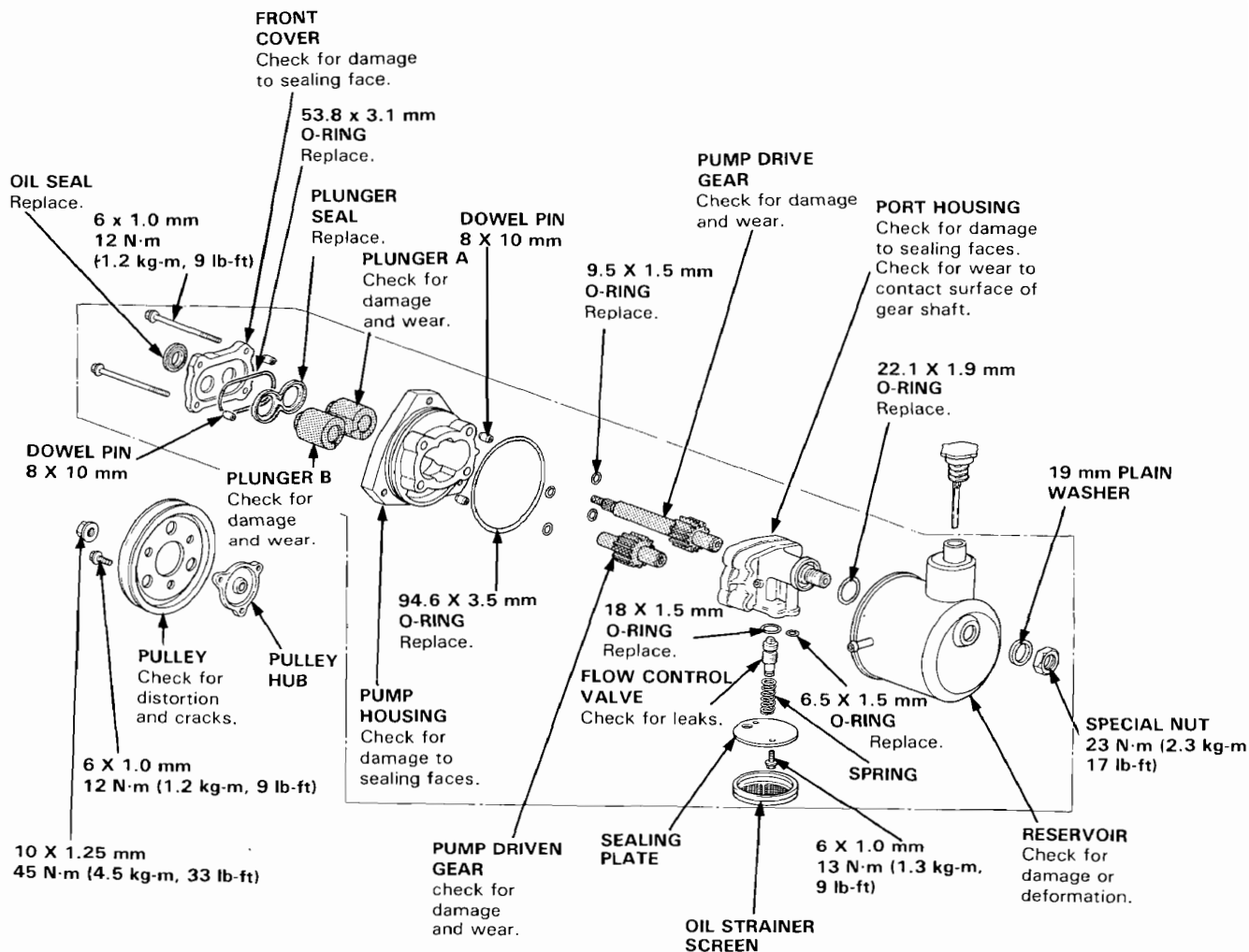
Index/Inspection



Use only STEERING GREASE (Honda part number 08740—99969) on seals and O-rings.

CAUTION: Pump components are made of aluminum. Be careful not to damage them when servicing.

- Clean all the disassembled parts thoroughly.
- Replace all O-rings and seals. Do not dip new O-rings and seals in solvent; coat O-rings with steering grease before installation, and make sure they stay in place during reassembly.
- The shaded parts are selectively fitted, and should not be disassembled except to replace seals. If any one of them is faulty, replace the whole pump as an assembly.



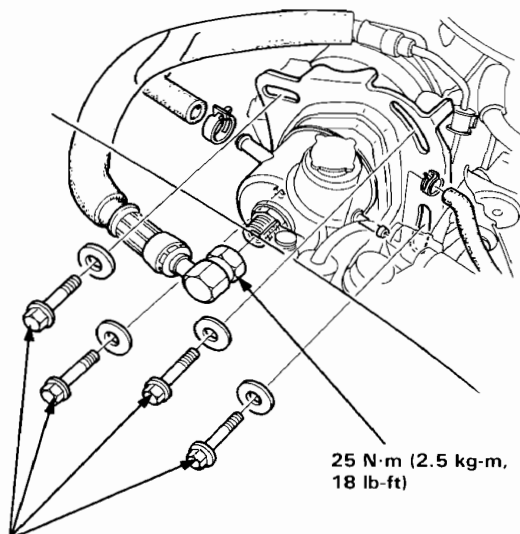


Removal/Installation

1. Disconnect the hoses from the reservoir.

NOTE: Cap the open fitting to keep dirt out of the system.

2. Remove the four bolts and pump assembly.



10 X 1.25 N·m
45 N·m (4.5 kg-m, 33 lb-ft)

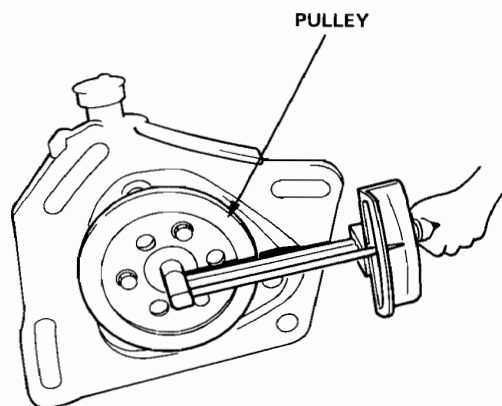
Install in reverse order, and:

- Connect hoses securely.
- Adjust the belt tension.
- Check fluid level and add if necessary.
- Bleed the air from the system.

Preload Check

Check pump preload with a torque wrench.

PRELOAD: 4 N·m (0.4 kg-m, 3 lb-ft) max.



Pump

Pump Bracket Removal

1. Remove the pulley nut using the special tool, then remove the three 6 mm bolts.

UNIVERSAL HOLDER
07725-0010101
or 07725-0030000

6 x 1.0 mm
12 N·m (1.2 kg-m
9 lb-ft)

PULLEY NUT
45 N·m (4.5
kg-m, 33 lb-ft)

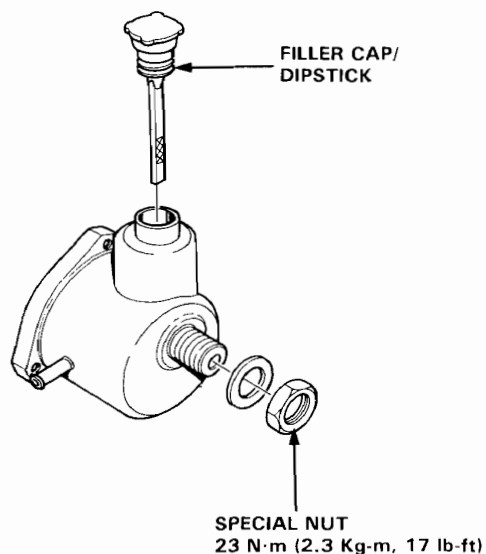
2. Remove the three 10 mm bolts and separate the pump from its bracket.

PUMP BRACKET

10 X 1.25 mm
45 N·m (4.5 kg-m, 33 lb-ft)

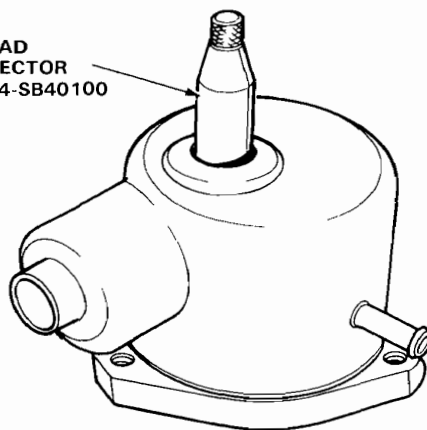
Reservoir Replacement

1. Remove the filler cap/dipstick.
2. Remove the special nut and 19 mm plain washer.



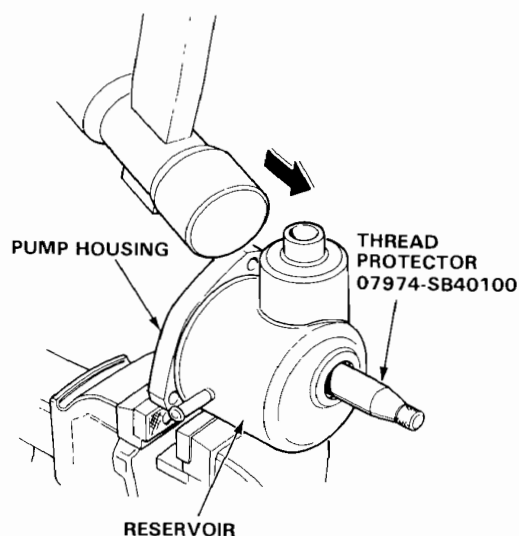
3. Attach the special tool on the high pressure fitting.

THREAD
PROTECTOR
07974-SB40100

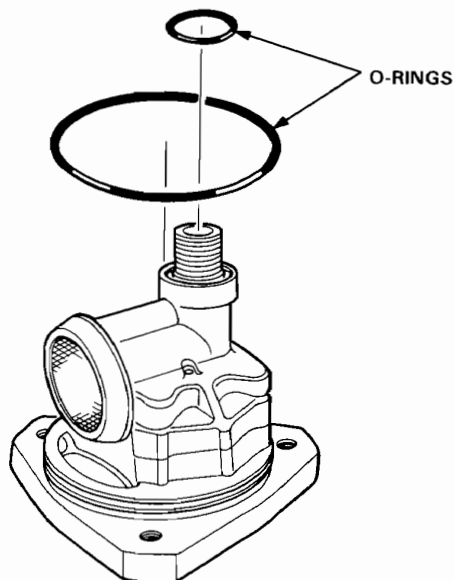


4. Tap the reservoir lightly with a plastic hammer to remove the reservoir from the pump housing.

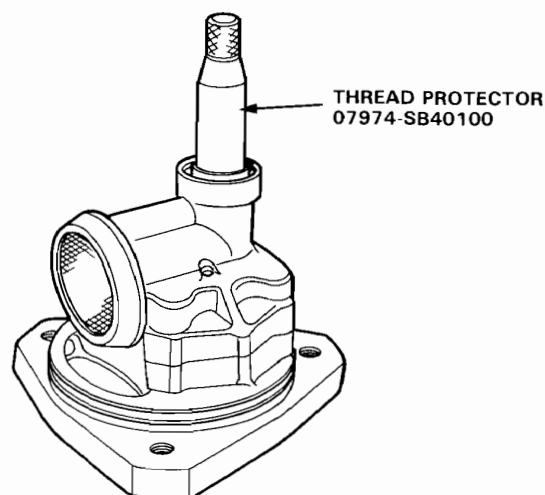
CAUTION: Be careful not to damage the reservoir.



5. Replace 94.6 x 3.5 mm and 22.1 x 1.9 mm O-rings.



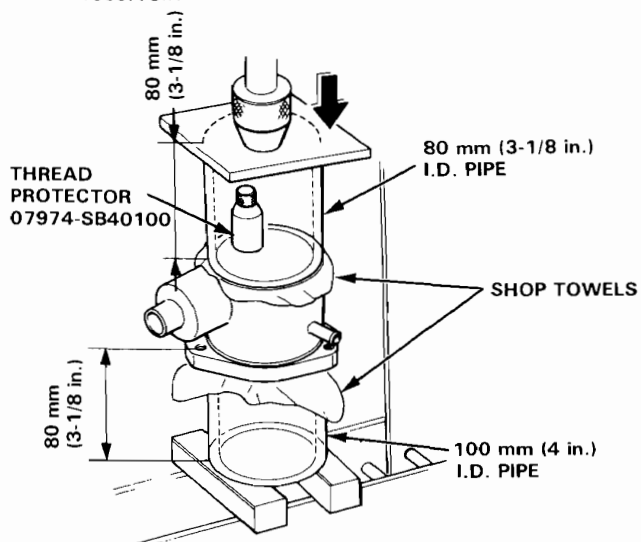
6. Install the thread adaptor as shown.



7. Press the reservoir into the pump housing using 80 mm (3-1/8 in.) and 100 mm (4 in.) diameter pipes until the end of the reservoir is seated on the flange of the pump housing.

CAUTION: Use shop towels to prevent damage to the reservoir or pump housing.

NOTE: The shop towels function as "gaskets" to keep the pipe sections securely positioned on the pump housing. This is particularly important against the rounded edge of the pump case/reservoir.

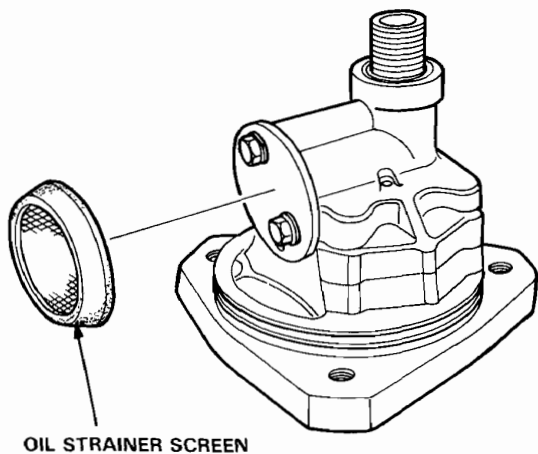


8. Install the 19 mm plain washer and special nut.

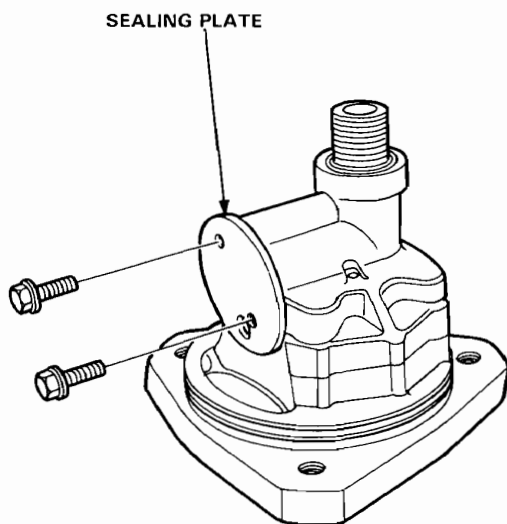
Pump

Flow Control Valve Inspection and Replacement

1. Remove and clean the oil strainer screen.

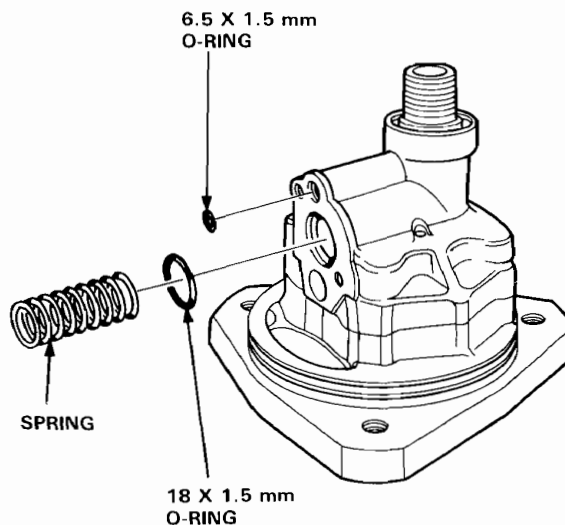


2. Remove the sealing plate.

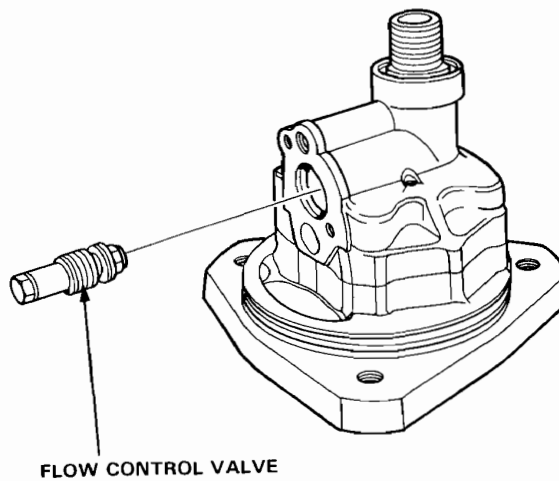


3. Remove the 18.5 × 1.5 mm and 6.5 × 1.5 mm O-rings.

4. Remove the spring.

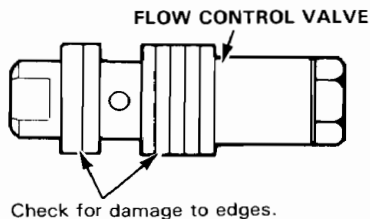


5. Remove the flow control valve.

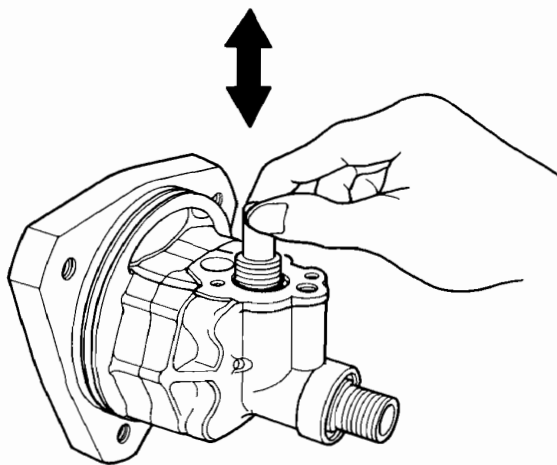




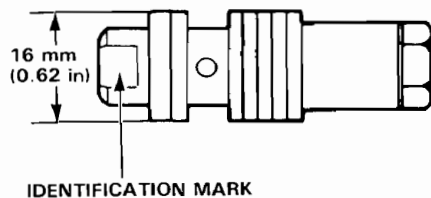
6. Check for wear, burrs, and other damage to the edges of the grooves in the valve.



7. Slip the valve back in the pump and check that it moves up and down smoothly.

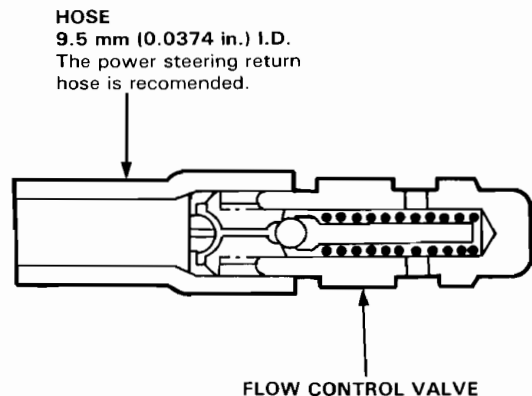


If OK, go to step 8. If not, replace the valve:
The original valve was selected for a precise fit in the port housing bore; make sure the new one has the same identification mark.

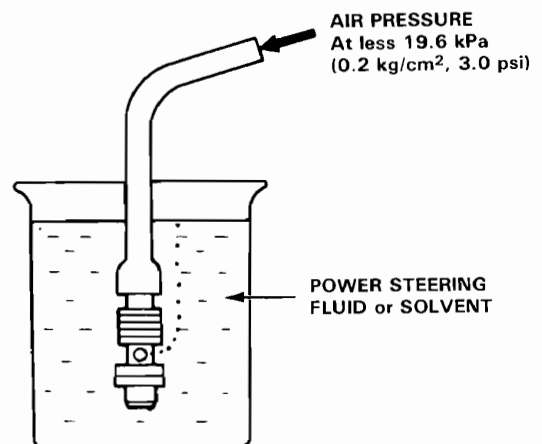


Mark	Part No.	Part Name	Size: mm
A	56350-PC1-000	Flow Control Valve Assembly A	16 less than 0 -0.005
Without mark	56360-PC1-000	Flow Control Valve Assembly B	16 +0.006 0

8. Attach a hose to the end of the valve.



9. Then submerge the valve in a container of power steering fluid or solvent, and blow on the hose. Replace the valve or repair it (next page) if you see air bubbles in the fluid.

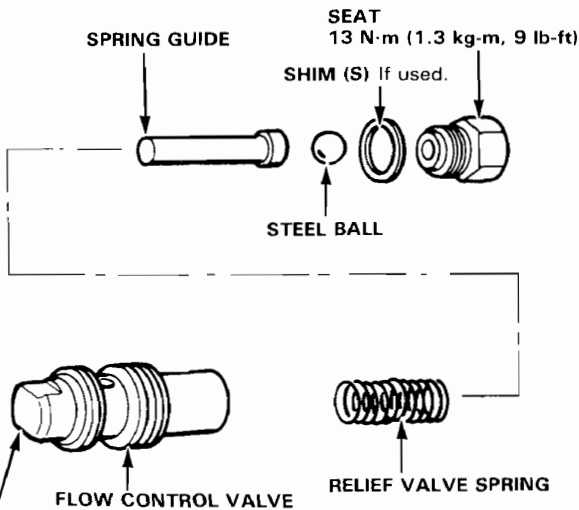


(cont'd)

Pump

Flow Control Valve Inspection and Adjustment (cont'd)

10. If the valve leaks, clamp the bottom end of it in a vise that has soft jaws.
11. Unscrew the seat in the top end of the valve, and remove any shims, the steel ball, the spring guide and its spring.



Clamp this end in a vise with soft jaws.

12. Clean all the parts in solvent, dry them off, then reassemble and re-test the valve.

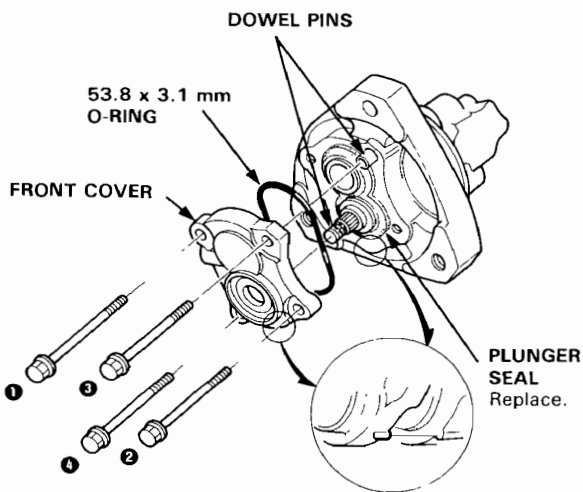
NOTE: If necessary, relief pressure is adjusted at the factory by adding shims under the check ball seat. if you found shims in your valve, be sure you reinstall as many as you took out.

13. Install in reverse order of removal. Also:

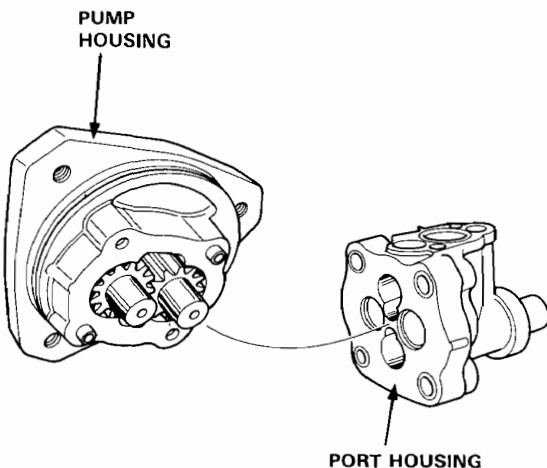
- Coat the new O-rings with steering grease to hold them.
- Coat the flow control valve with power steering fluid.

Housing Disassembly

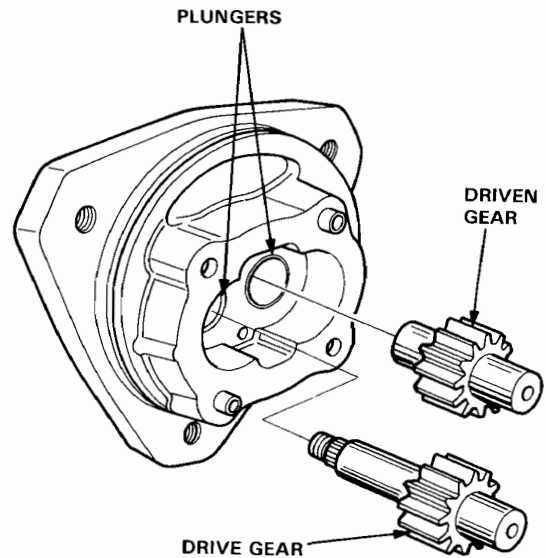
1. Separate the reservoir from the pump housing (page 19-17).
2. Remove the flow control valve if necessary (page 19-18).
3. Loosen the four 6 mm bolts in the sequence shown, then remove them and the front cover.
4. Remove the 53.8 x 3.1 mm O-ring from the front cover.
5. Remove the plunger seal.



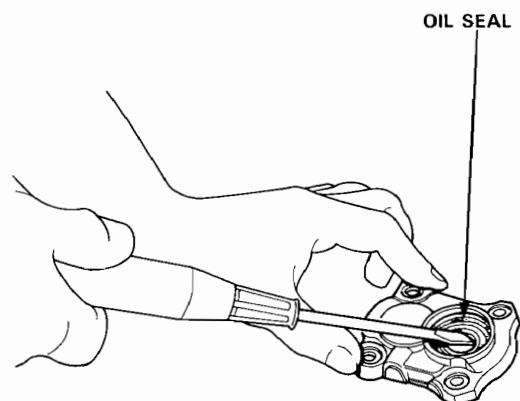
6. Separate the port housing from the pump housing.



7. Remove the drive and driven gears from the pump housing.



8. Remove the four O-rings.
9. Remove the plungers if necessary.
10. Pry the oil seal out from the front cover.

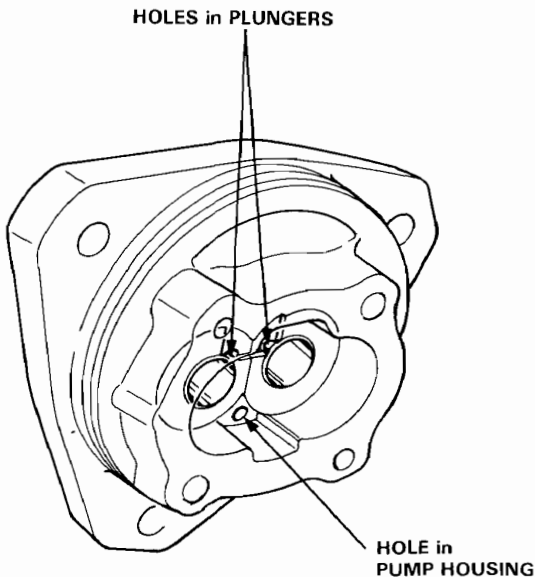


Pump

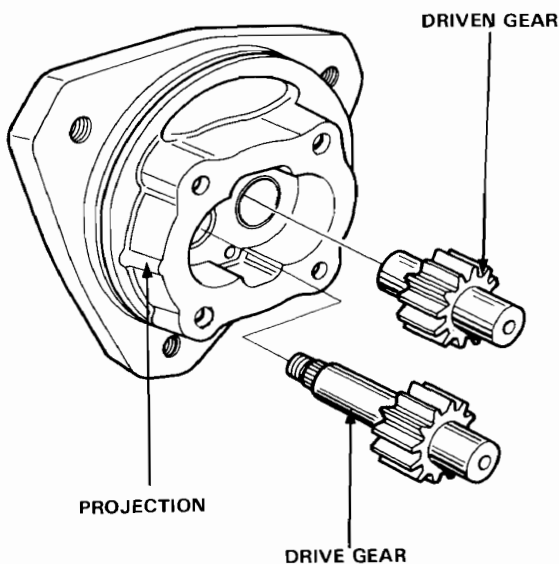
Housing Assembly

1. If the plungers are removed, be sure to position them correctly, as shown.

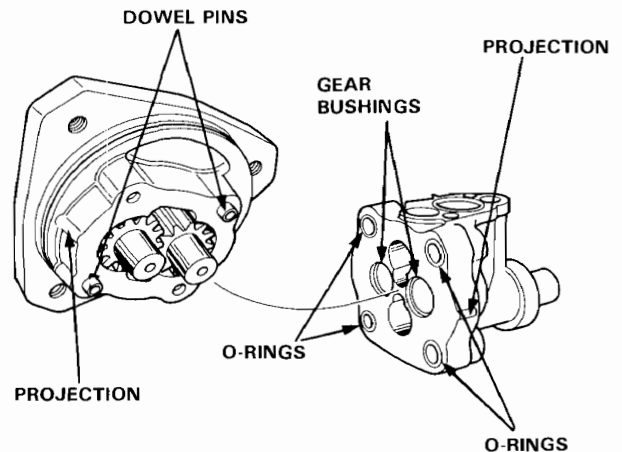
NOTE: Lubricate the plunger housing with power steering fluid before installing the plungers.



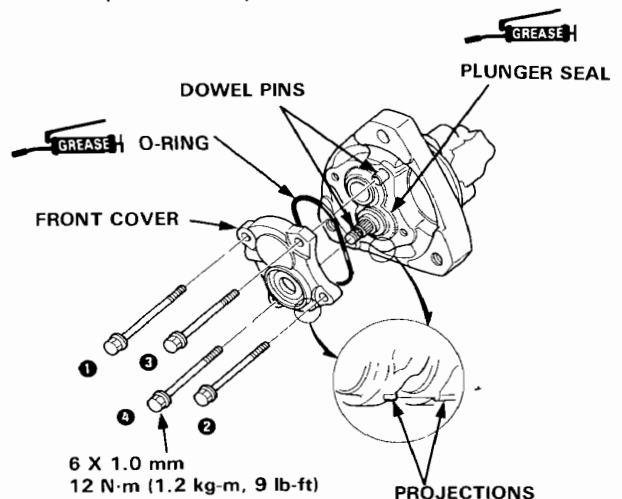
2. Lubricate the inside surface of the plungers and gears with power steering fluid.
3. Install the gears in the pump housing with the drive gear toward the projection on the housing.



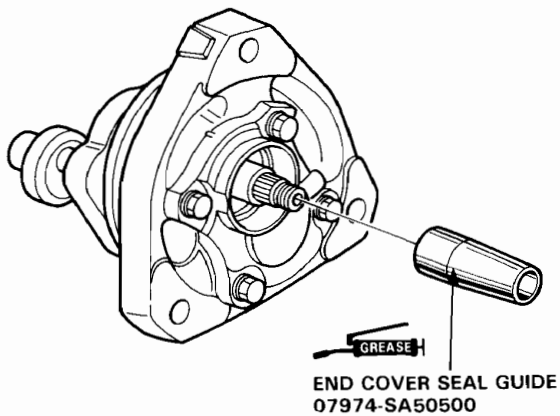
4. Lubricate the gear bushing of the port housing.
5. Install the two dowel pins and four O-rings.
6. Install the port housing on the pump housing with the projections on the housings aligned.



7. Grease the plunger seal, then install it over the plungers.
8. Install the dowel pins in the pump housing.
9. Put grease in the groove of the front cover first, then position a new O-ring on the cover.
10. Install the front cover on the pump housing with the projections aligned.
11. Install the four bolts. Torque them to the specified torque in the sequence shown.

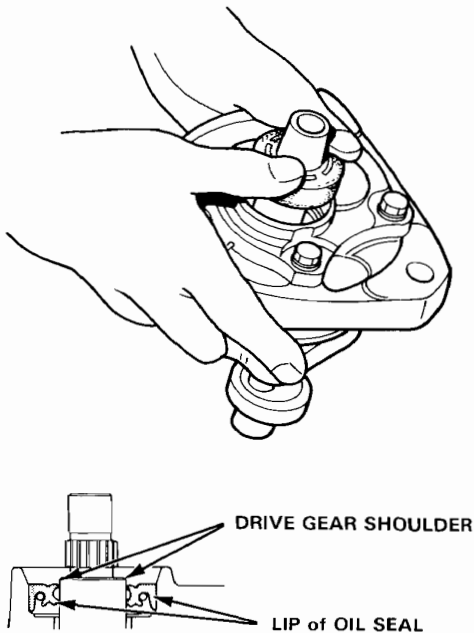


12. Grease the end cover seal guide then install it over the drive gear.

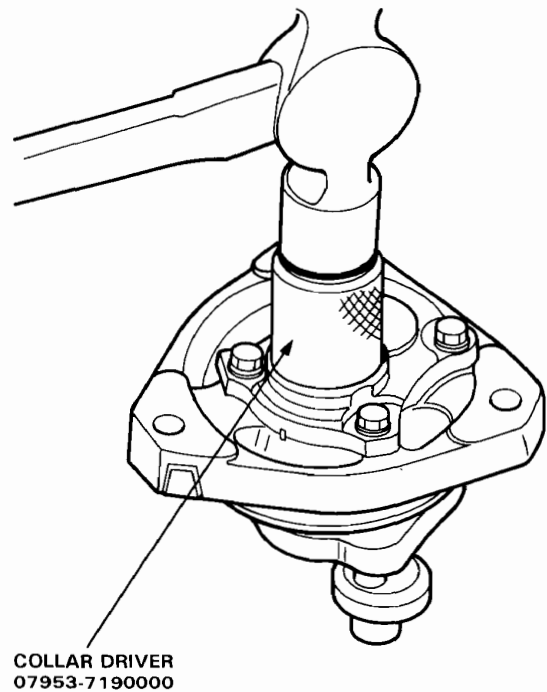


13. Install the oil seal by hand.

NOTE: Remove the special tool if the lip of the oil seal is over the drive gear shoulder.



14. Drive in the oil seal with special tool.

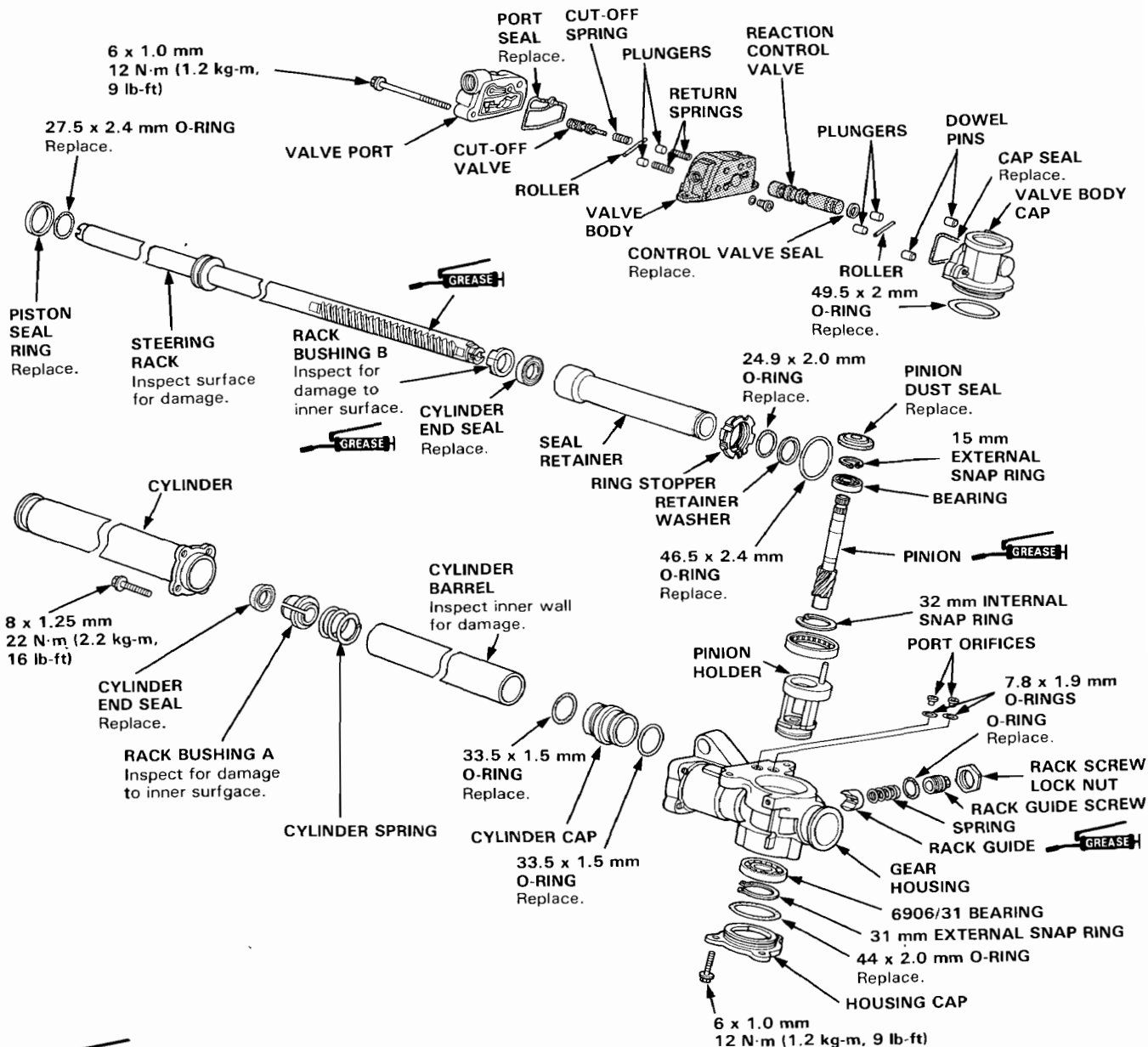


Gearbox

Index

CAUTION:

- Before disassembling the gearbox, wash it off with solvent.
- Thoroughly clean all disassembled parts.
- Always replace O-rings and seals.
- Replace parts with damaged sliding surfaces.
- Do not dip seals and O-rings in solvent; coat O-rings with grease, and make sure they stay in position during reassembly.
- The shaded parts (valve body, reaction control valve, cut-off valve) are a matched set; if the valve body is faulty, replace the complete valve body unit.



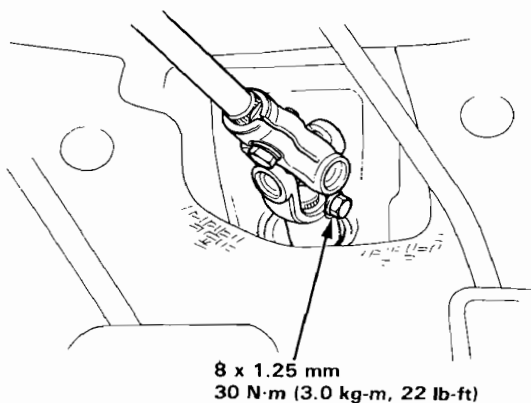
STEERING GREASE...Honda parts number 08740-99969

NOTE: L/H shown, R/H is symmetry.

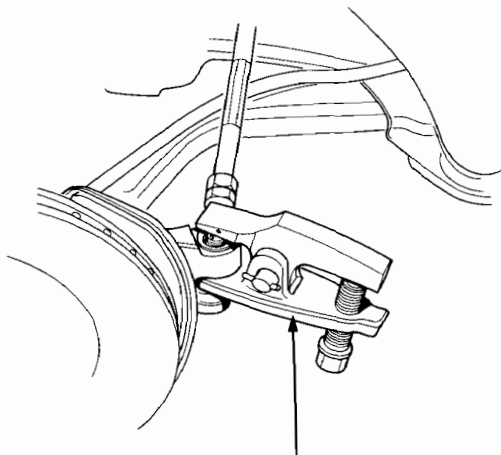


Removal/Installation

1. Turn the steering wheel all the way to the left.
2. Remove the bottom bolt in the steering shaft connector, and pull the connector up off the pinion shaft.



3. Drain the power steering fluid as described on 19-11.
4. Raise the front of car on jack stands and remove the front wheels.
5. Remove the cotter pins, and unscrew the tie-rod end ball joint nuts halfway.
6. Break the ball joints loose using the Ball Joint Remover.
7. Then, remove the nuts, and lift the tie-rod ends out of steering knuckles.



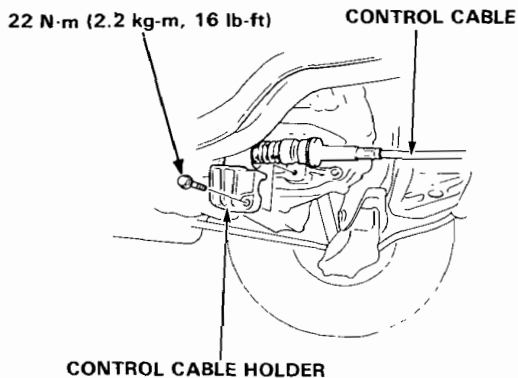
BALL JOINT REMOVER
07941-6920001

8. Manual Transmission Only:

- Disconnect the shift lever torque rod from the clutch housing.
- Slide the pin retainer out of way, drive out the spring pin with pin punch, then disconnect the shift rod.

9. Honda matic Transmission Only:

- Remove the shift cable holder and pull the shift cable down by hand.

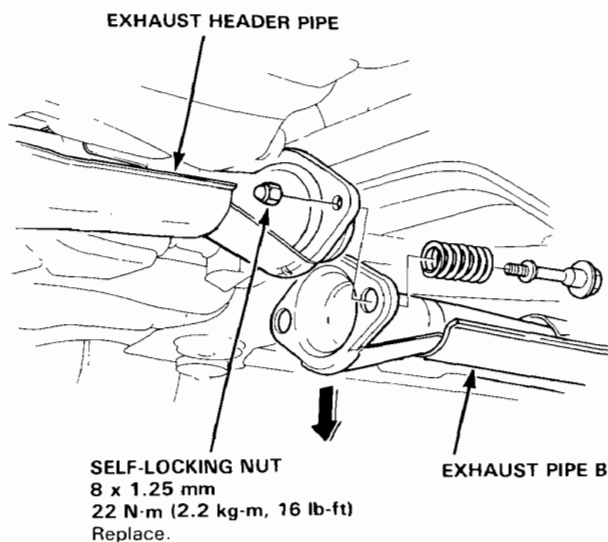


(cont'd)

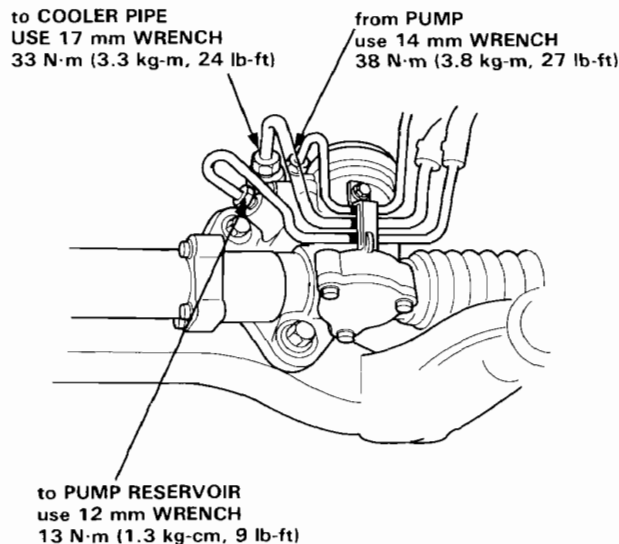
Gearbox

Removal/Installation (cont'd)

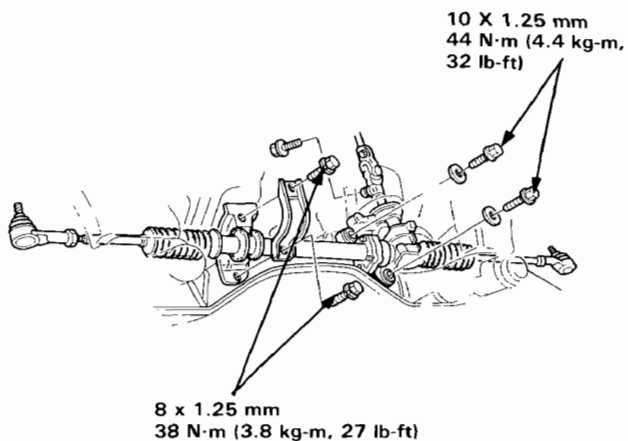
9. Remove the self-locking nuts connecting the exhaust header pipe to exhaust pipe B, then separate exhaust pipe B from the header pipe.



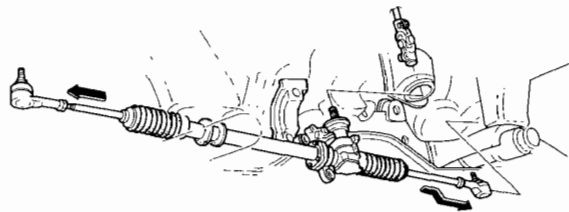
10. Remove exhaust pipe B.
11. Clean the gearbox and surrounding area thoroughly.
12. Disconnect fluid lines from the valve body unit.



13. Remove the gearbox mounting bolts.



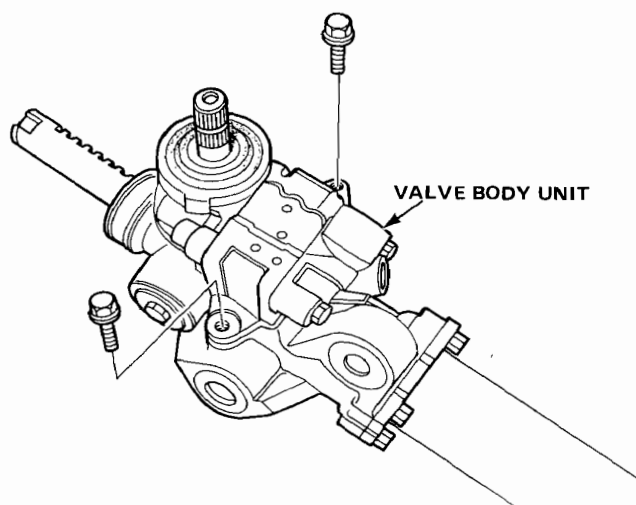
14. Drop the gearbox far enough so the end of the pinion shaft comes out of its hole in the frame channel, then rotate it forward until the shaft is pointing to the rear. Slide the gearbox to the right (L/H) or left (R/H) until the left (L/H) or right (R/H) tie rod clears the rear beam, then drop it down and out of the car to the left.



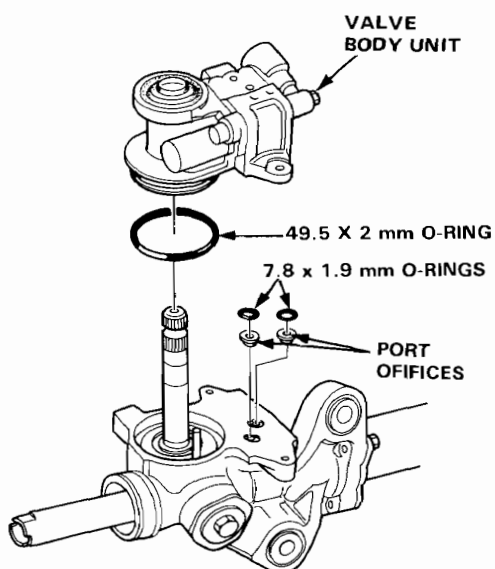
15. Gearbox installation is in the reverse order of removal.

Valve Body Unit Disassembly

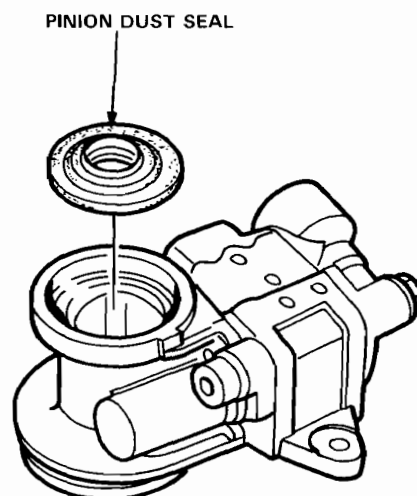
1. Remove the two 6 mm bolts holding the valve body unit to the gear housing.



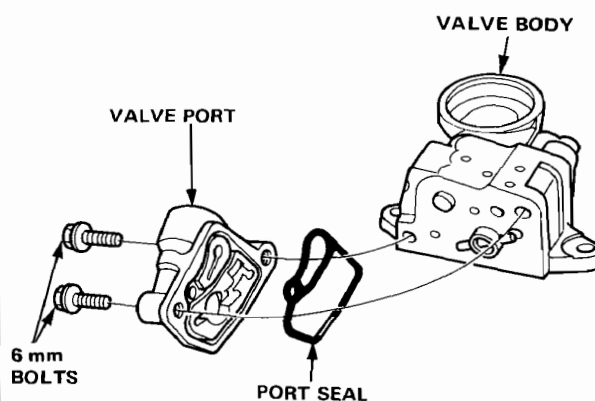
2. Remove the 7.8 x 1.9 mm O-rings and port orifices from the gear housing.
3. Remove the 49.5 x 2 mm O-ring from the valve body unit.



4. Remove the pinion dust seal from the valve body unit.



5. Remove the two 6 mm bolts and remove the valve port from the valve body.
6. Remove the port seal.

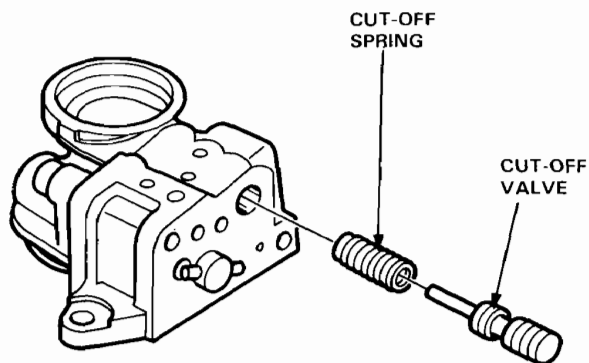


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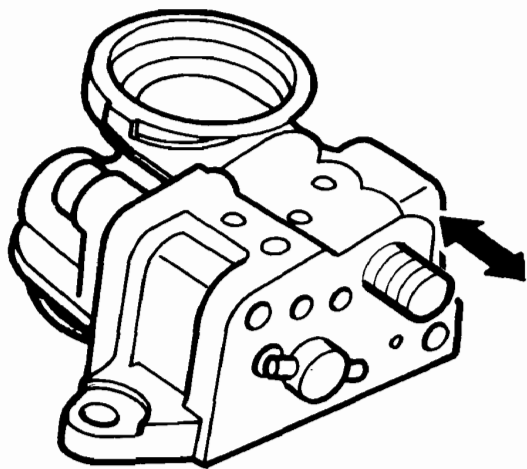
Gearbox

Valve Body Unit Disassembly (cont'd)

7. Remove the cut-off valve and spring from the valve body.

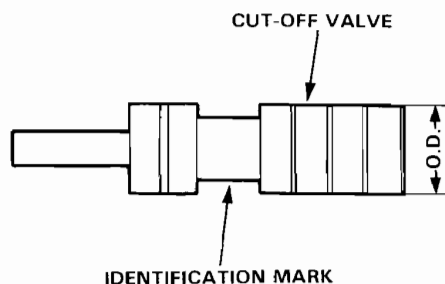


8. Check the cut-off valve:
 - Inspect its surface for scoring or scratches.
 - Slip it back into the valve body, and make sure it slides smoothly without drag and without side play.



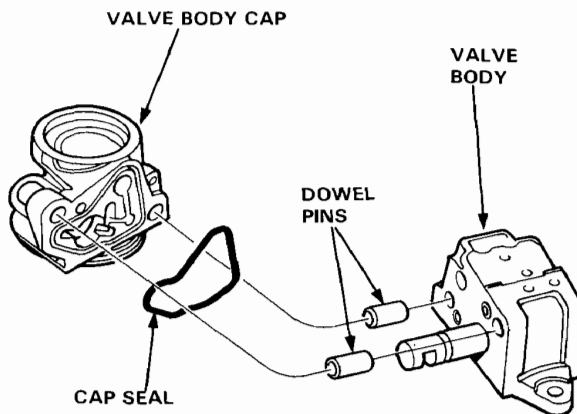
NOTE:

- The cut-off valve is sized to fit the valve body, so, if you replace it, make sure the new valve has the same identification mark on it.
- If the valve body is damaged, replace the valve body unit as a set.



Identification Mark	Outside Diameter	Part Number
A	10.000 – 10.005 mm (0.3937 – 0.3939 in.)	53650 – SB4 – 950
B	9.995 – 10.000 mm (0.3935 – 0.3937 in.)	53651 – SB4 – 950
C	9.990 – 9.995 mm (0.3933 – 0.3935 in.)	53652 – SB4 – 950

9. Remove the valve body cap from the valve body.
10. Remove the cap seal.
11. Remove the dowel pins.

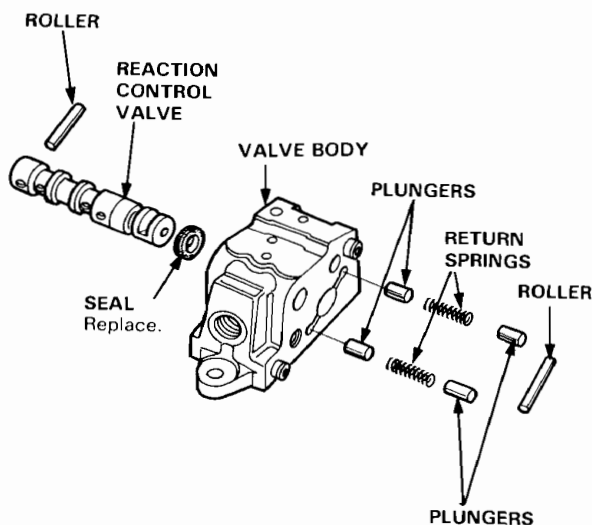


12. Remove the rollers from the valve body by pushing the valve out one side of the valve body, and then the other.

NOTE: When removing the rollers, hold the plungers with your fingers to keep them from popping out.

13. Remove the plungers, return springs and reaction control valve.

14. Remove the control valve seal from the reaction control valve.



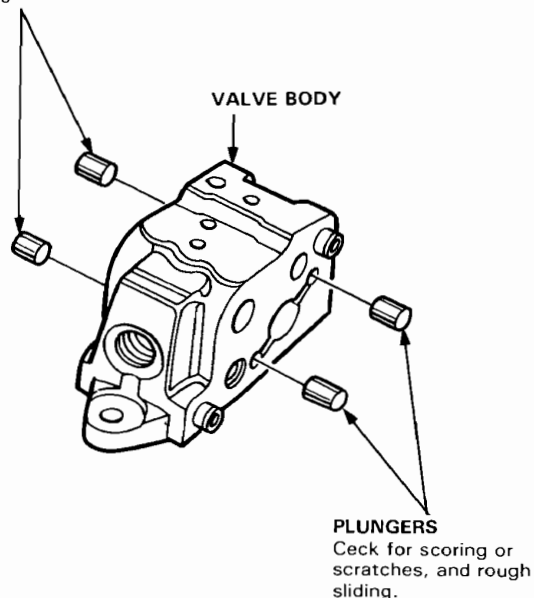
15. Check the plungers.

- Inspect their surface for scoring or scratches.
- Slip each plunger into the valve body, and make sure it slides smoothly, without drag or side play. If any plunger is damaged, replace it.

NOTE: If the valve body is damaged, replace the valve body unit as a set.

PLUNGERS

Check for scoring or scratches, and rough sliding.

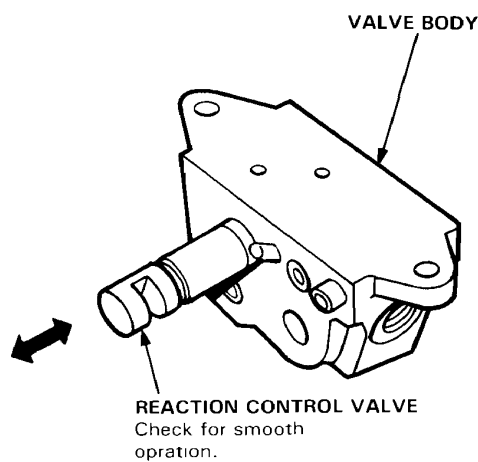


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Gearbox

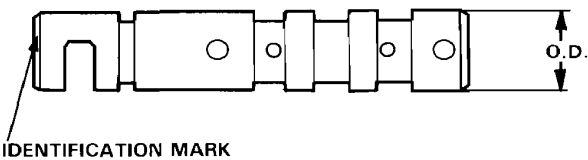
Valve Body Unit Disassembly (cont'd)

16. Check the reaction control valve.
- Inspect its surface for scoring or scratches.
 - Slip it into the valve body, and make sure it slides smoothly, without drag or side play.



NOTE:

- The reaction control valve is sized to fit the valve body, so, if you replace it, make sure the new valve has the same identification mark on it.
- If the valve body is damaged, replace the valve body unit as a set.



Identifica- tion Mark	Outside Diameter	Part Number
X	13.998 – 14.003 mm (0.5511 – 0.5513 in.)	53646 – SB4 – 950
Y	13.993 – 13.998 mm (0.5509 – 0.5511 in.)	53647 – SB4 – 950
Z	13.988 – 13.993 mm (0.5507 – 0.5509 in.)	53648 – SB4 – 950



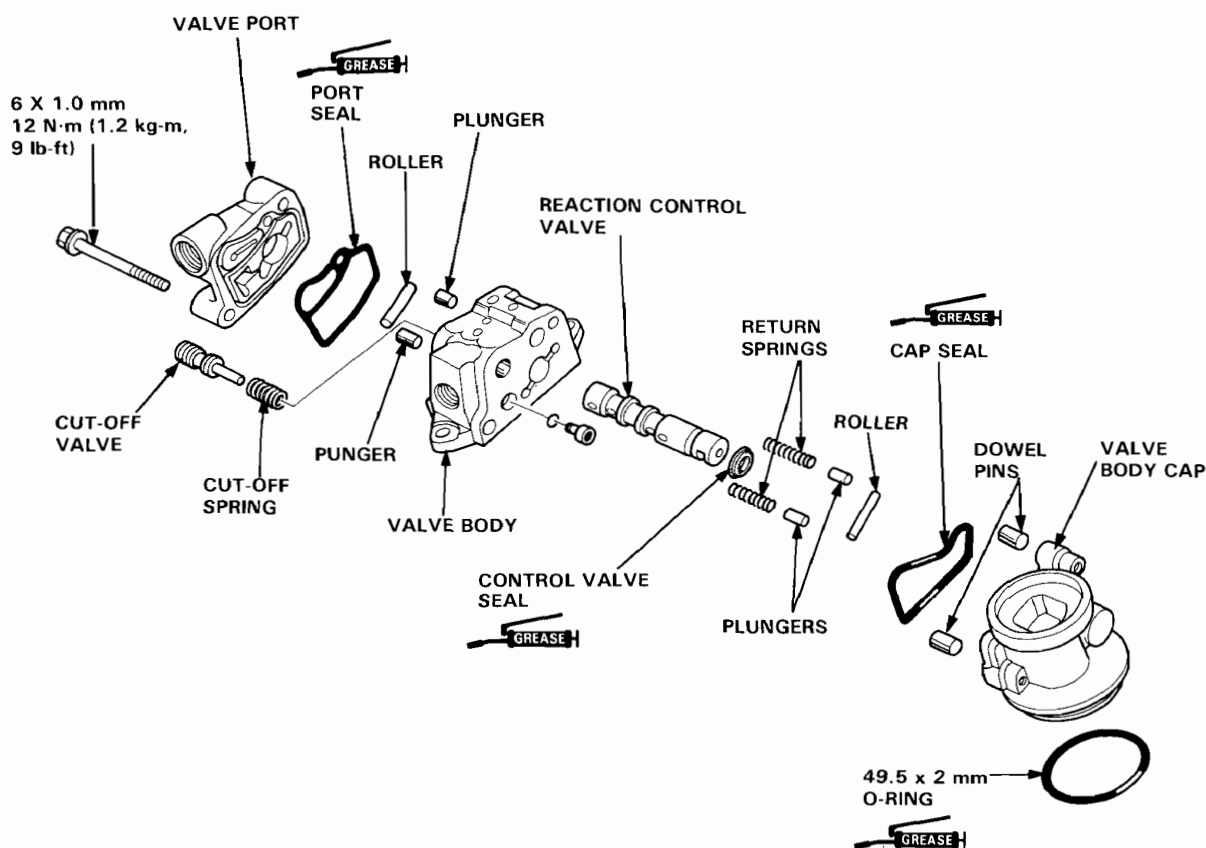
Valve Body Unit Assembly

1. Thoroughly clean the disassembled parts shown below.
2. Coat the plungers, cut-off valve and reaction control valve surfaces with power steering fluid.
3. Reassemble the parts in reverse order of disassembly.

CAUTION:

- Replace the O-rings and seals with new ones.
- Do not dip the O-rings and seals in solvent.
- Apply grease in the cap seal and port seal grooves to keep the seals in place.
- Apply grease to the O-ring for valve body cap and reaction control valve seal to keep them in place.

NOTE: L/H shown, R/H similar



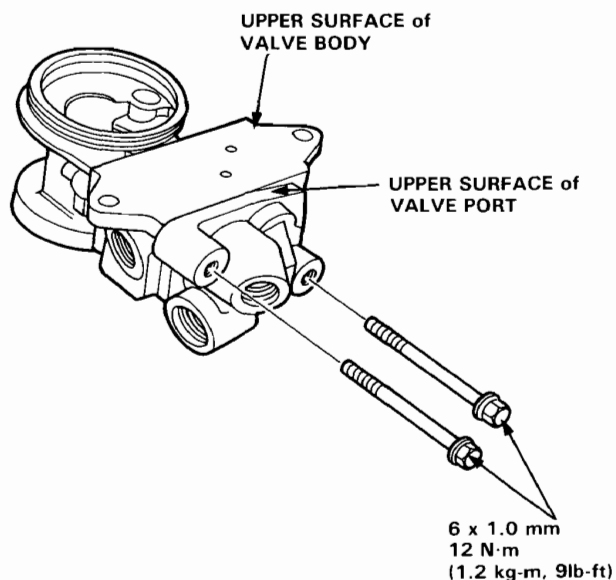
(cont'd)

Gearbox

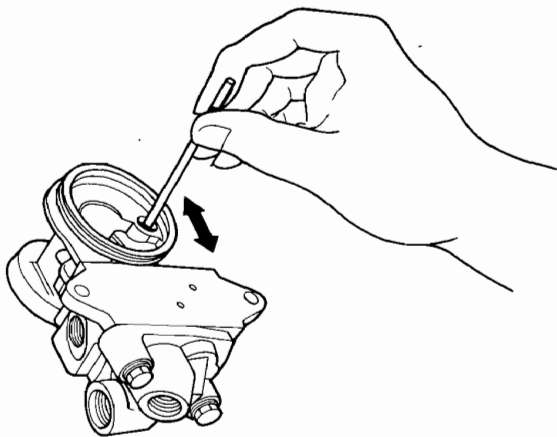
Valve Body Unit Assembly (cont'd)

4. Tighten the 6 mm bolts in the valve body unit.

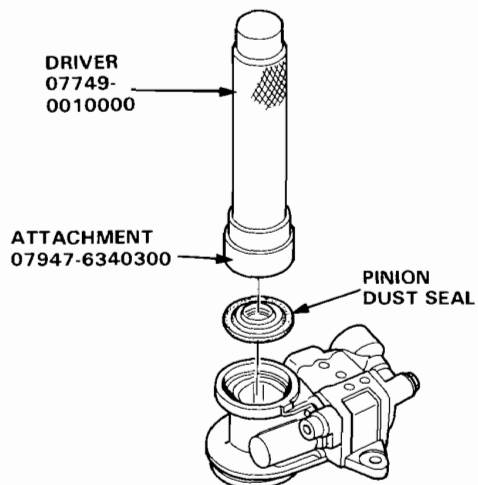
CAUTION: Make sure that the upper surface of the valve port does not exceed the upper surface of the valve body.



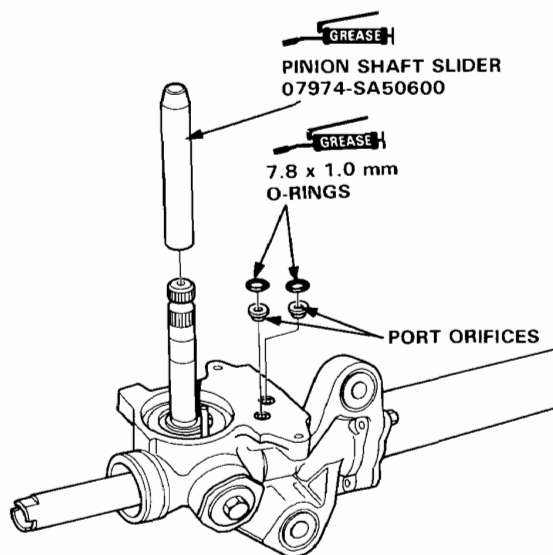
5. Make sure the reaction control valve moves smoothly, and returns to a neutral position.



6. Install the pinion dust seal with the special tool.



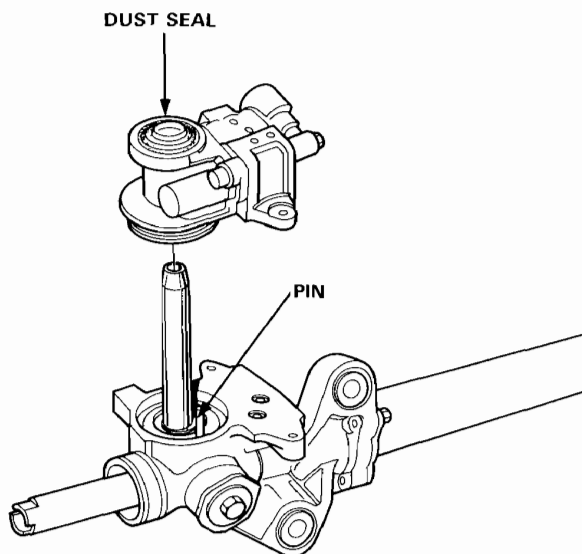
7. Apply grease in the O-ring grooves of the gear housing.
8. Install the port orifices and 7.8 x 1.0 mm O-rings.
9. Apply grease to surface of special tool and install the special tool on the pinion.



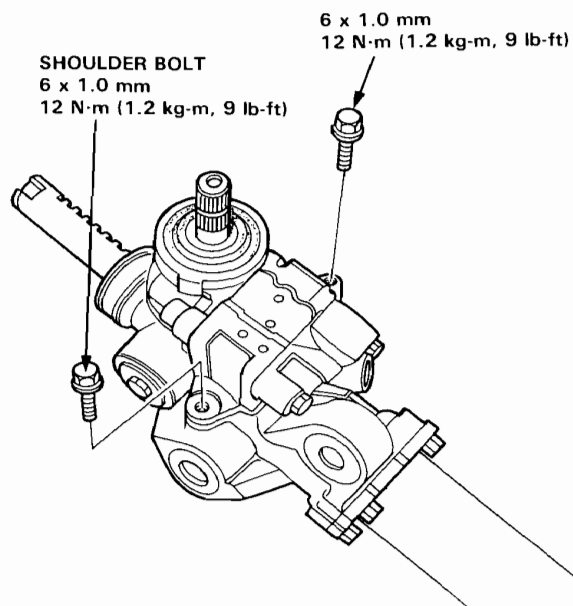
10. Install the valve body unit on the gearbox housing.

CAUTION:

- Be careful not to damage or distort the lip of dust seal, or it may be dislodged.
- Install the valve body unit carefully to avoid damaging the pin in the pinion holder
- Make sure the 49.5 x 2.0 mm O-ring is positioned properly and is not pinched when installing the valve body unit.



11. Tighten the 6 mm bolts.

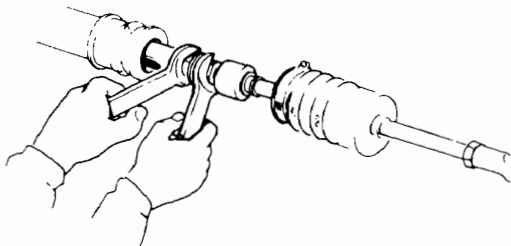


CAUTION: For proper assembly, the shoulder bolt must be installed in the rear hole; toward the rear of the car.

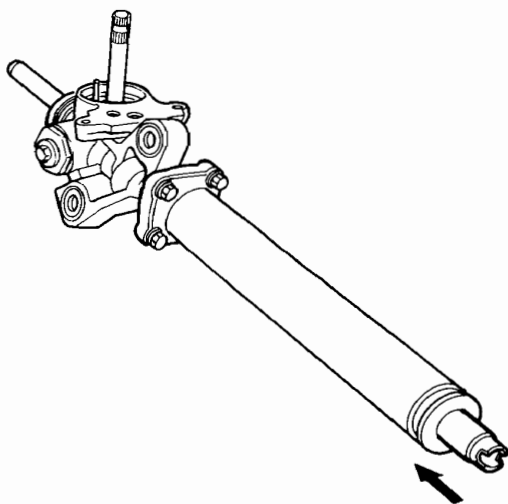
Gearbox

Gearbox Disassembly/Reassembly

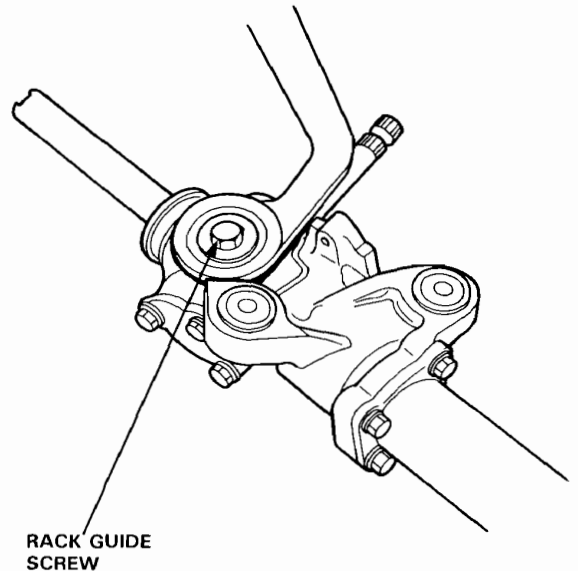
1. Remove the gearbox assembly from the rear beam (page 19-26).
2. Remove the valve body unit from the gearbox housing (page 19-27).
3. Bend back the tie-rod lock washers.
4. Hold the rack with a 19 mm wrench and unscrew the tie-rods with a 17 mm wrench.



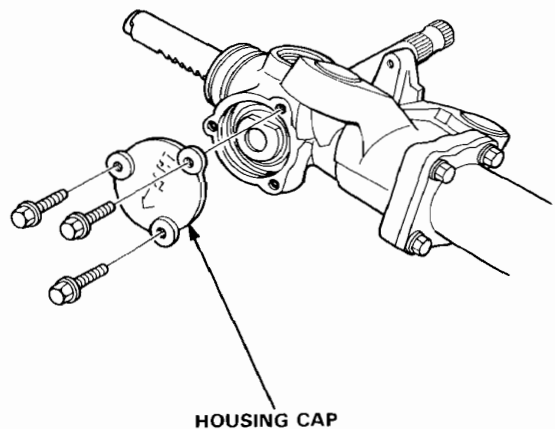
5. To prevent damage to the sealing surface of the steering rack, push the rack into its cylinder.



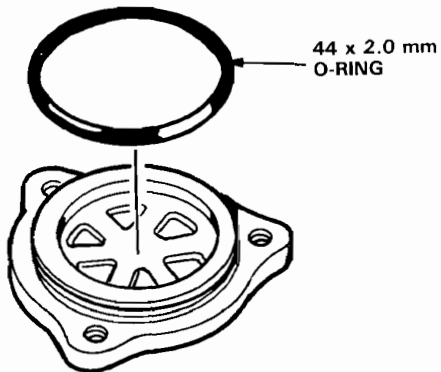
6. Loosen the rack screw locknut and remove the rack guide screw, spring and rack guide.



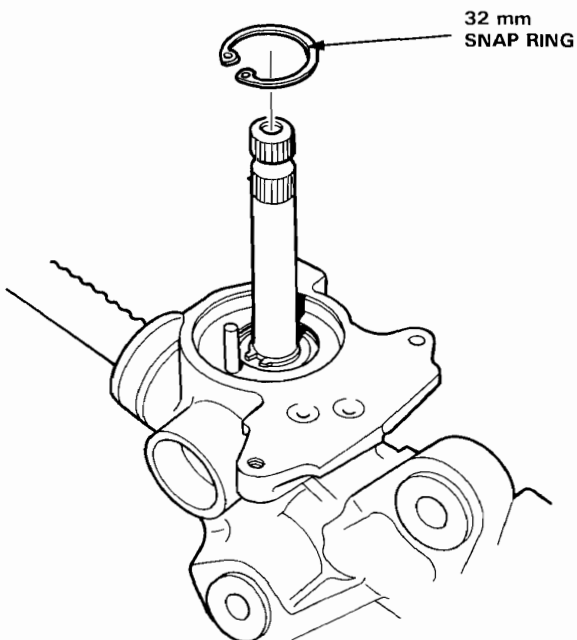
7. Remove the three 6 mm bolts and housing cap.



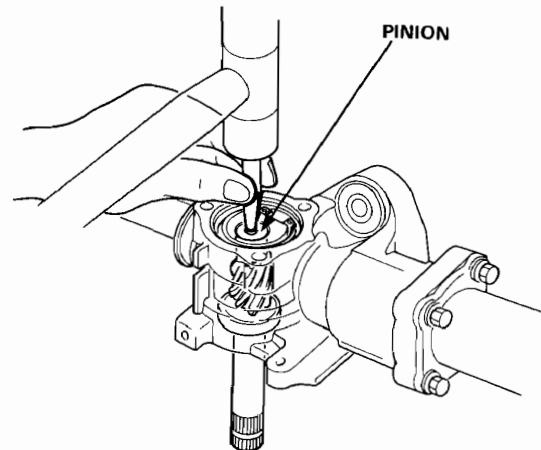
8. Remove the 44 x 2.0 mm O-ring from the housing cap.



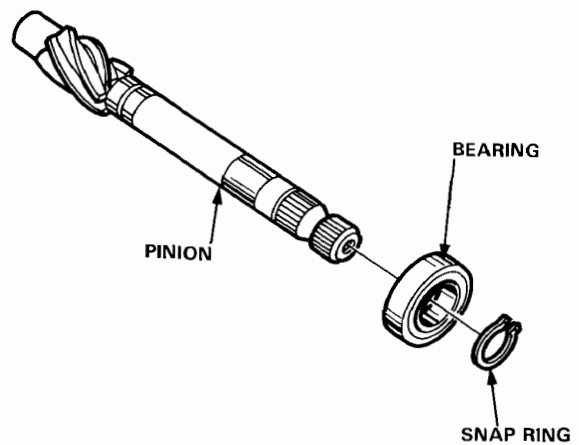
9. Remove the 32 mm snap ring.



10. Remove the pinion from the gear housing by tapping it lightly.



11. Remove the 15 mm snap ring and bearing if necessary.

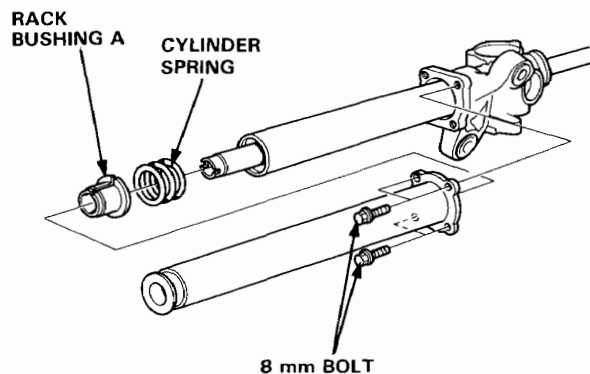


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Gearbox

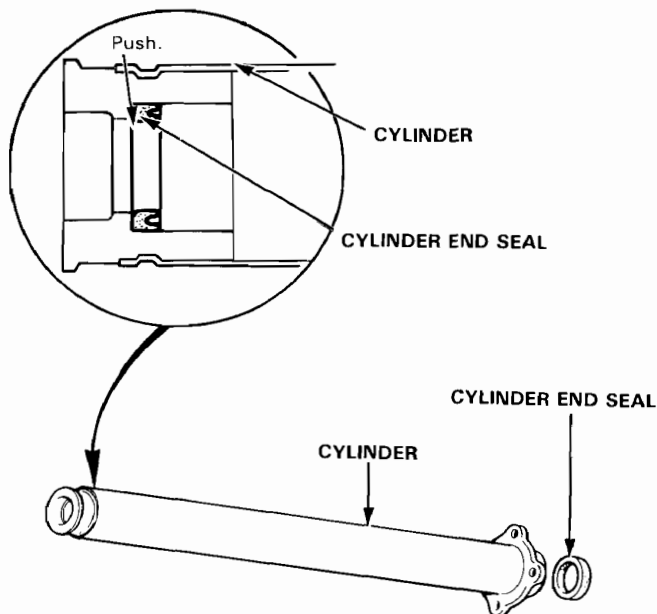
Gearbox Disassembly/Reassembly (cont'd)

12. Remove the four 8 mm bolts.
13. Remove the cylinder.
14. Remove the rack bushing A and cylinder spring.

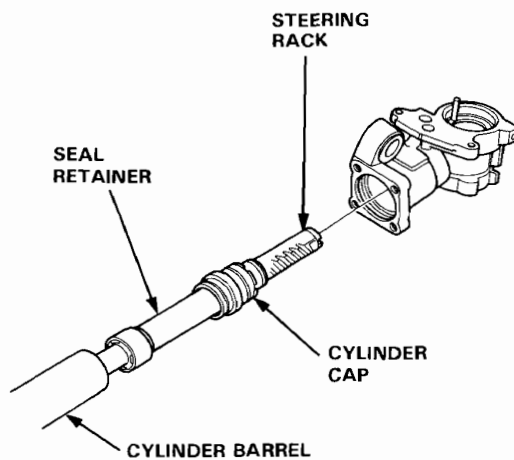


15. Remove the cylinder end seal from the cylinder.

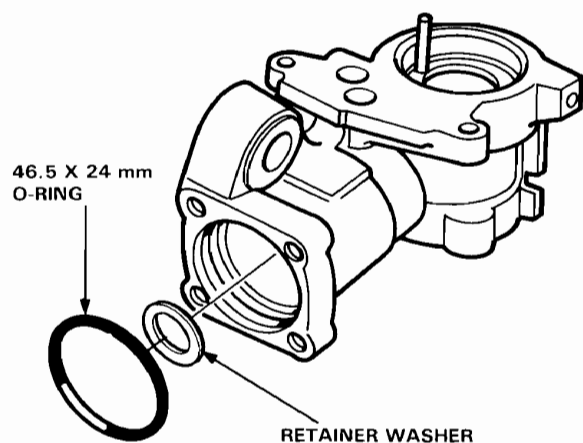
CAUTION: Do so by hand to avoid damaging the housing.



16. Remove the cylinder barrel, seal retainer, cylinder cap and steering rack from the gear housing.

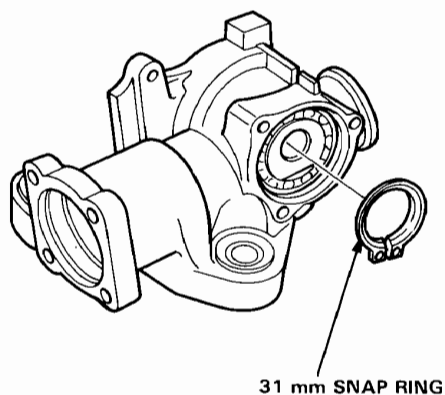


17. Remove 46.5 x 2.4 mm O-ring and retainer washer from the gear housing.

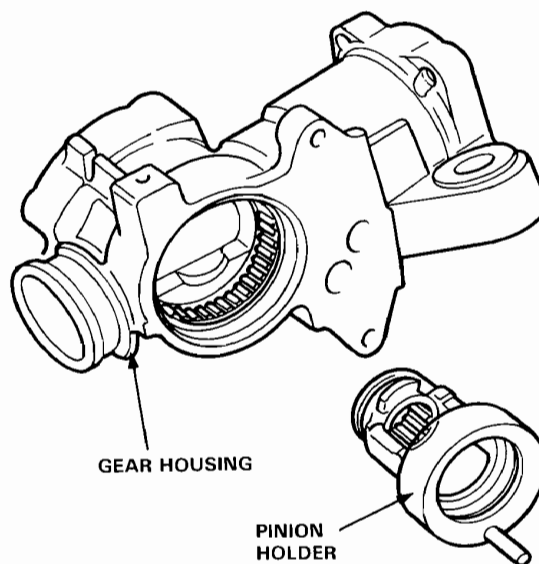


18. Check the bearing and pinion holder for faulty movement and excessive play. If they are OK and the grease in the bearing is clean, go on step 19. If they are damaged, or if there is dirt in the grease, proceed as follows:

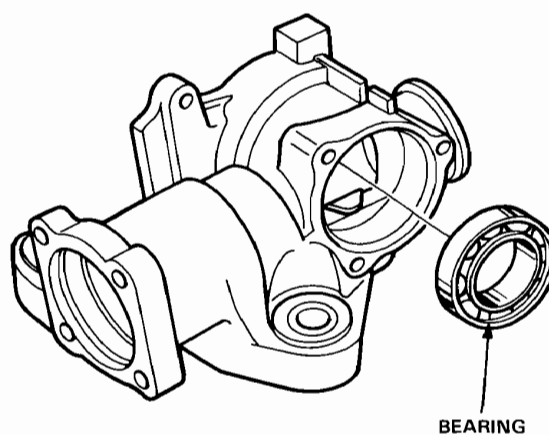
- Remove the 31 mm snap ring.



- Remove the pinion holder from the gear housing.



- Remove the bearing from the gear housing.

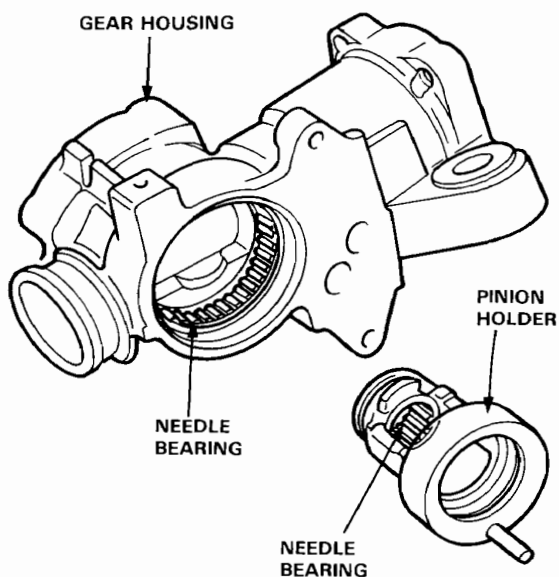


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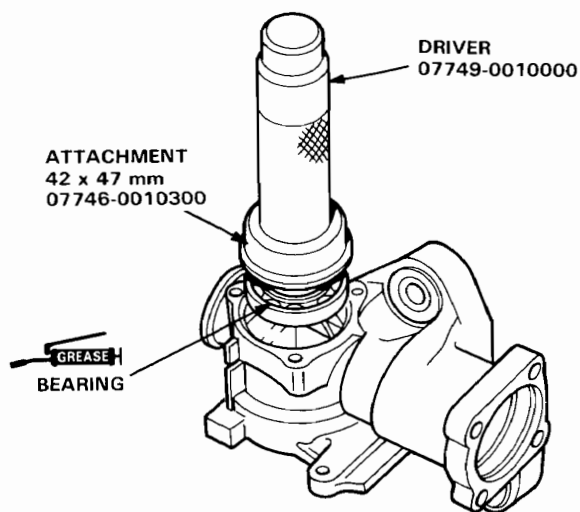
Gearbox

Gearbox Disassembly/Reassembly (cont'd)

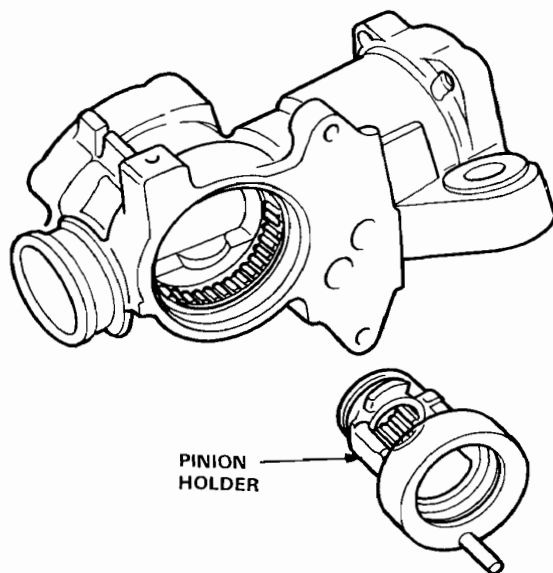
- Check the needle bearings in the pinion holder and in the gear housing for damage. If they are OK, pack with grease. If the bearings are damaged replace the gearbox assembly as a set.



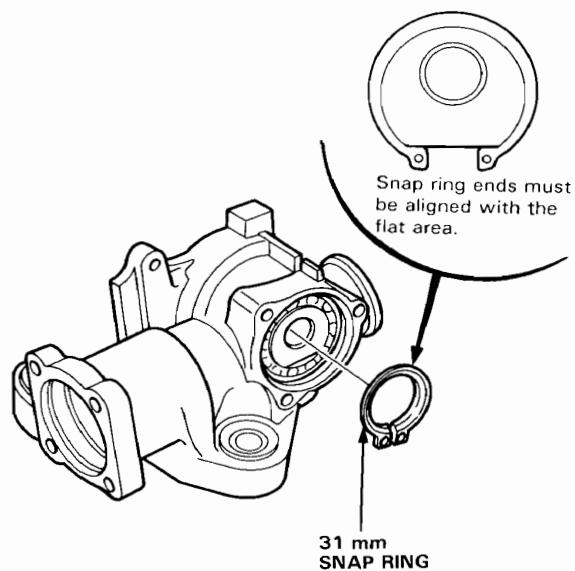
- Pack a new bearing with grease, then install with special tool, as shown.



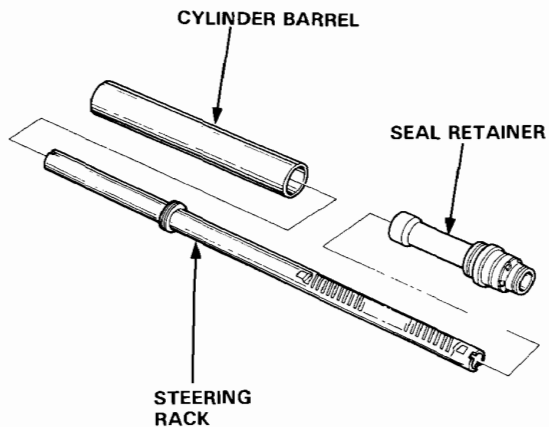
- Install the pinion holder in the gear housing.



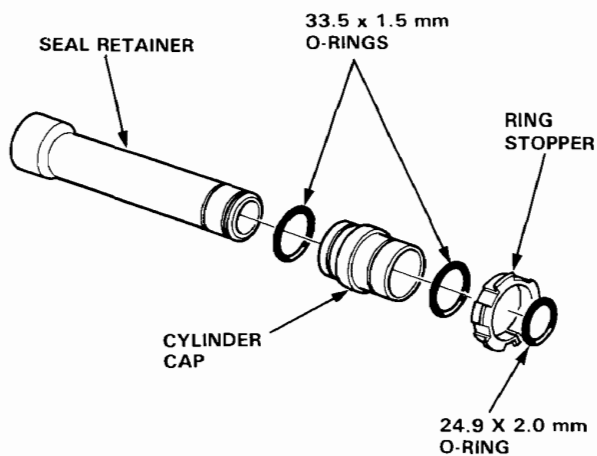
- Install the 31 mm snap ring with the taper side facing out.



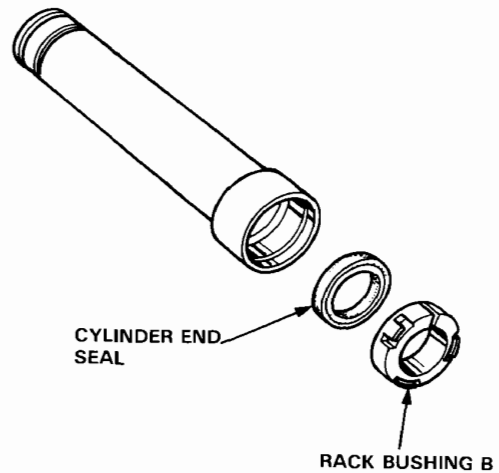
19. Remove the cylinder barrel and seal retainer from the steering rack.



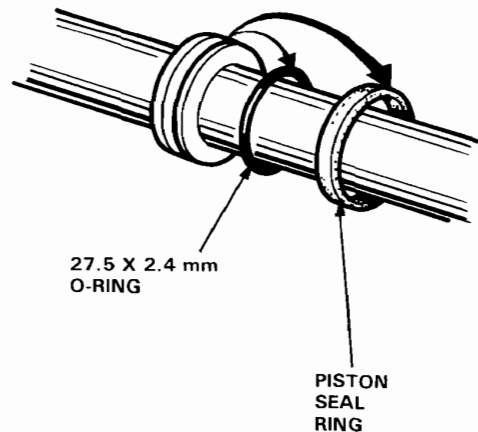
20. Remove the 24.9 x 2.0 mm O-ring, the ring stopper, and the cylinder cap from the seal retainer. Remove 33.5 x 1.5 mm O-rings from the cylinder cap.



21. Remove rack bushing B from the seal retainer. Then remove the cylinder end seal.



22. Carefully pry off the piston seal ring with a small screwdriver, then remove the 27.5 x 2.4 mm O-ring.

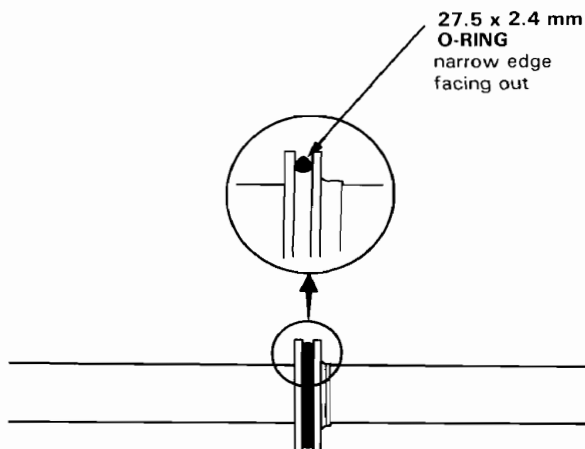


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Gearbox

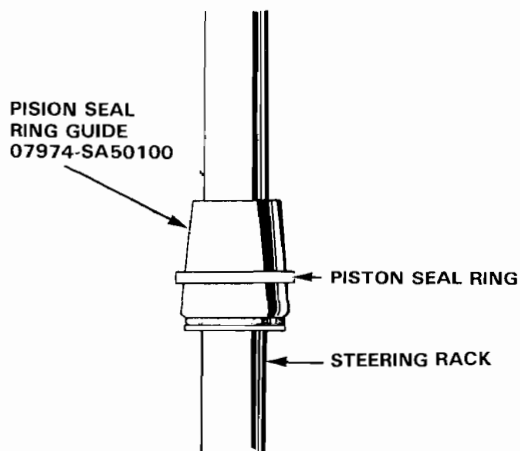
Gearbox Disassembly/Reassembly (cont'd)

23. Install a new O-ring on the rack.



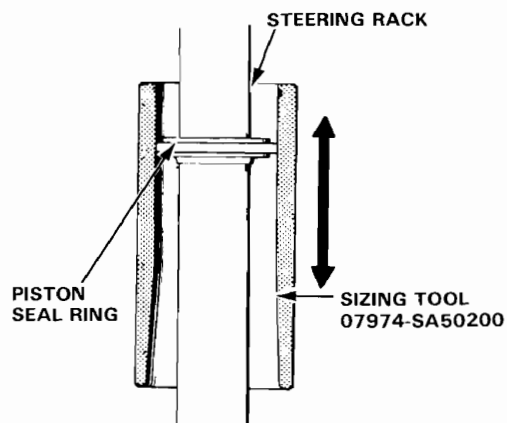
24. Install a new piston seal ring.

- Coat the piston seal ring guide tool with power steering fluid, and slide it onto the rack, big end first.

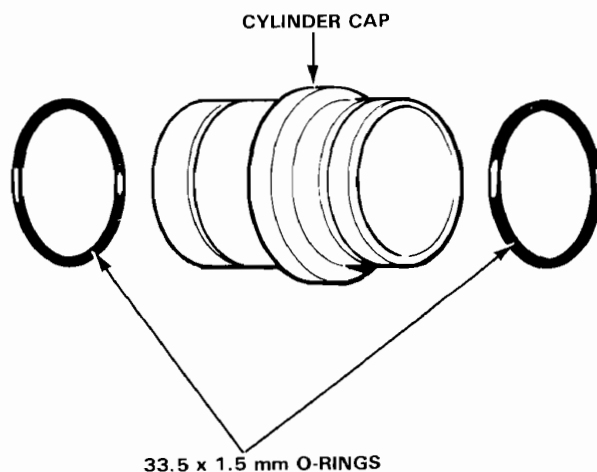


- Slide the new piston seal ring onto the guide tool, work it down to the big end of the tool, and then pull it off into the piston groove, on top of the O-ring.

25. Coat the piston seal ring and the inside of the sizing tool with power steering fluid. Carefully slide the tool onto the rack and over the piston seal ring, then rotate the tool as you move it up and down to break-in the piston ring.

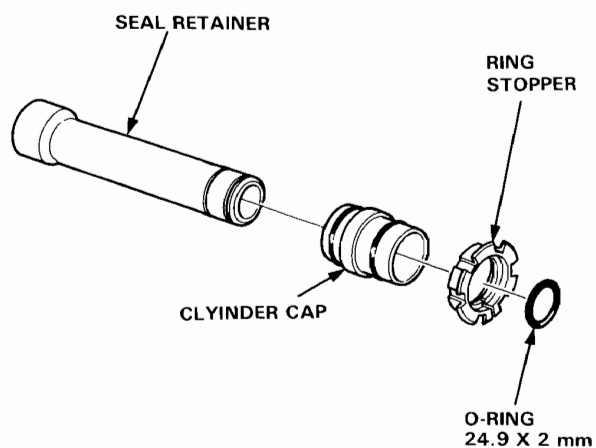


26. Coat new O-rings with grease and install them on the cylinder cap.

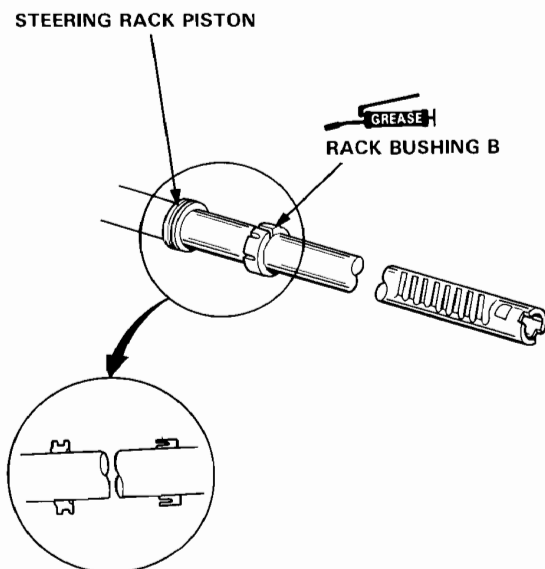




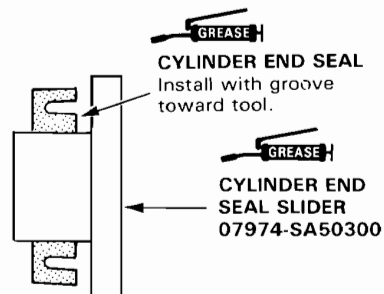
27. Slide the cylinder cap onto the seal retainer.
28. Install the ring stopper on the seal retainer.
29. Install a new 24.9 x 2 mm O-ring on the seal retainer.



30. Install the rack bushing B with the groove toward the steering rack piston.

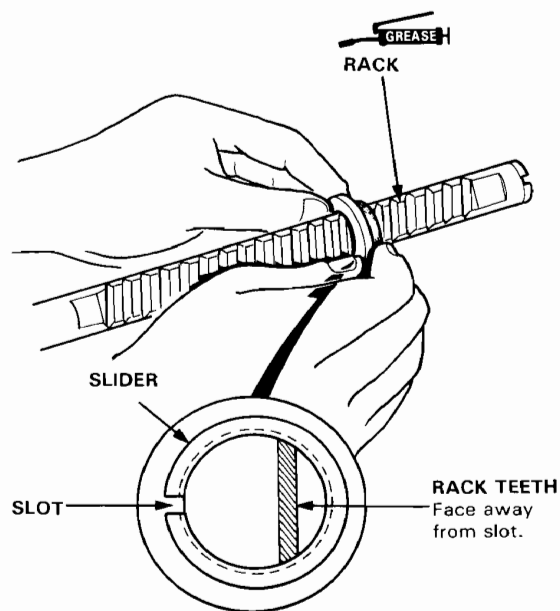


31. Coat the end seal slider tool and cylinder end seal with grease, and install the cylinder end seal on the seal slider tool with the groove toward the tool.



32. Coat the rack and fill its teeth with grease then slide the tool onto the steering rack.

CAUTION: Make sure the rack teeth do not face the slot in the slider tool.

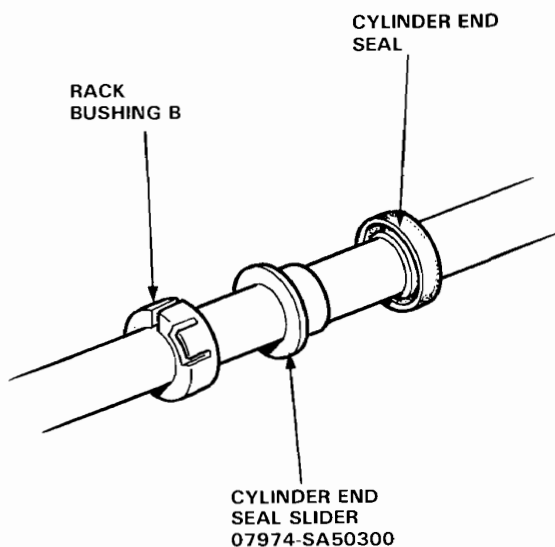


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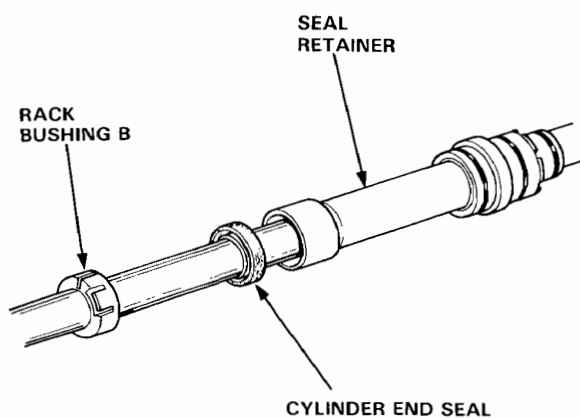
Gearbox

Gearbox Disassembly/Reassembly (cont'd)

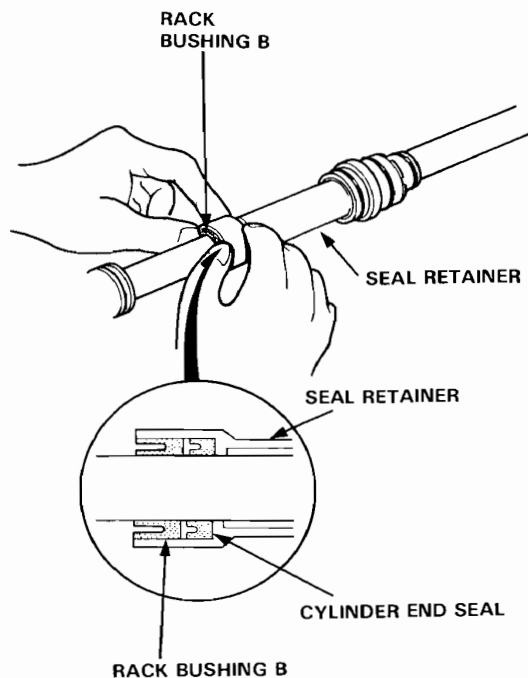
33. Remove the cylinder end seal slider from the steering rack.



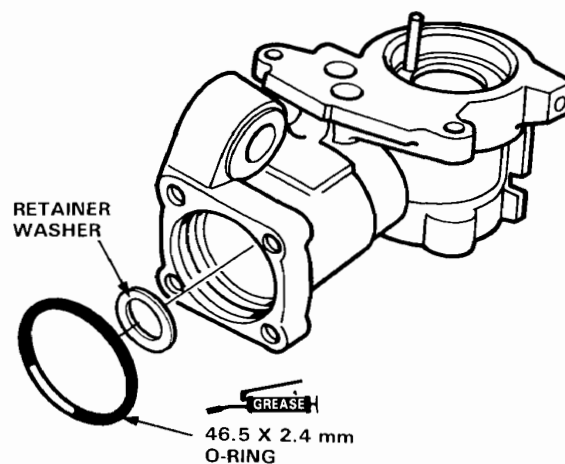
34. Slide the seal retainer onto the steering rack.



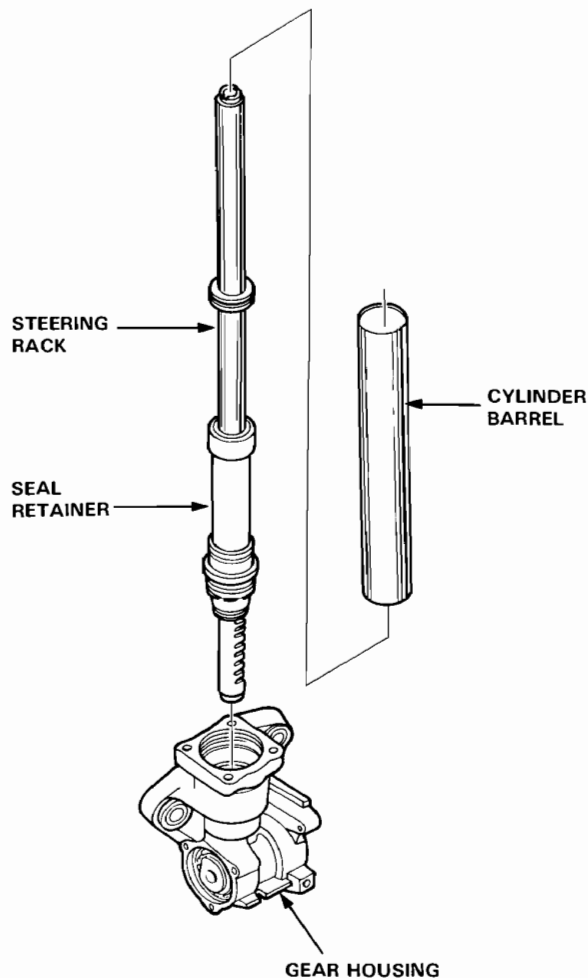
35. Push the rack bushing B into the seal retainer so as to contact the cylinder end seal with the seal retainer.



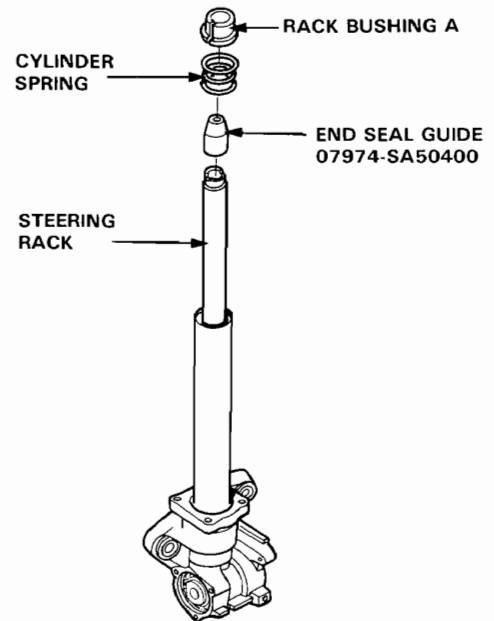
36. Install the 46.5 x 2.4 mm O-ring and retainer washer into the gear housing.



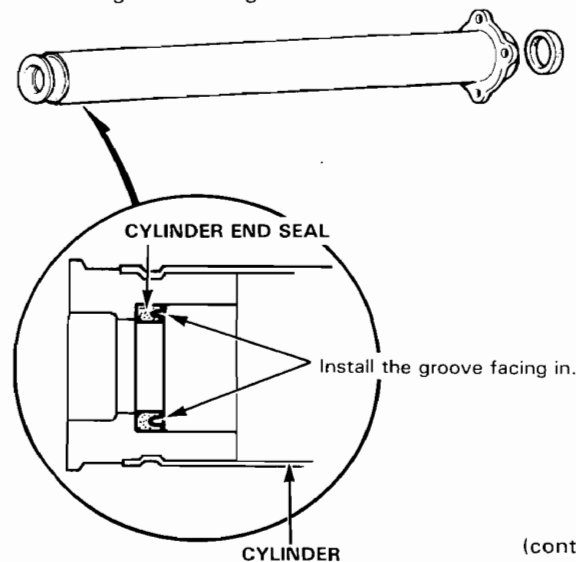
37. Stand the gear housing on the work bench and press the seal retainer and steering rack into the gear housing.
38. Coat the inside surface of the cylinder barrel with power steering fluid, slide it over the rack and into the gear housing; press it into the housing until it seats.



39. Install the end seal guide tool on the steering rack.
40. Install the cylinder spring over the rack, then coat the rack bushing A with power steering fluid, and install it on the spring.



41. Coat the inside surface of the cylinder with power steering fluid and install the cylinder end seal with the groove facing in.



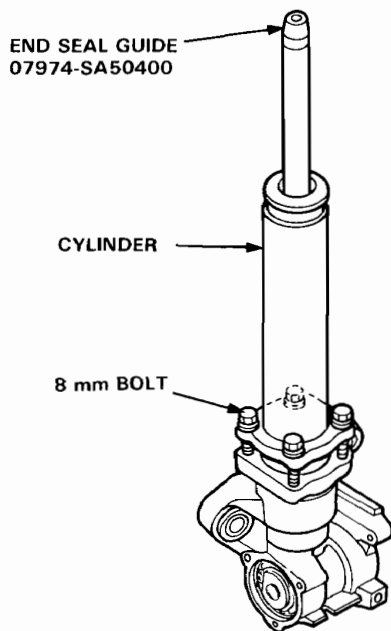
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Gearbox

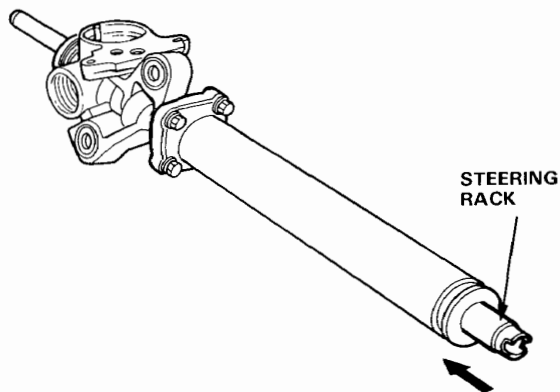
Gearbox Disassembly/Reassembly (cont'd)

42. Coat the end seal guide tool with grease.
43. Carefully slide the cylinder over the rack and install it on the gear housing.

CAUTION: Be carefull not to damage the cylinder end seal in the cylinder.

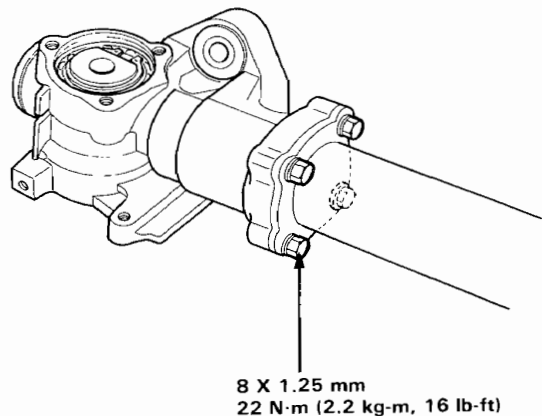


44. Push the steering rack until the rack is in the cylinder to prevent damage to the sliding surface of the steering rack.

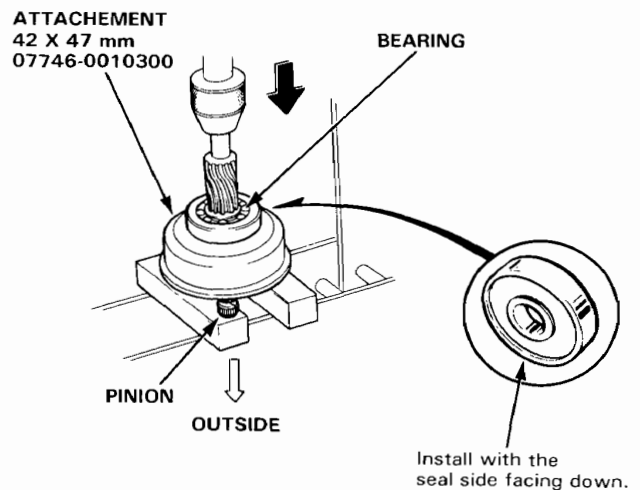


45. Secure the cylinder to the gear housing with the four 8 mm bolts.

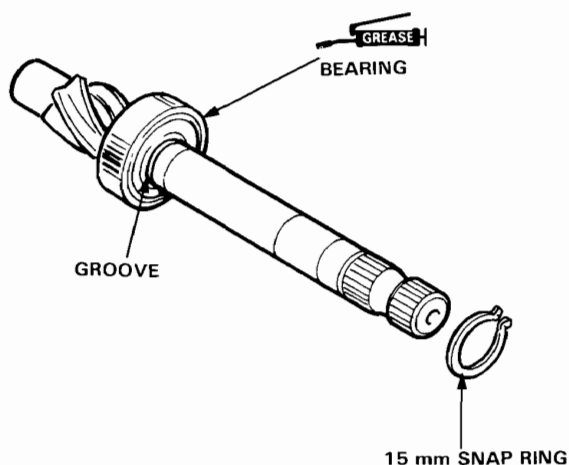
NOTE: Before tightening, make sure the mating surface of the cylinder and gear housing fit properly. Push and hold them together while you tighten the bolts.



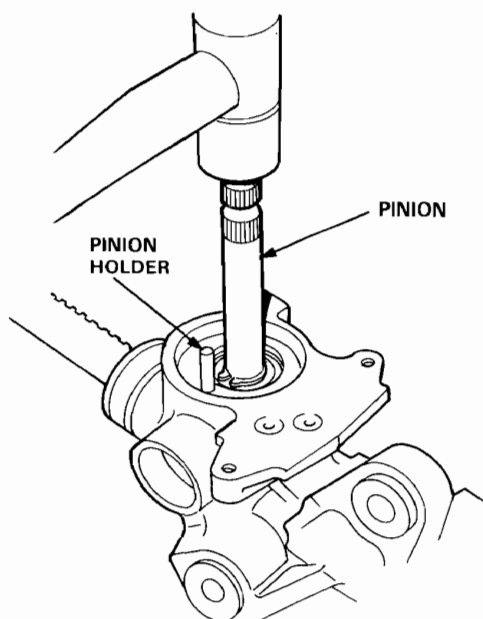
46. Using a press, install the bearing on the pinion, with its seal side facing down (out).



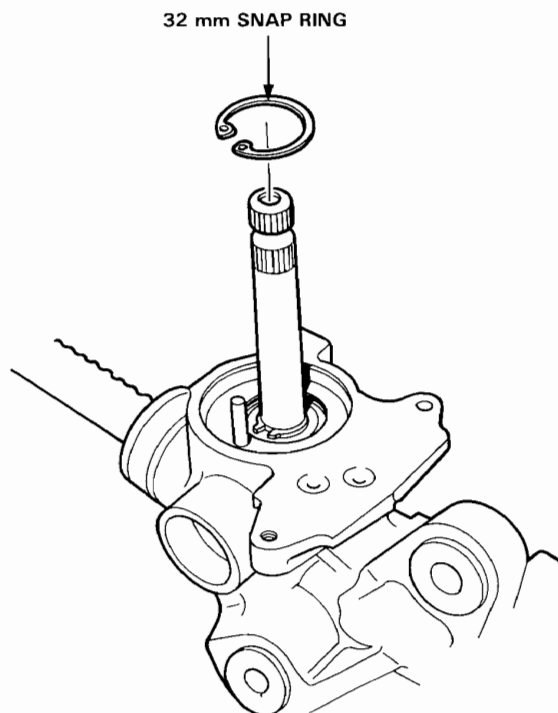
47. Install the 15 mm snap ring on the pinion.



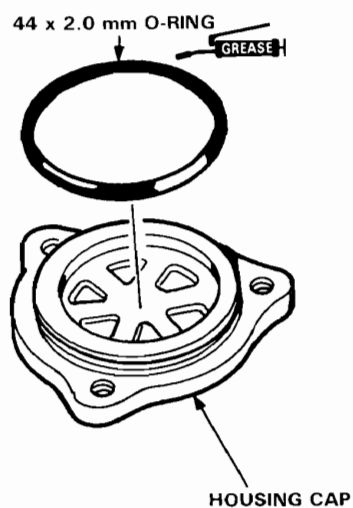
48. Grease the bearing and make sure that it turns smoothly.
49. Insert the pinion into the pinion holder in the gear housing.



48. Install 32 mm snap ring in the groove in the pinion holder, with its taper side facing out.



49. Coat the 44 x 2.0 mm O-ring with grease and install it in the groove in the housing cap.



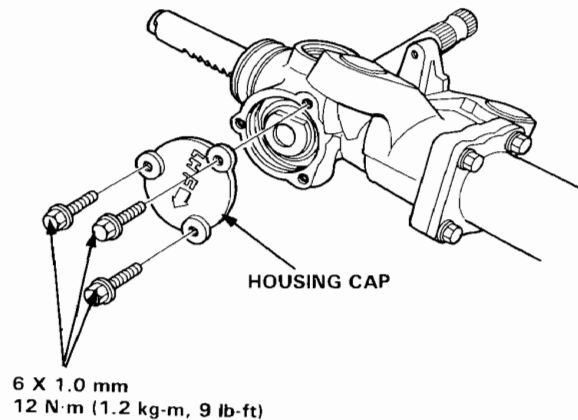
(cont'd)

Gearbox

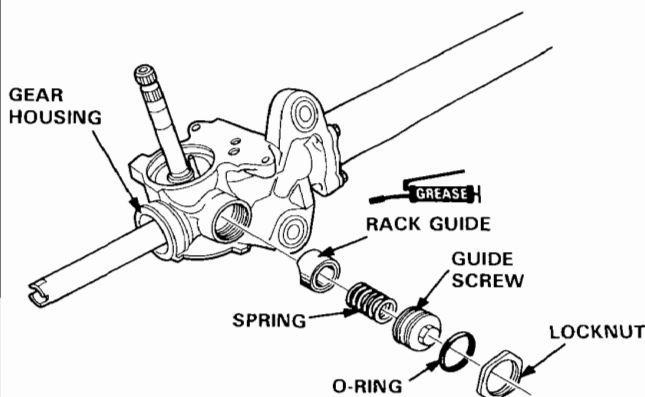
Gearbox Disassembly/Reassembly (cont'd)

50. Tighten the three 6 mm bolts to secure the housing cap to the gear housing.

NOTE: The arrow on the housing cap must face toward the front of the car when the gearbox is installed.

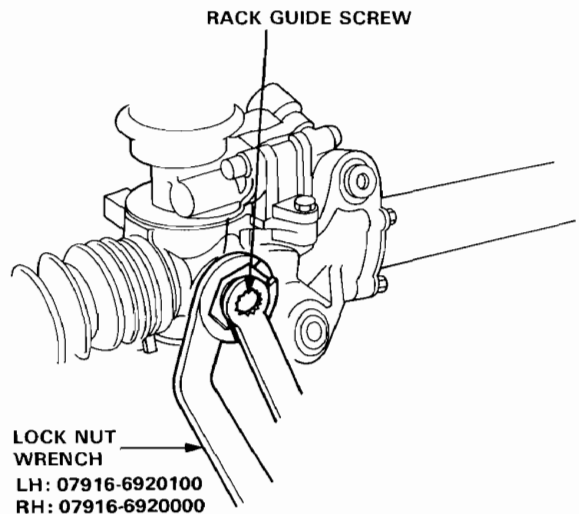


51. Coat the rack guide with grease.
52. Install the rack guide, spring, O-ring and rack guide screw in the gear housing.
53. Install the locknut on the rack guide screw, finger tight.

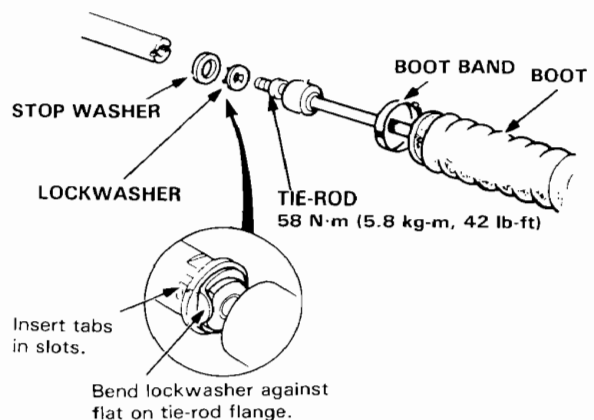


54. Tighten the guide screw until it compresses the spring and seats against the guide, then loosen it. Retighten it to 4 N·m (0.4 kg-m, 3 lb-ft) and back off about 35° (about 1/10 of a turn).

While holding the guide screw in its position, tighten the locknut to 25 N·m (2.5 kg-m, 18 lb-ft) with the locknut wrench.



55. Screw each tie-rod into the rack while holding the lockwasher so its tabs are in the slots in the rack end. Tighten the tie-rod securely, then bend the lockwasher back against the flat on the flange as shown. Install the boots and bands.



56. Make sure that the boots move smoothly while sliding the steering rack.

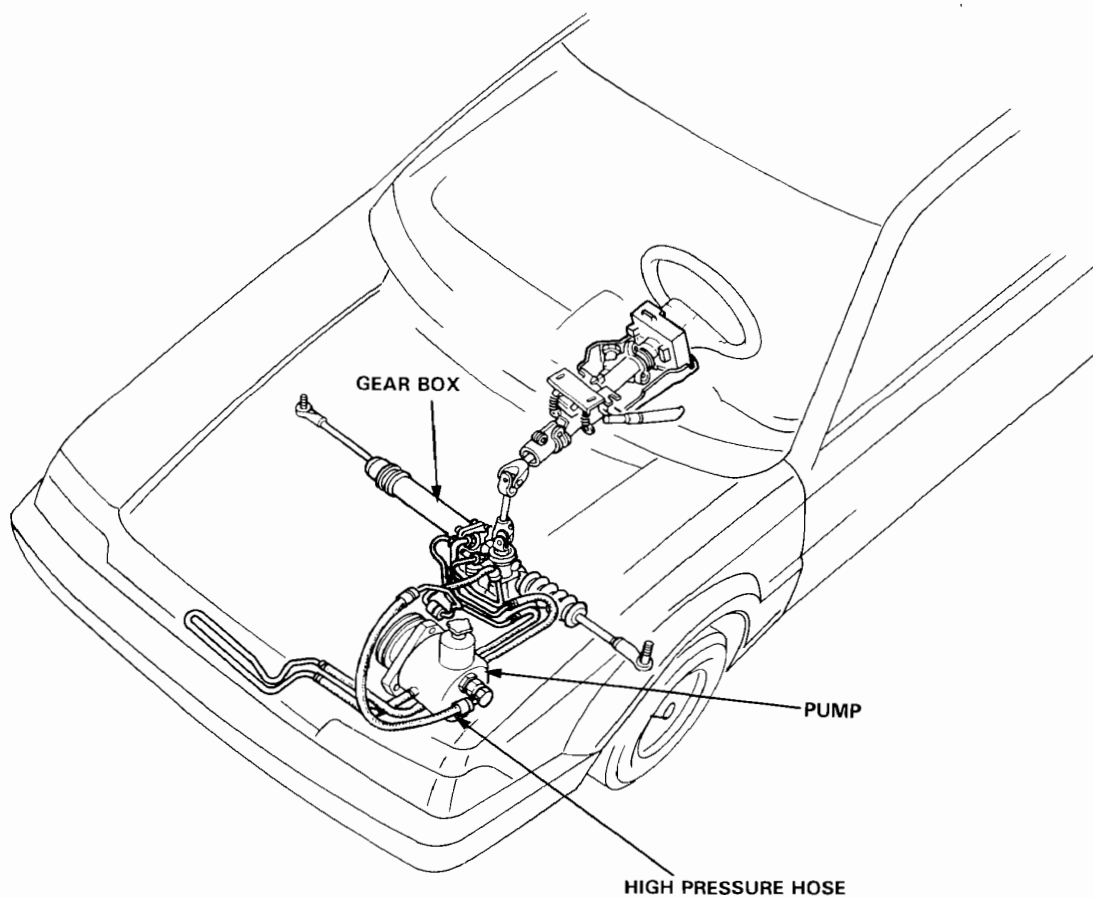


Operation

Description

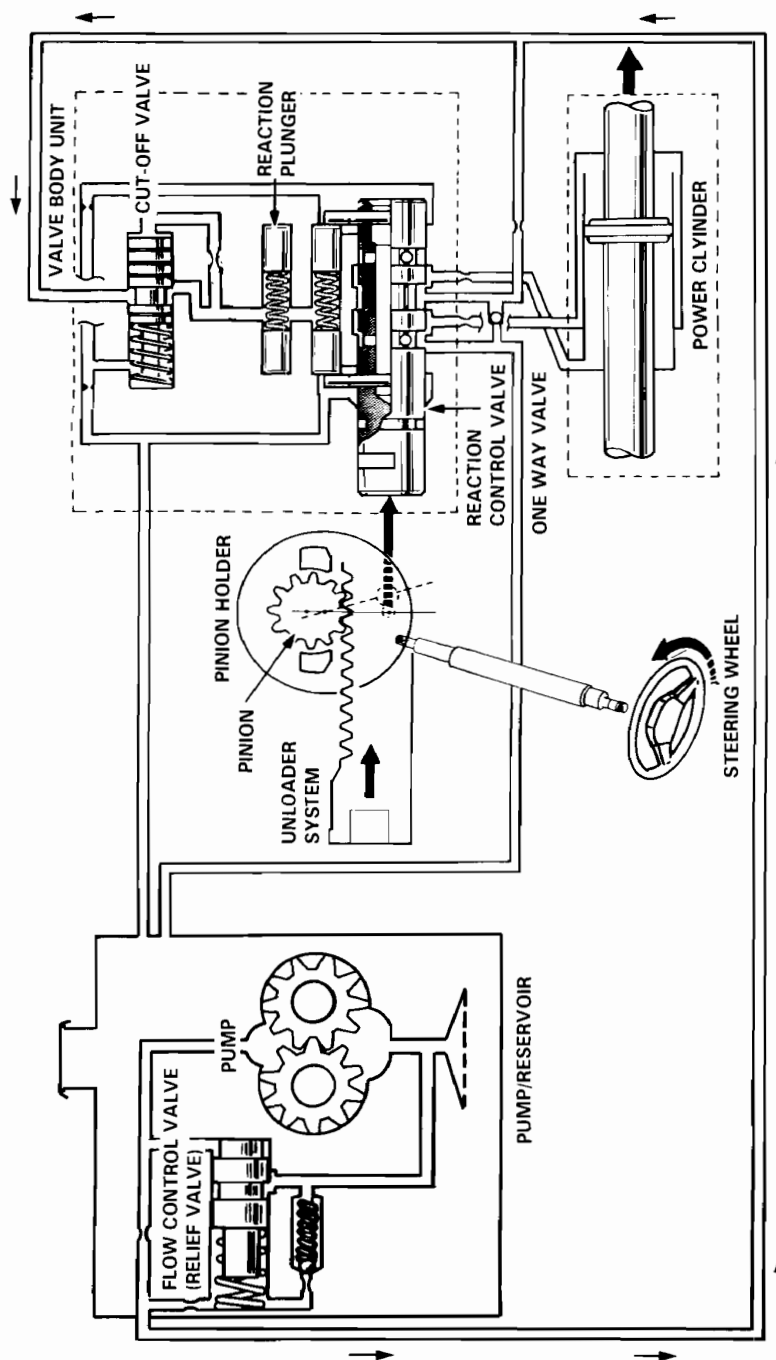
The power steering is rack and pinion type. The power operating assembly is integral with the steering gear. Road feel is maintained throughout the entire speed range of the vehicle.

NOTE: L/H shown, R/H similar



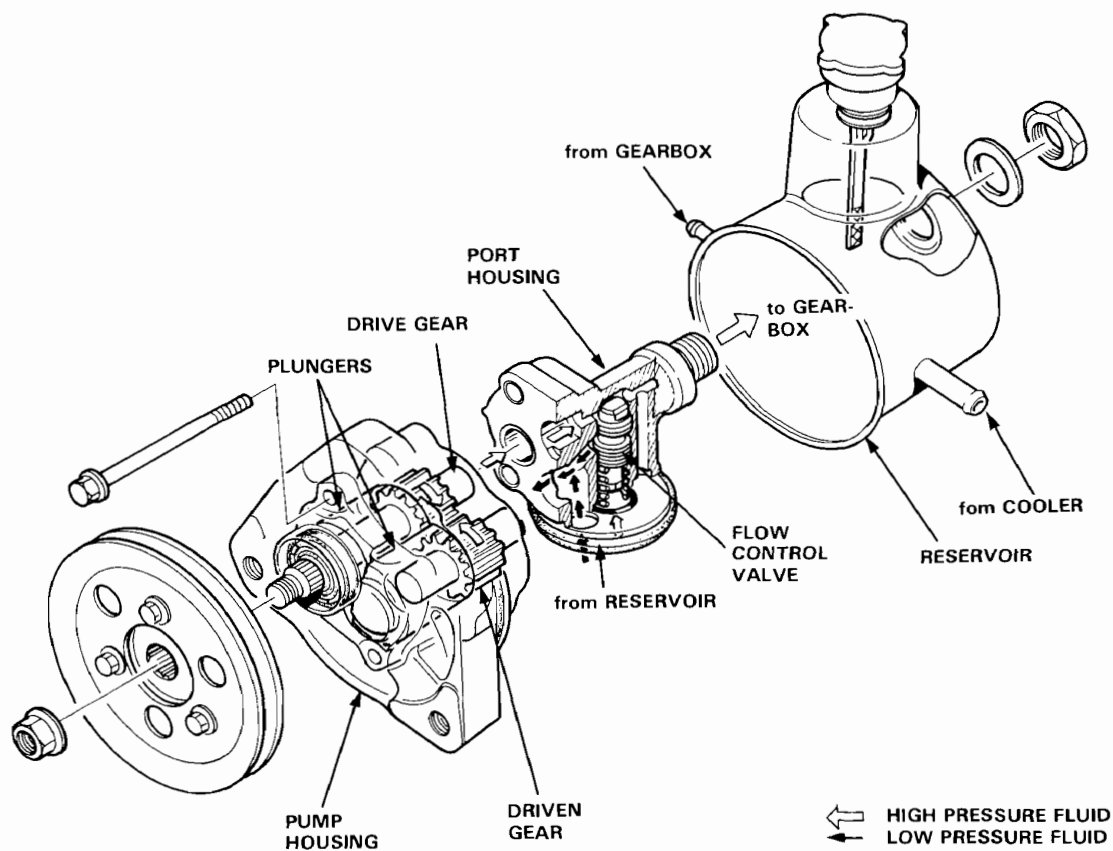
Operation

Flow Circuit



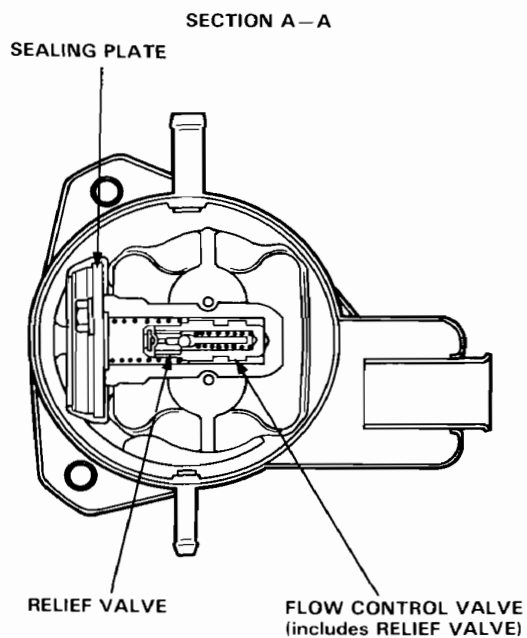
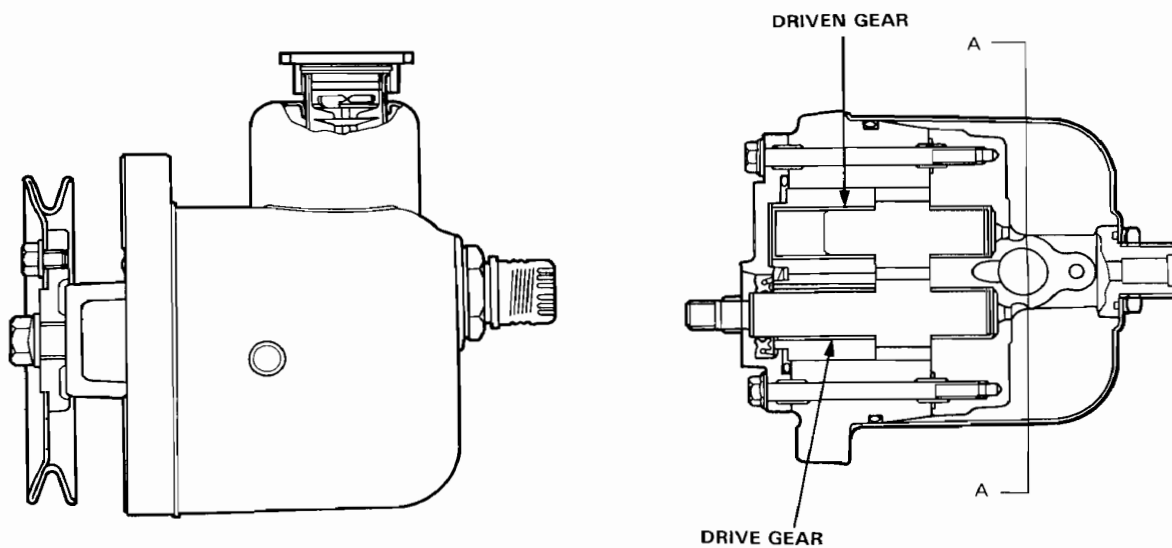
Pump

The pump is located at the left side of the engine cylinder head. It is a constant displacement spur gear type integral with the reservoir, and driven by the crankshaft through a V-belt. A relief valve in the pump housing returns excess fluid to the pump inlet. The pump housing and port housing are made of aluminum for light-weight construction.



Operation

Pump (cont'd)





Pump

Flow Control

Fluid from the pump runs through a metering orifice to the valve body. This creates a pressure differential between the pump and valve body sides of the orifice.

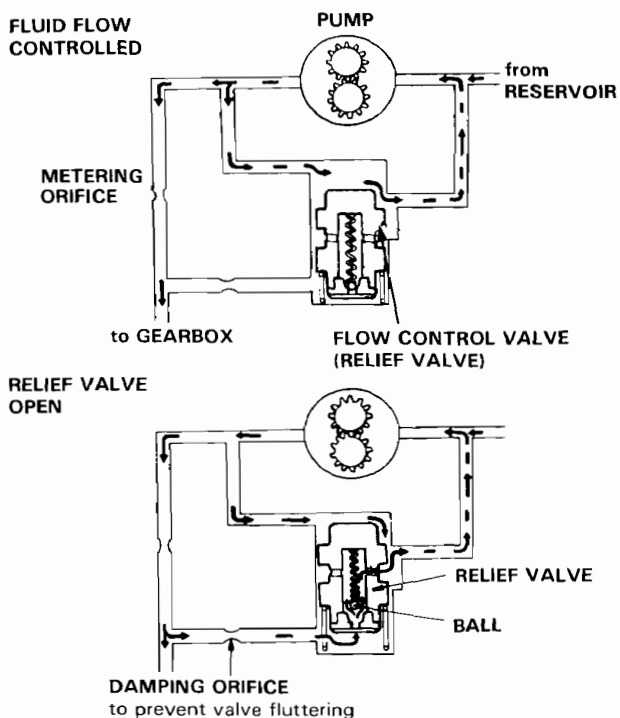
When pressure in the pump side is higher than the force of the spring holding the flow control valve closed, it pushes the valve down (open), and excess fluid returns to the pump inlet. The combined effect of the metering orifice and the flow control valve provides a relatively constant flow of fluid to the valve body.

Pressure Relief

As pressure on the valve body side builds up, it pushes the relief valve ball (inside the flow control valve) up against its spring.

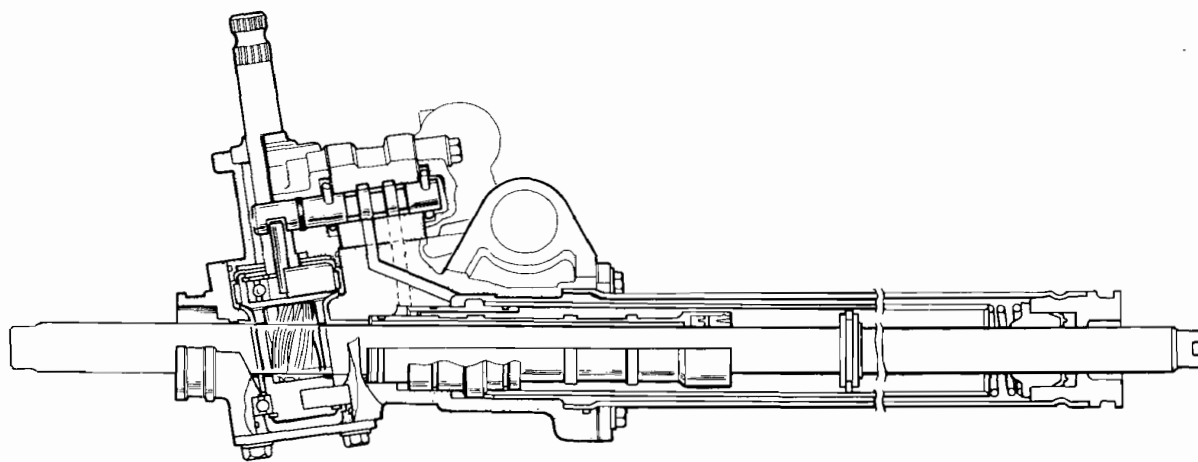
As the pressure under the flow control valve drops, the relief valve ball is closed by its spring, and the flow control valve is forced down again, allowing excess fluid from the pump side to return to the inlet.

This flow control valve-relief valve cylinder keeps pump output pressure between 80—90 kg/cm².



Gearbox

The steering gear is a self-contained rack and pinion type with the power cylinder placed in line with the steering rack. The power cylinder has an outer wall which serves as a fluid passage to eliminate external piping. The control valve is located at the top of the steering gearbox.



(cont'd)

Operation

Gearbox (cont'd)

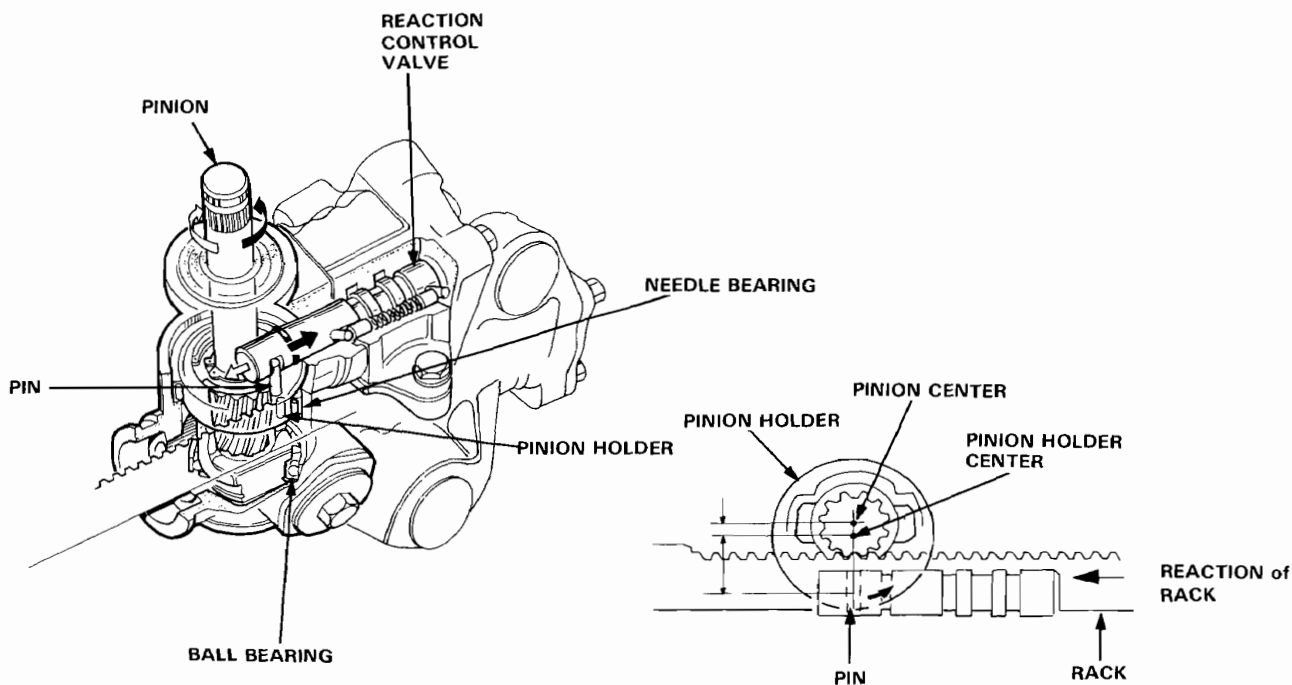
Valve Body Unit

In the power steering unit, the method used to direct a single source of fluid pressure in either of two directions (for left or right turns) involves the pinion gear transferring a "message" of direction to the fluid control valve.

The pinion is mounted slightly off-center in a pair of bearings, which are in turn mounted in a pinion holder cylinder that rotates, centered in its own outer bearings. At the bottom of the pinion holder is a pin, which fits in a slot in the control valve.

As the pinion is turned (to turn left or right), because it is off-centered it also moves slightly along the rack. This movement is transferred to the holder. The pin in the holder then moves the control valve, to direct fluid pressure to either side of the rack power cylinder.

The back edges of the pinion holder hit stops cast into both sides of the gear housing to avoid pushing the control valve too far in either direction. The front edge of the pinion holder cuts off assist at full lock.

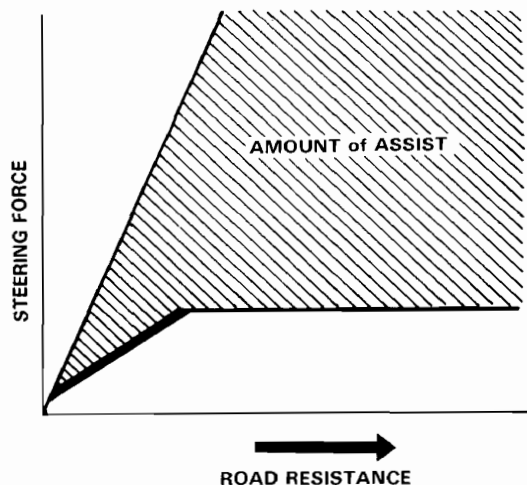
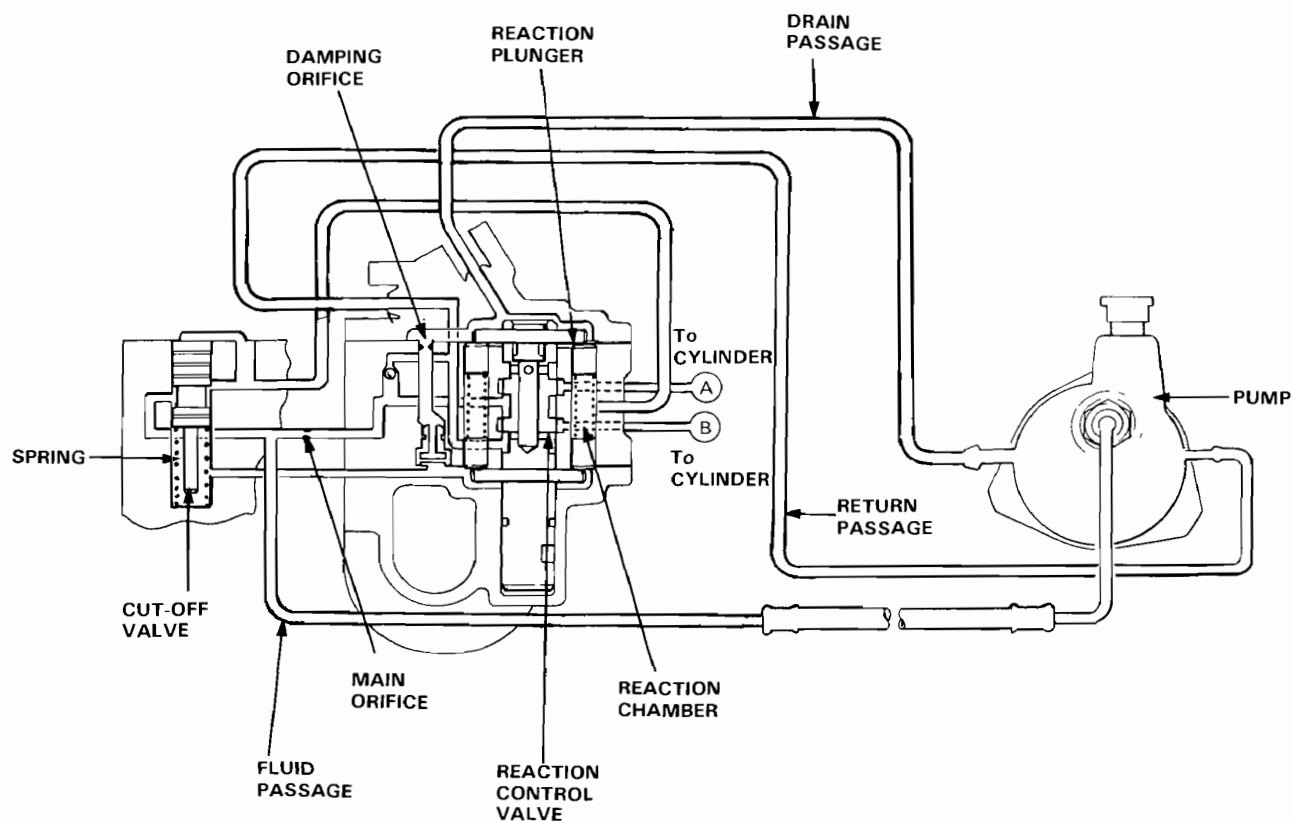


Operation

Gearbox (cont'd)

Turns at Medium or High Speed

At medium or high speed, pump pressure is not high enough to force the cut-off valve to rise. Fluid from the pump will then work its way around the cut-off valve and flow directly into the reaction chambers. This gives essential "road feel."



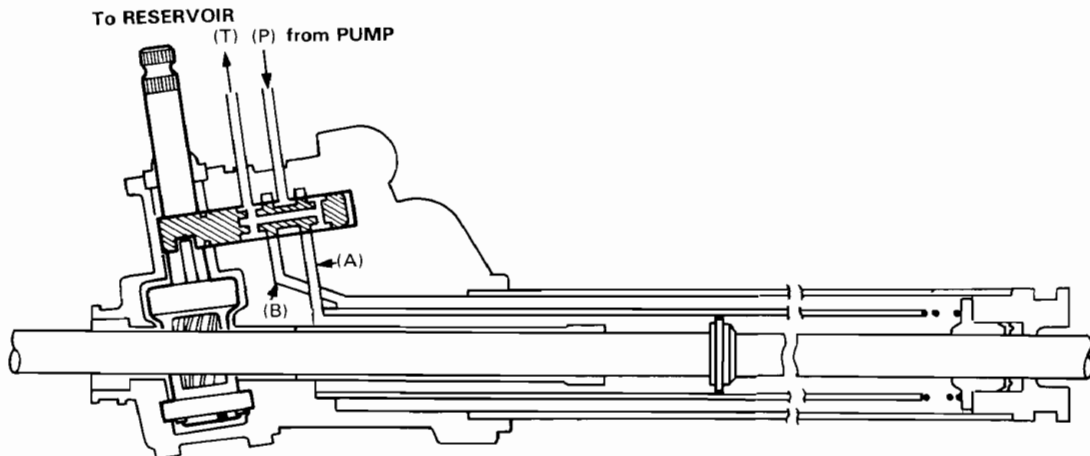
Power Cylinder

Straight Ahead (Steering Wheel not Turned)

When the car is stationary, or when it is moving straight ahead, fluid from the pump will expand the reaction plungers, assisted by the plunger springs, allowing the reaction control valve to remain in the neutral position.

When the reaction control valve is in neutral, it opens the ports, connecting the pump passage (P) to the reservoir through the right and left turn passages (A and B) of the power cylinder.

In this position, the only pressure built up in the pump passage is that produced by the main orifice.



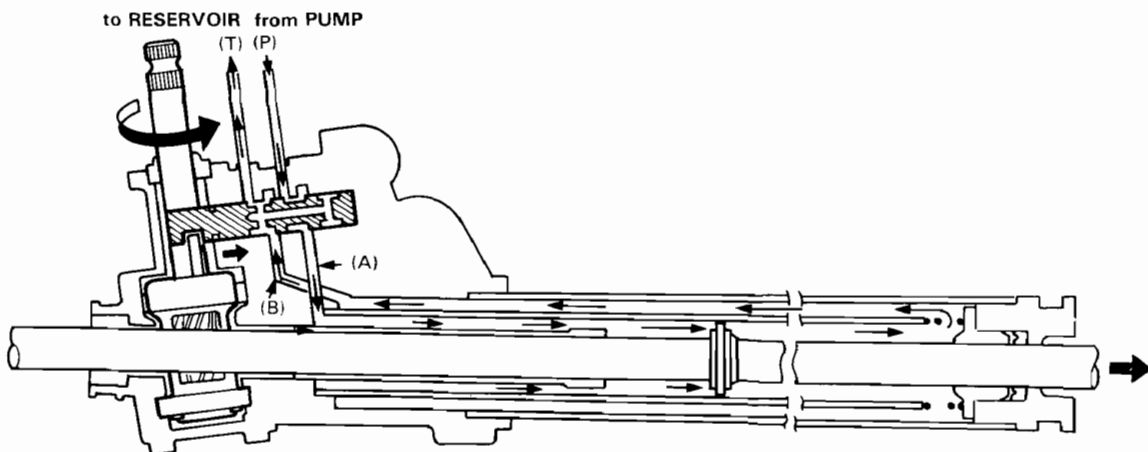
Left Turn

When the steering wheel is turned to the left, the reaction control valve is moved to the right, compressing the reaction plungers.

This connects the left turn passage (A) of the power cylinder to the pump passage (P), and connects the right turn passage (B) of the power cylinder to the reservoir passage (T).

Fluid now flows into the power cylinder. Thus, the fully hydraulic pressure developed by the pump is applied to the left turn side of the piston in the cylinder.

As the piston is moved to the right the fluid in the right side of the piston is returned to the reservoir.



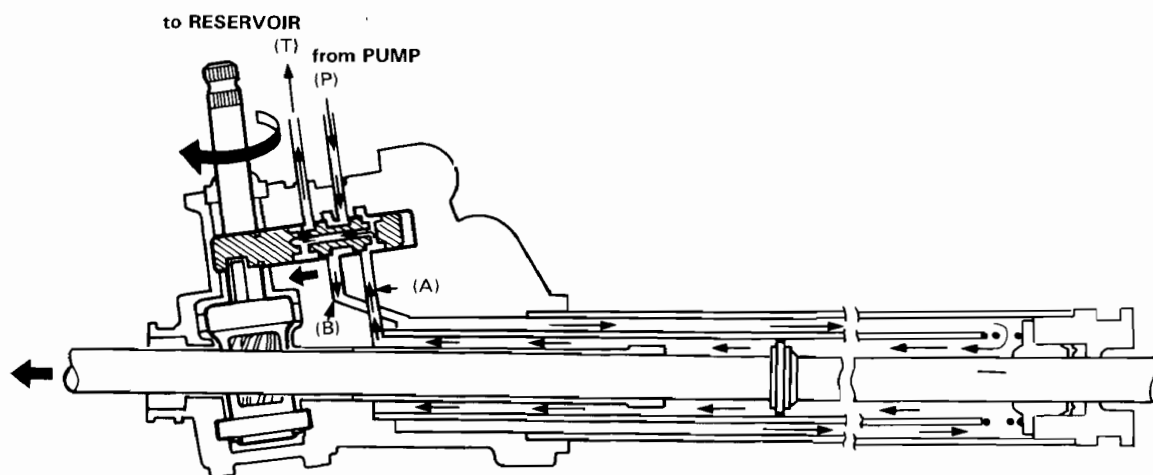
(cont'd)

Operation

Gearbox (cont'd)

Right Turn

A right turn will produce the same action using the right turn chamber of the power cylinder.



Return to Straight Ahead

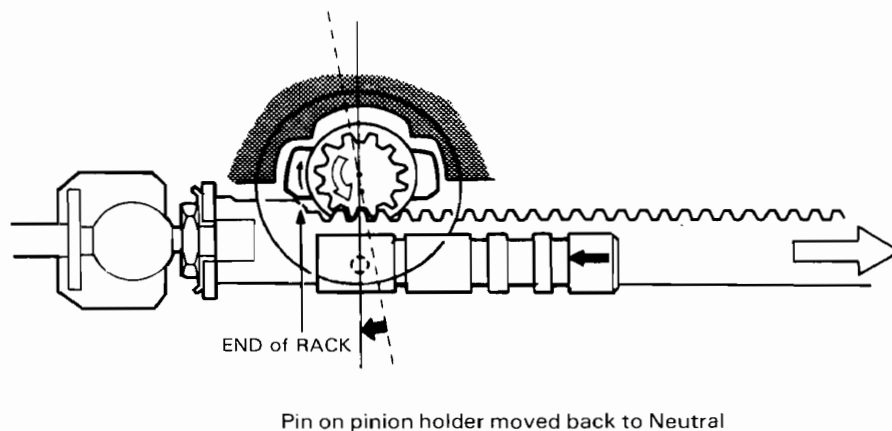
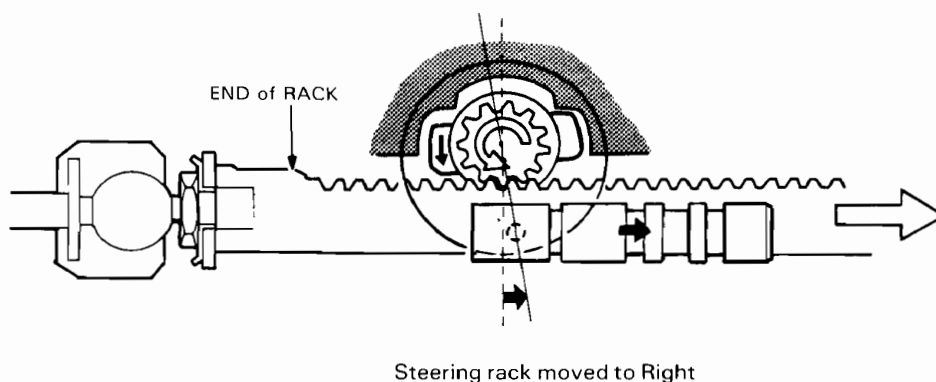
When the steering wheel is returned to the straight ahead position, the reaction control valve returns to its neutral position, and the reaction plungers are expanded by the power steering fluid and the tension of the plunger springs.

Unloader System

The control valve shifts the direction of fluid flow when the steering wheel is turned right or left.

However, when the wheel is turned to right or left lock at parking speed, the edge of the pinion holder rides up on the end of the rack, moving the pin in the opposite direction which pulls the control valve back to neutral.

This keeps pump pressure from building up (which could cause idle speed to drop), and improves steering feel by increasing resistance at left and right lock.



(cont'd)

Operation

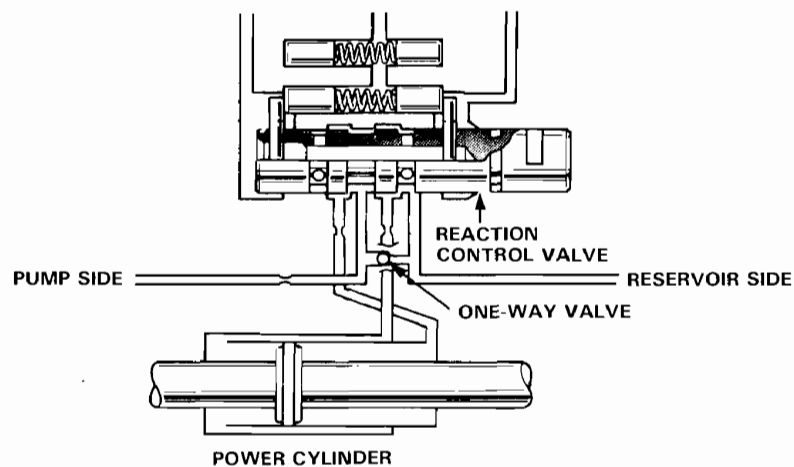
Gearbox (cont'd)

One-way Valve (Vacuum Check Valve)

When the wheel is turned with the pump stopped (engine not running), negative pressure develops in the pump circuit. To compensate for this, a one-way check valve is provided in the circuit between the pump and low pressure side of the control valve.

As the wheel is turned, the valve lets the oil recirculate through the control valve and power cylinder without returning to the reservoir.

This prevents fluid in the reservoir from rising and allows adequate steering effort.



Suspension

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FRONT DAMPER

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FRONT STABILIZER BAR

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FRONT KNUCKLE, HUB

Removal, page 20-10

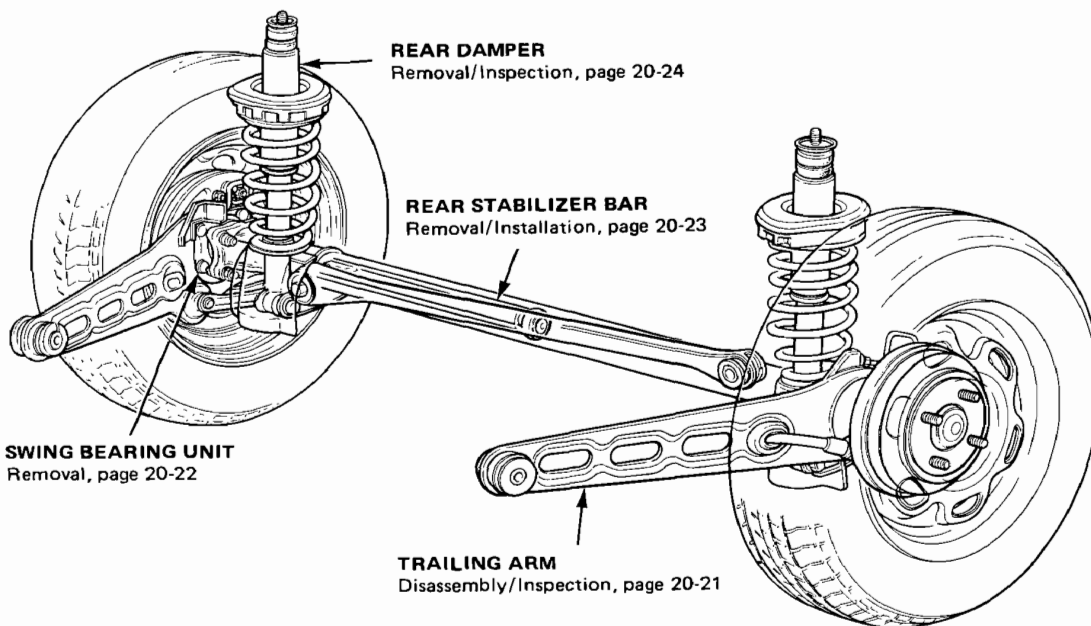
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Suspension

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FRONT KNUCKLE, HUB

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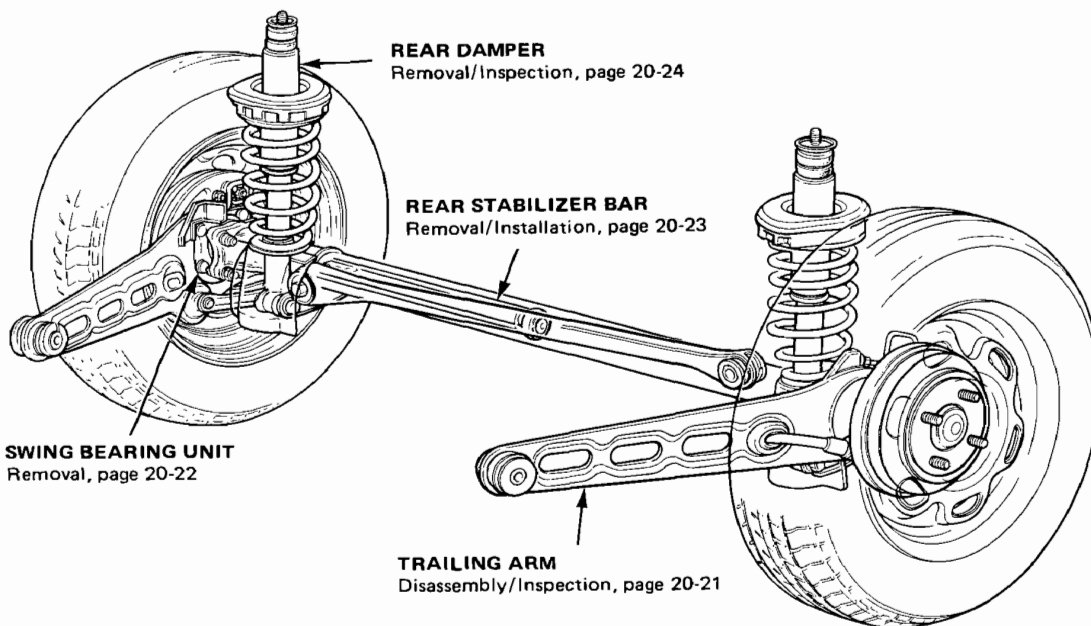
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TORSION BAR SPRING

Removal/Installation, page 20-15



Wheel Alignment



Front Toe Adjustent

1. Center steering wheel spokes.

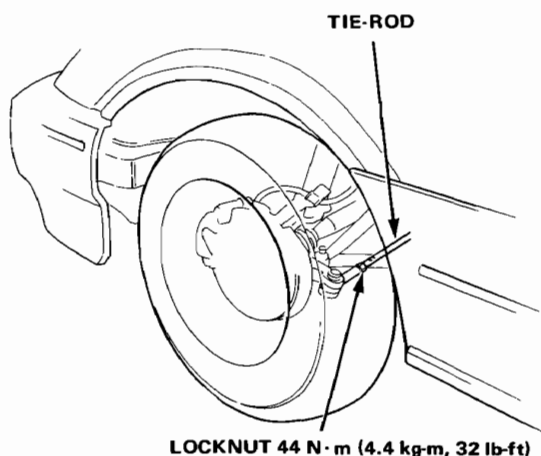
NOTE: Measure difference in toe measurements with the wheels pointed straight ahead.

Front Toe: $0 \pm 3 \text{ mm}$ ($0 \pm 0.118 \text{ in.}$)

- If adjustment is required, go on to step 2.
- If no adjustment is required, remove alignment equipment.

2. Loosen the tie-rod locknuts and turn both tie-rods in the same direction until the front wheels are in straight ahead position.
3. Turn both tie-rods equally until the toe reading is correct.
4. After adjusting, tighten the tie-rod locknuts.

NOTE: Make sure the tie-rod boots are not twisted or otherwise displaced after adjustment.



Rear Toe Inspection

1. Release parking brake.

NOTE: If the parking brake is engaged, you may get an incorrect reading.

Rear toe: $IN 2 \pm 2 \text{ mm}$ ($0.079 \pm 0.079 \text{ in.}$)

NOTE: Rear wheel toe is not adjustable. If out of specification, check suspension for damage and replace parts as necessary, then recheck alignment.

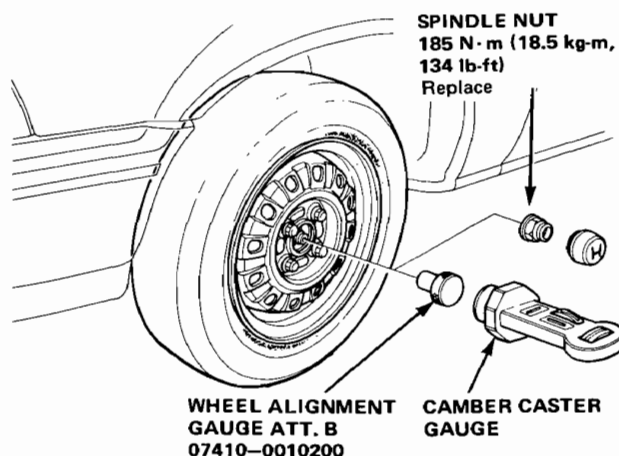
Camber Inspection

1. With the wheels in a straight ahead position, remove the spindle nut and install the special tool on the spindle as shown.
2. Set up the camber/caster gauge.
3. Read the camber on the gauge with the bubble at the center of the gauge.

Front Camber

	KC	KB,KE,KF KG,KS,KW KX	KQ	KY	Others
Coupe	$0^{\circ}00' \pm 1^{\circ}$	←	←	←	←
2D H/B	$0^{\circ}00' \pm 1^{\circ}$	←	←	$0^{\circ}20' \pm 1^{\circ}$	$0^{\circ}00' \pm 1^{\circ}$
4D	$0^{\circ}00' \pm 1^{\circ}$	←	←	$0^{\circ}30' \pm 1^{\circ}$	$0^{\circ}00' \pm 1^{\circ}$
4D H/B	$0^{\circ}20' \pm 1^{\circ}$	$0^{\circ}26' \pm 1^{\circ}$	$0^{\circ}30' \pm 1^{\circ}$	←	←

Rear Camber : $-0^{\circ}45' \pm 15'$



NOTE:

- If your alignment equipment must be mounted at axle centerline, use Honda front and rear wheel alignment attachments as shown.
- Camber is not adjustable. If out of specification, check suspension for damage and replace parts as necessary, then recheck alignment.

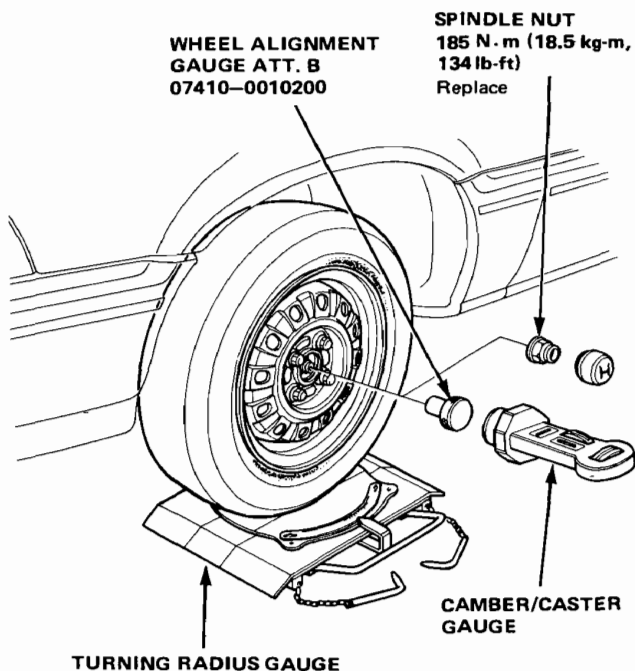
Wheel Alignment

Caster Inspection

1. Jack up the front of the car and set the turning radius gauges beneath the front wheels, then lower the car.
2. Remove the spindle nut and install Wheel Alignment Gauge Attachment.
3. Install Camber/Caster Gauge on the Attachment and apply the front brake. Turn the wheel 20° inward.
4. Turn the adjust screw so that the bubble in the caster gauge is at 0°. Return the wheel to the straight ahead position.

Caster angle:

	KC	KB,KE,KF KG,KS,KW KX	KQ	KY	Others
Coupe	2°25'±1°	←	←	←	←
2D H/B	2°20'±1°	←	←	2°10'±1°	2°20'±1°
4D	2°20'±1°	←	←	2°15'±1°	2°20'±1°
4D H/B	2°05'±1°	1°49'±1°	2°00'±1°	←	←



NOTE:

- If your alignment equipment must be mounted at axle centerline, use Honda front and rear wheel alignment attachments as shown.
- Caster is not adjustable. If out of specification, check suspension for damage and replace parts as necessary, then recheck alignment.

Spring Height

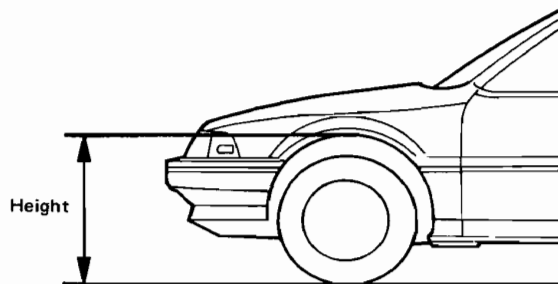
Inspection

NOTE:

- The car must not be occupied; the tires must be properly inflated and in good condition (i.e., the tread wear indicators must not be showing); and the fuel tank must be full.
- Bounce the front or rear of the car up and down several times before measuring.

Front

1. Measure the torsion bar spring height between the ground and the wheel arch.



		Standard
Coupe	KC	644±15mm (25.35±0.59 in)
	Others	639±15mm (25.16±0.59 in)
2D H/B	All	646±15mm (25.43±0.59 in)
	Arias for higher Ground clearance	671±15mm (26.42±0.59 in)
4D	All	651±15mm (25.63±0.59 in)
	Arias for higher Ground clearance	676±15mm (26.61±0.59 in)
4D H/B	KB,KE,KF,KG,KS,KW KX,KD,KP,KT,KU,KQ	659±15mm (25.94±0.59 in)
	KC	649±15mm (25.55±0.59 in)
	Arias for higher Ground clearance	674±15mm (26.54±0.59 in)

Higer Ground Clearance:

Standard for KY

Factory Option for KD, KE, KF, KP, KT, KU, KW

2. Adjust the height if the reading is not within specifications. (Page 20-5)

Wheel Measurement

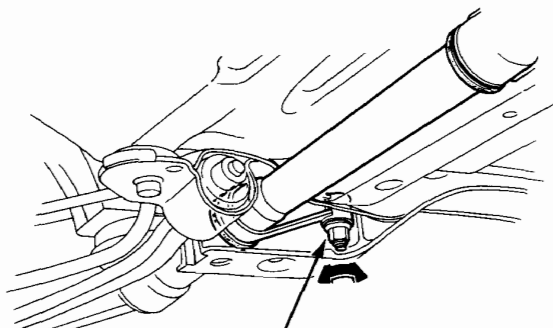


Torsion Bar Adjustment

1. Raise the front wheels off the ground.
2. Adjust the height by turning the height adjusting nut.

Height adjusting nut	Height
Tighten (Turned right)	Up
Loosen (Turned left)	Down

Height varies 5 mm (0.20 in) per turn of the adjusting nut.



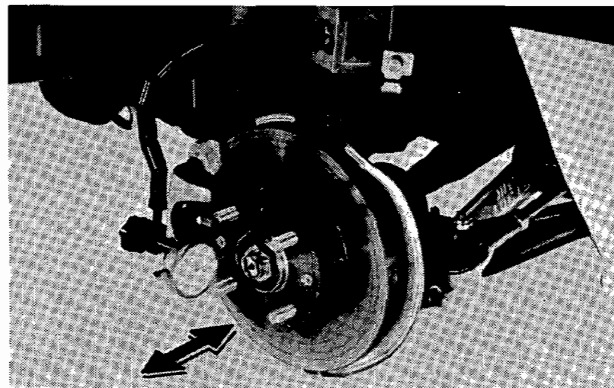
HEIGHT ADJUSTING NUT

3. Lower the front wheels to the ground, push the car up and down and back and forth several times, then confirm that the spring height is within specifications.

Bearing End Play

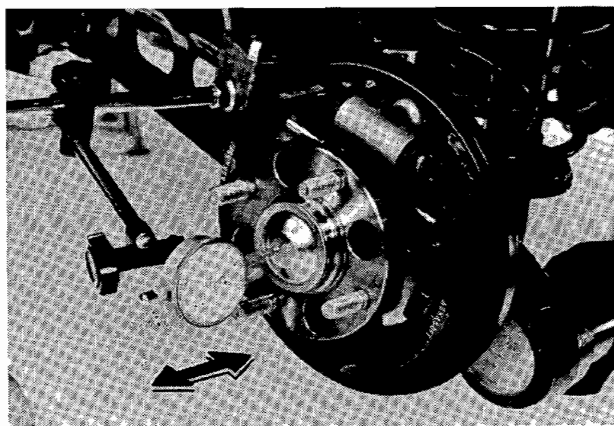
Front Wheel End Play

Standard: 0–0.05 mm (0–0.002 in.)



Rear Wheel End Play

Standard: 0–0.05 mm (0.002 in)

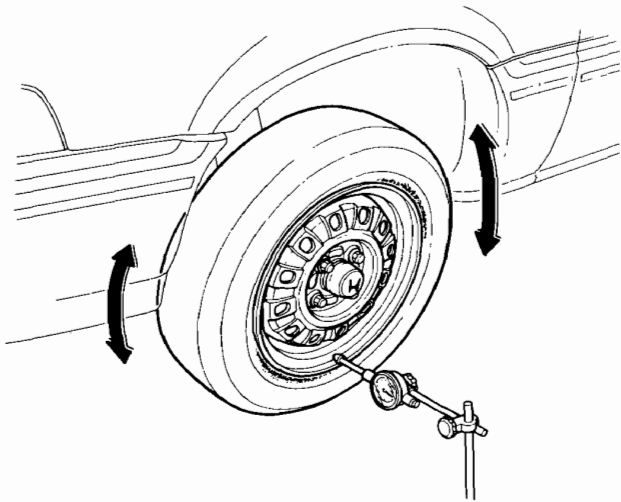


Wheel Measurement

Runout

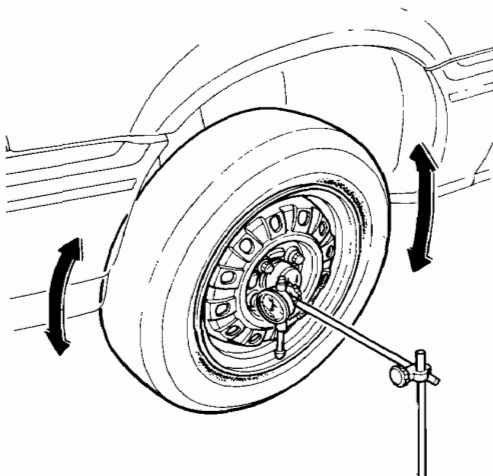
Front and Rear Wheel Axial Runout

Standard: Steel—
0–1.0 mm (0.04 in.)
Aluminum—
0–0.7 mm (0.028 in.)



Front and Rear Wheel Radial Runout

Standard: Steel— 0–1.0 mm (0.039 in.)
Aluminum— 0–0.7 mm (0.028 in.)

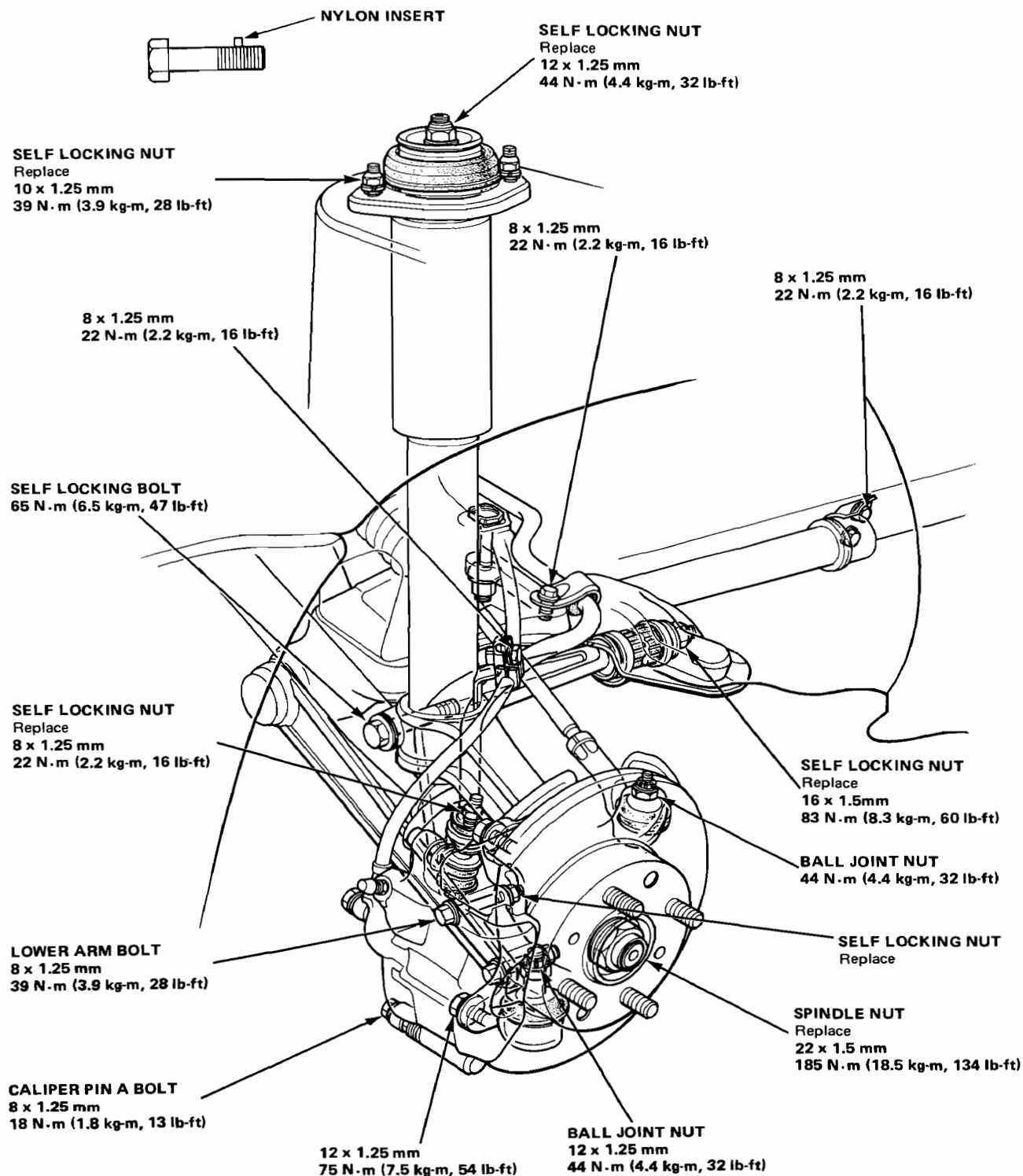


Front Suspension



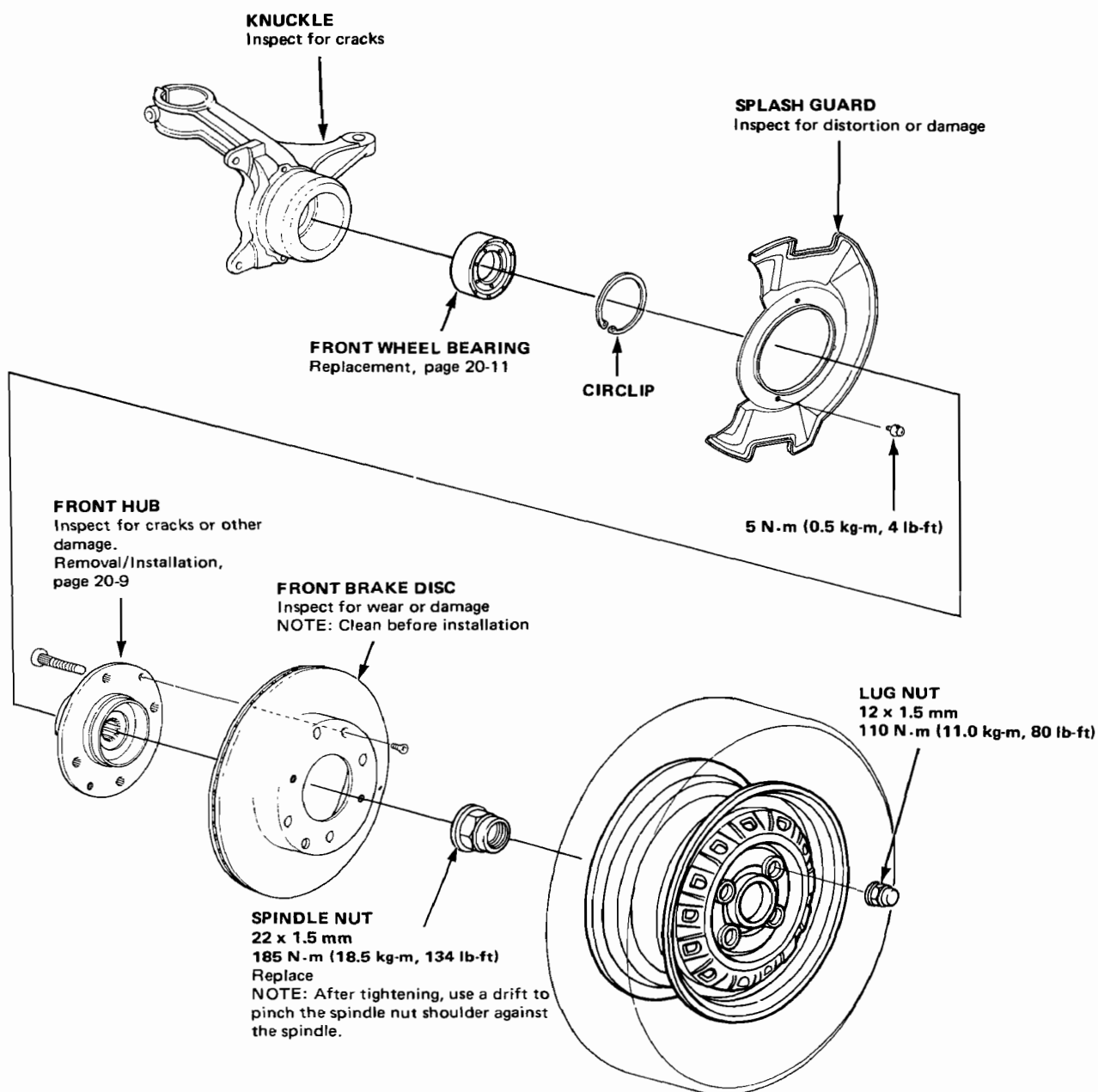
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NOTE: Replace the self-locking bolts if you can easily thread a nut past their nylon locking inserts.



Knuckle/Hub

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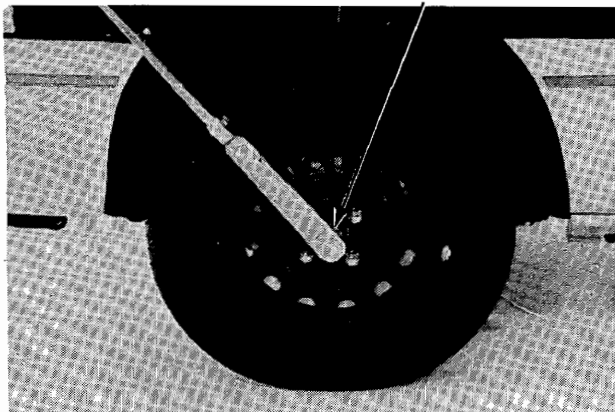
Replacement

1. Pry the spindle nut lock tab away from the spindle, then loosen the nut using a 32 mm socket.

SPINDLE NUT

Replace

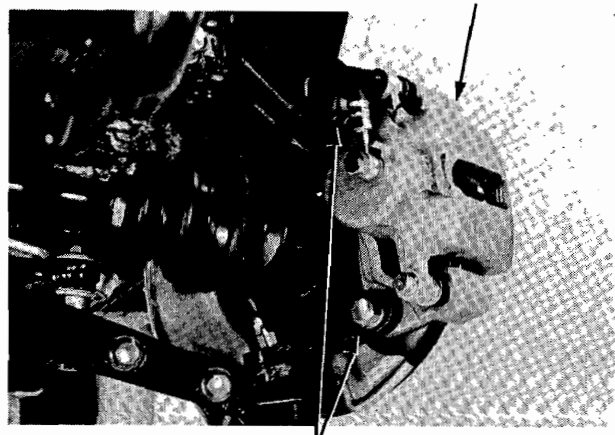
185 N·m (18.5 kg-m, 133 lb-ft)



2. Loosen the lug nuts slightly.
3. Raise the front of car and support with safety stands in the proper locations.
4. Remove the lug nuts, wheel, and spindle nut.
5. Remove the caliper mounting bolts and hang the cliper assembly to one side.

CAUTION: To prevent accidental damage to the caliper assembly or brake hose, use a short piece of wire to hang the caliper assembly from the under-carriage.

CALIPER ASSY

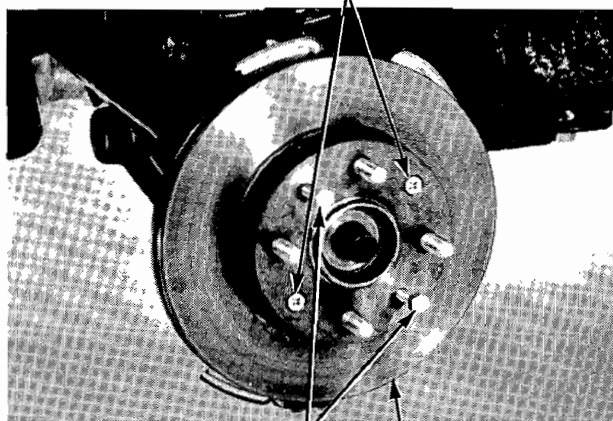


78 N·m (7.8 kg-m, 56 lb-ft)

6. Remove the 6 mm brake disc retaining screws.
7. Screw two 8 x 1.25 x 12 mm bolts into the disc to push it away from the hub.

NOTE: Turn each bolt two turns at a time to prevent cocking disc excessively.

DISC RETAINING SCREWS 6 mm

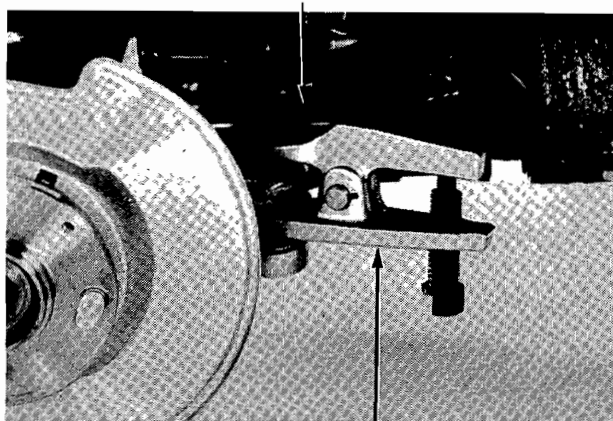


BOLTS FOR DISC REMOVAL
8 x 1.25 mm

BRAKE DISC

8. Remove the cotter pin from the tie-rod castle nut then remove the nut.
9. Break loose the tie-rod ball joint using Ball Joint Remover, then lift the tie-rod out of the knuckle.

TIE-ROD



BALL JOINT REMOVER
07941-6920001

44 N·m (4.4 kg-m, 32 lb-ft)

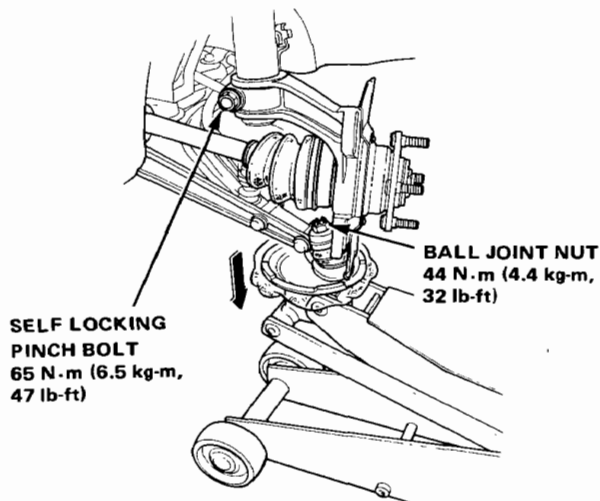
Knuckle/Hub

Replacement (cont'd)

10. Use a floor jack to support the lower control arm, then remove the ball joint cotter pin and nut.

CAUTION: Make sure the floor jack is positioned securely under the lower control arm, at the ball joint. Otherwise, torsion bar tension on the lower control arm may cause the arm to "jump" suddenly away from the steering knuckle as the ball joint is being removed.

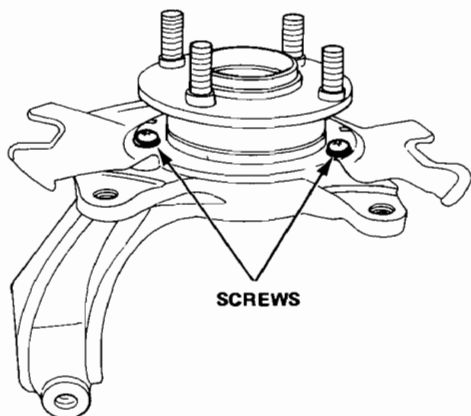
11. Pry the ball joint out of the steering knuckle. If necessary use the ball joint remover.



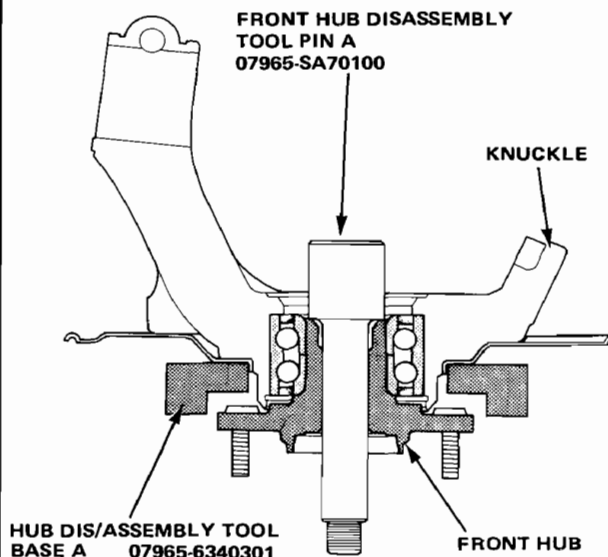
12. Remove the self locking pinch bolt, then use a brass or lead hammer to tap the knuckle down until it clears the damper.

13. Pull the driveshaft out of the knuckle, then remove the hub/knuckle assembly.

14. Remove the two screws holding the splash guard on the steering knuckle.



15. Remove the hub from the knuckle with a hydraulic press and the special tools shown below.



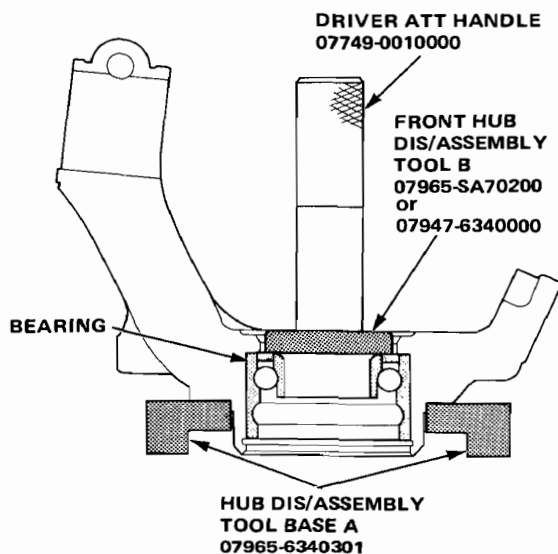
CAUTION:

- Make sure the knuckle is securely mounted on the base.
- Take care not to distort the splash guard.



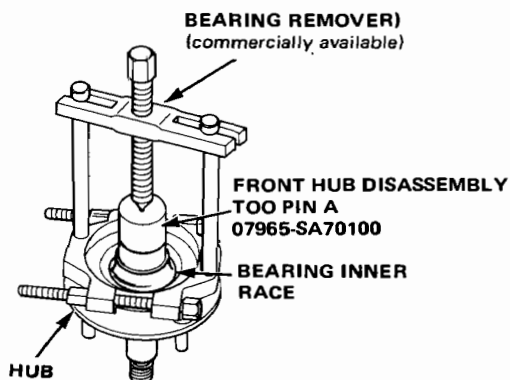
Bearing Removal

1. Remove the circlip.
2. Remove the bearing from the steering knuckle with the hydraulic press and special tools shown below.



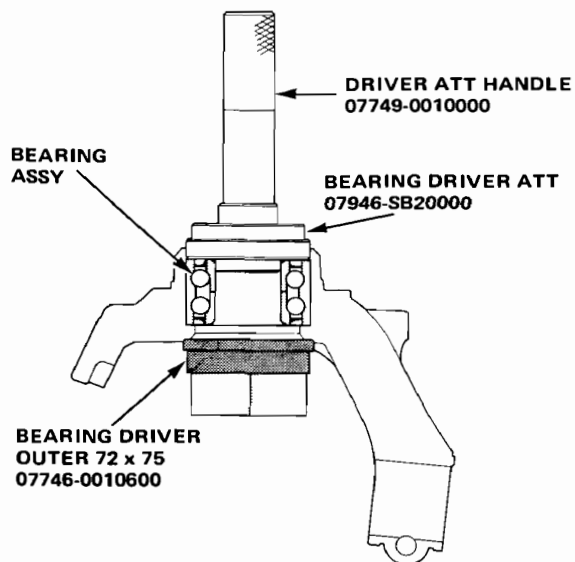
3. Remove the outboard bearing inner race from the hub with the special tool and a bearing remover.

NOTE: Wash the knuckle and hub thoroughly before re-assembly.

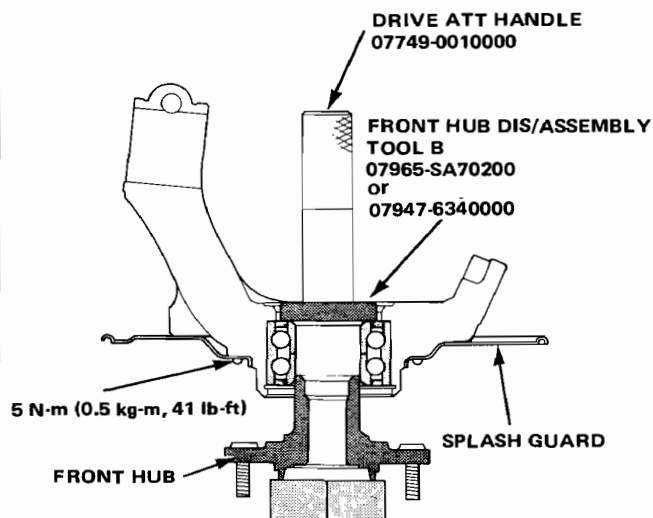


Knuckle/hub Reassembly

1. Push the bearing assy into the knuckle with a hydraulic press and the special tools shown below.



2. Install the circlip.
3. Install the splash board.
4. Push the front hub into the knuckle with a hydraulic press and the special tools shown below.

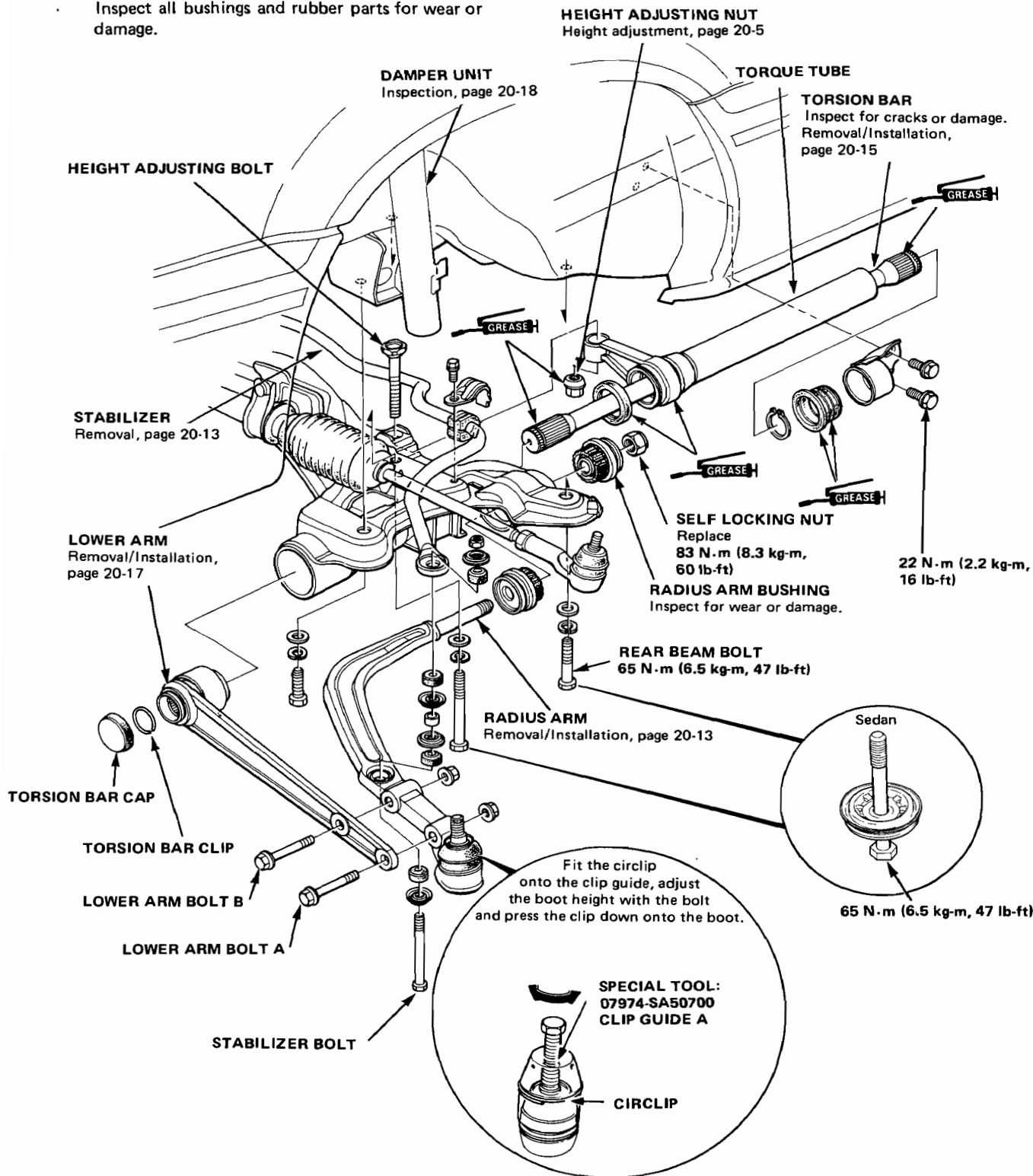


Lower Arm/Radius Arm/Stabilizer/Torsion Bar

Index

NOTE: Replace the locking nut after removal.

Inspect all bushings and rubber parts for wear or damage.



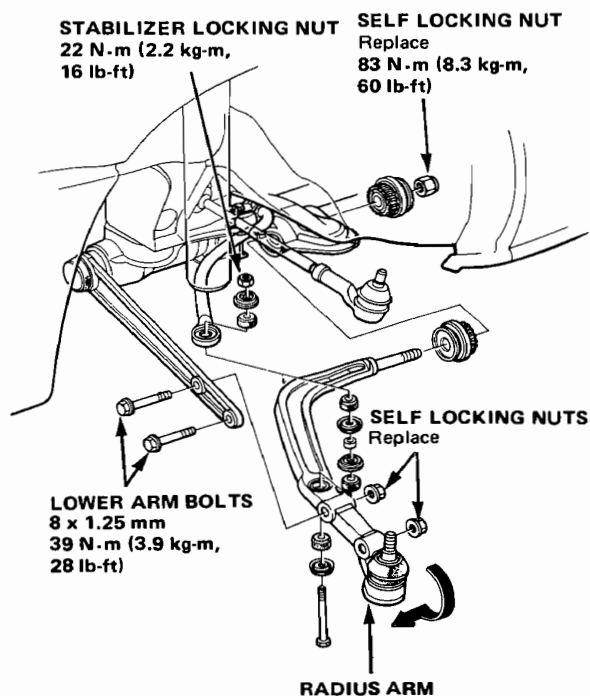
Radius Arm

Removal/Installation

1. Raise the front of the car off the ground and support it with safety stands (see page 1-7 for the proper location of the safety stands).
2. Remove the front wheels.
3. Remove the radius arm ball joint from the knuckle. (Page 20-10)

CAUTION: Make sure the floor jack is positioned securely under the lower control arm at the ball joint. Otherwise, torsion bar tension on the lower control arm may cause the arm to "jump" suddenly away from the steering knuckle as the ball joint is being removed.

4. Remove the radius arm self locking nuts.



5. Remove the stabilizer locking nut and separate the radius arm from the stabilizer spring.
6. Remove the lower arm bolts.
7. Remove the radius arm by pulling it down and then forward.
8. Installation is the reverse order of removal.

CAUTION: Tighten all bushings and rubber damped parts only after the car is back on the ground.

Stabilizer

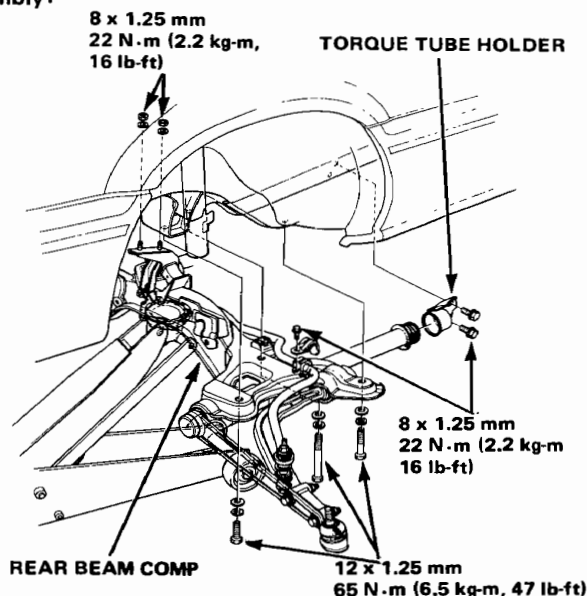
Removal

1. Raise the front of the car off the ground and support it with safety stands (see page 1-7 for the proper location of the safety stands).
2. Remove the front wheels.
3. Support the weight of the engine with a chain hoist or similar device.
4. Remove the steering gearbox. (Page 18-4)
5. Separate the radius arm ball joint from the knuckle. (Page 20-10)

CAUTION: Make sure the floor jack is positioned securely under the lower control arm at the ball joint. Otherwise, torsion bar tension on the lower control arm may cause the arm to "jump" suddenly away from the steering knuckle as the ball joint is being removed.

6. Remove the torque tube holder.
7. Remove exhaust pipes A and B. (Page 9-4)
8. Manual Transmission:
Disconnect the shift rod and extension from the transmission. (Page 5-9, 5-10)
Automatic Transmission:
Remove the shift cable guide from the floor and pull the shift cable down by hand.
9. Remove the engine mount bracket nuts.
10. Pry off the rear beam by placing a jack at the center and removing the six 12 mm bolts.

CAUTION: Take care not to drop the rear beam assembly.



11. Remove the stabilizer bracket and bolt, and then remove the stabilizer spring.
12. Remove the rear mounting bracket.

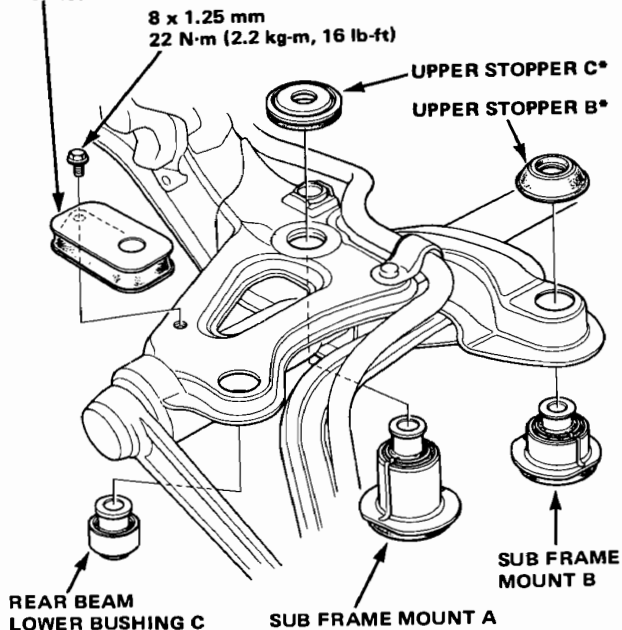
Sub Frame Mount/Lower Bushing(4D)

Installation

Whenever the rear beam is replaced, the rear beam lower bushing C and sub frame mounts A and B must be replaced.

SUB FRAME* MOUNT

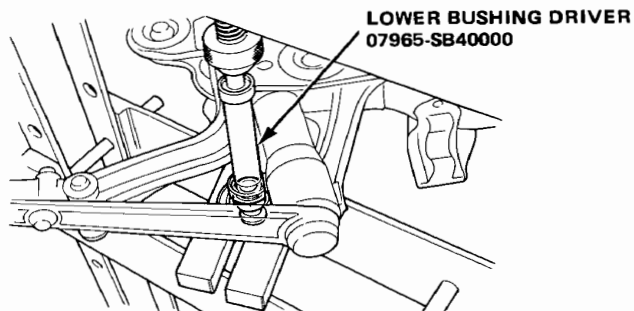
* reuse, unless damaged



• Lower bushing C

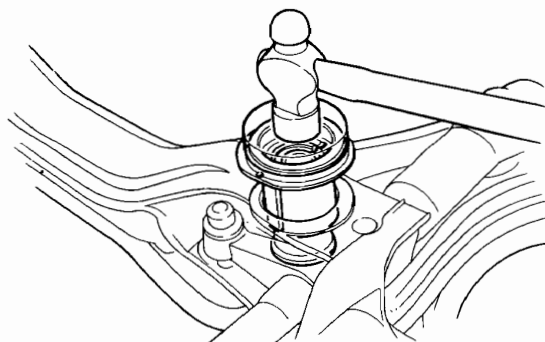
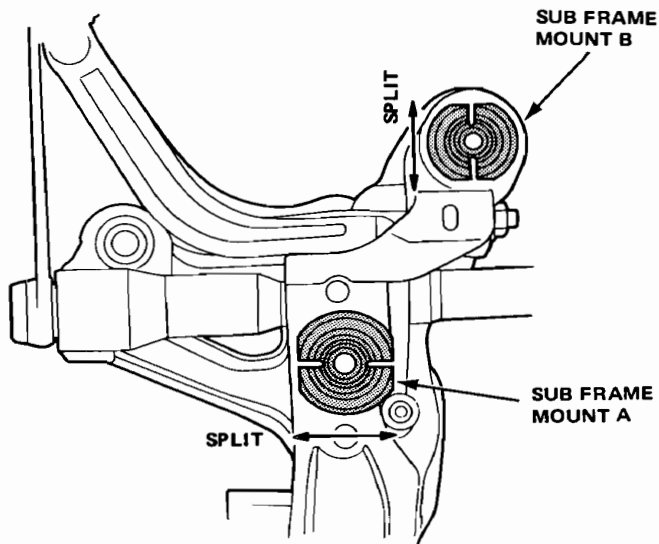
Press a new bushing into the replacement rear beam until the outer edge is flush with the end of the rear beam.

Press-fit load:
500—1,000 kg (1,102—2,205 lb)



• Sub frame mounts A and B

Drive new mounts into place in the rear beam with the splits in the mounts positioned as shown.



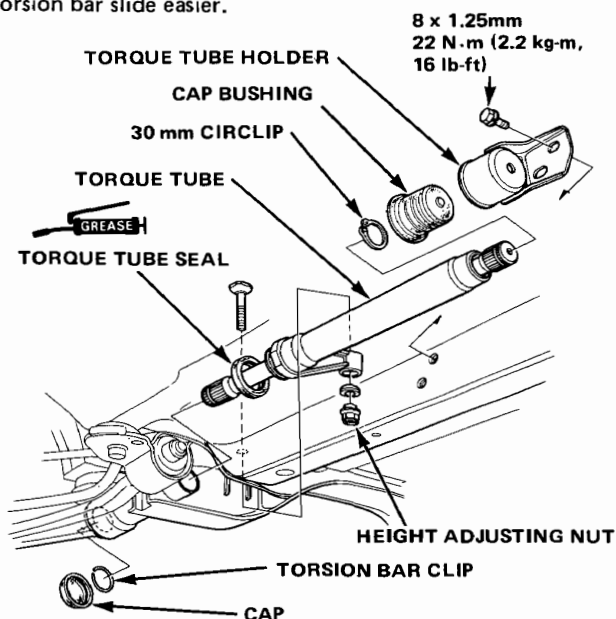
NOTE: Be sure to use a suitable pad when driving the new mounts into the rear beam as shown.

Torsion Bar Assembly

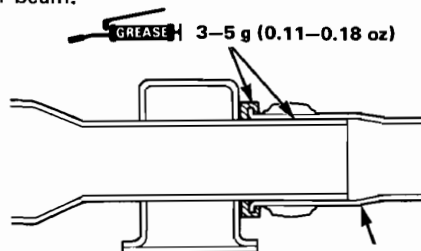
Removal/Installation

1. Jack up the front of the car and support it with safety stands (see page 1-7 for the proper location of safety stands)
2. Remove the height adjusting nut.
3. Remove the torque tube holder.
4. Remove the 30 mm circlip.
5. Remove the torsion bar cap and then remove the torsion bar clip by tapping the torsion bar out of the torque tube.

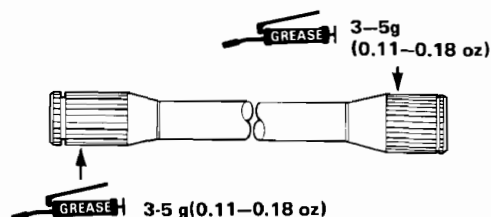
NOTE: Move the lower arm up and down to make the torsion bar slide easier.



6. Tap the torsion bar backward, out of the torque tube.
7. Remove the torque tube.
8. Inspect the torsion bar for cracks or damage.
9. Install a new torque tube seal onto the torque tube.
10. Coat the torque tube seal and the torque tube sliding surface with grease, then install them on the rear beam.

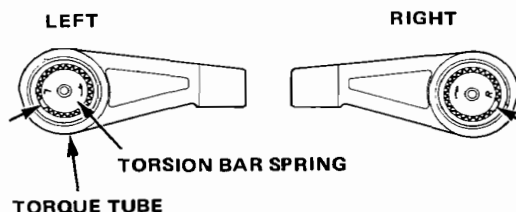


11. Grease the spline at each end of the torsion.



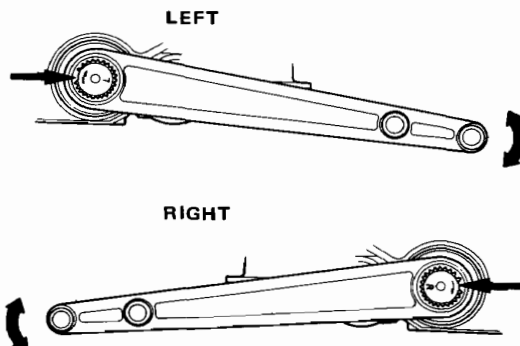
12. Insert the torsion bar into the torque tube from the back.

13. Align the projection or punch mark on the torque tube splines with the cutout or paint mark in the torsion bar splines and insert the torsion bar approximately 10 mm (0.394 in).



NOTE:

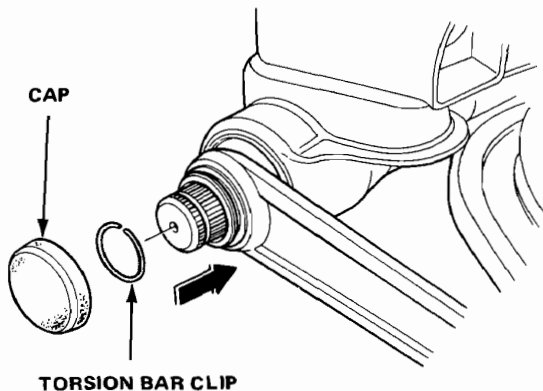
- Move the lower arm up and down for easier installation.
- There are two types of torsion bars and torque tube; torque tubes with and without raised lugs and torsion bars with and without lug reliefs. The torque tube with the raised lug will not fit over a torsion bar without a lug relief. But all other combinations of torque tube and torsion bar will fit together and work properly.



Torsion Bar Assembly

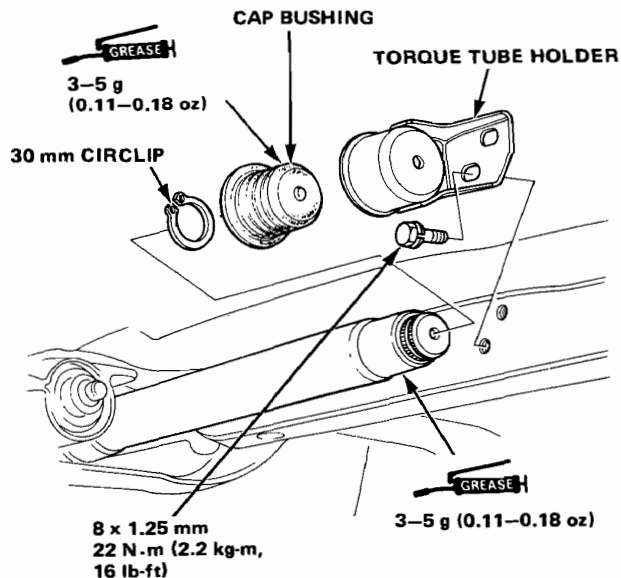
Removal/Installation (cont'd)

14. Install the clip on the torsion bar, and the cap over it.



15. Install the 30 mm circlip and the torque tube cap.

NOTE: Push the torsion bar forward so that there is no clearance between the 30 mm circlip and the torque tube.



18. Tighten the height adjusting nut temporarily.

NOTE: Coat the height adjusting nut and the torque tube sliding surface with grease.

19. Set the car on the ground and adjust the torsion bar spring height. (Page 20-5)

16. Coat the cap bushing with grease and install it on the torque tube.

17. Install the torque tube holder.

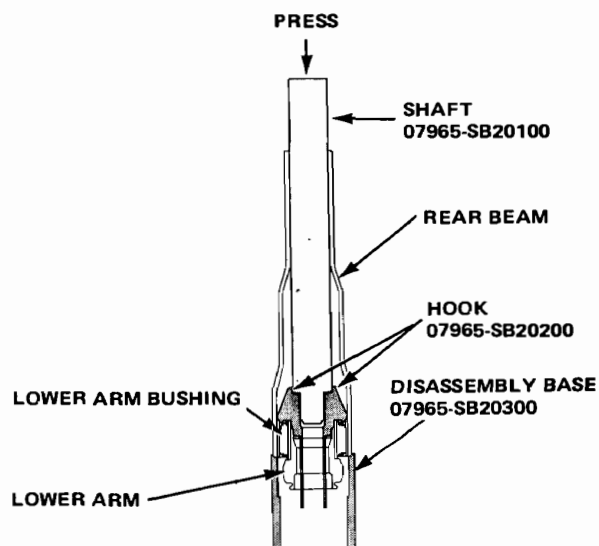
Lower Arm



Replacement

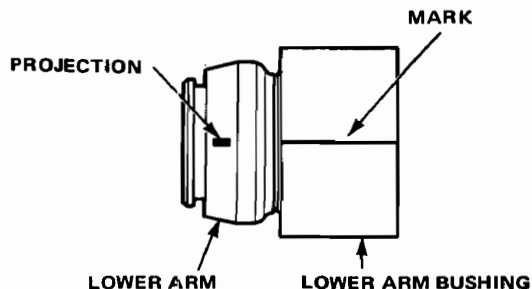
NOTE: Lower arm replacement should be done with the rear beam removed from the frame.

1. Remove the lower arm with a hydraulic press and the special tools.
- 1) Set the special tool B in the lower arm bushing outer pipe as shown.
- 2) Insert special tool A into the rear beam to set special tool B.
- 3) Place special tool C under the rear beam and remove the lower arm by pressing tool A.

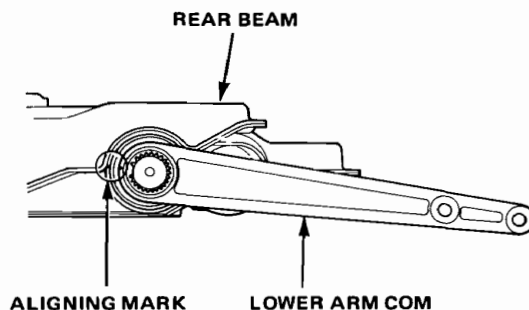


LOWER ARM COMP DIS/ASSEMBLY TOOL SET 07965-SB20000

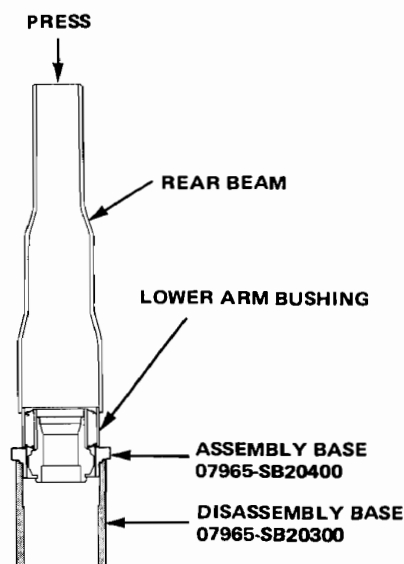
2. Mark the lower arm bushing so that the lower arm can be aligned easily with the rear beam.



3. Align the index mark on the rear beam with the mark on the bushing and then install the rear beam by tapping it with a soft hammer.



4. Install the lower arm with a hydraulic press and the special tools.
- 1) Place special tool D on special tool C. Align special tool D with the lower arm bushing outer pipe.

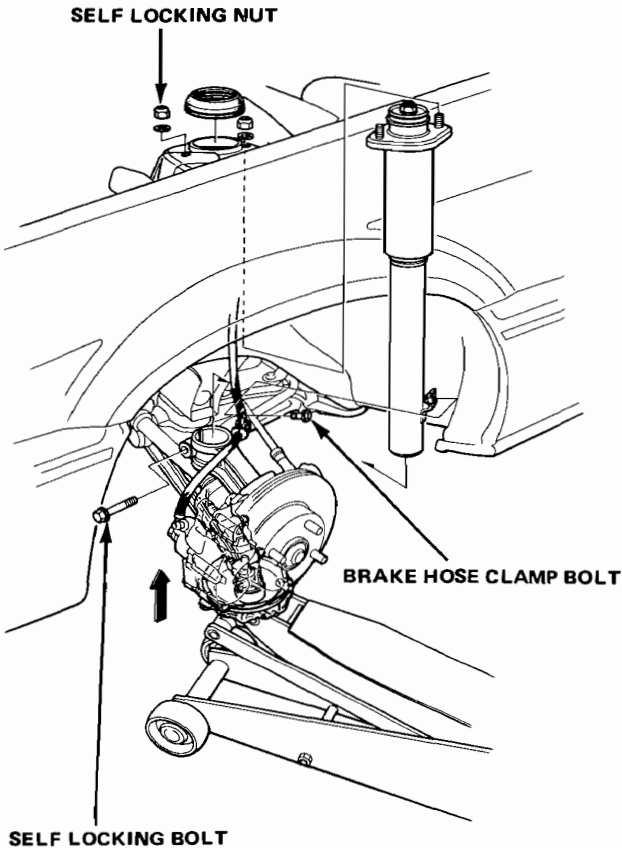


- 2) Insert the rear beam so that the lower arm bushing outer pipe aligns with the rear beam pipe, then press the rear beam.

Front Damper

Removal

1. Jack up the front of a car and remove the front wheels.
2. Remove the brake hose clamp bolt.
3. Use a floor jack to support the lower control arm.



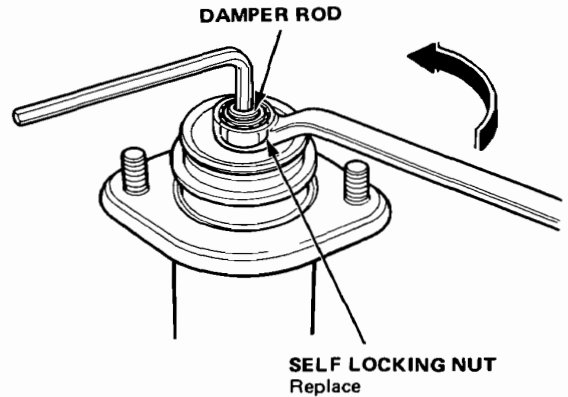
4. Remove the self locking bolt, then lower the jack gradually and remove it from the knuckle.

CAUTION: Make sure the floor jack is positioned securely under the lower control arm at the ball joint. Otherwise, torsion bar tension on the control arm may cause the knuckle to "jump" suddenly away from the front shock absorber as the pinch bolt is being removed.

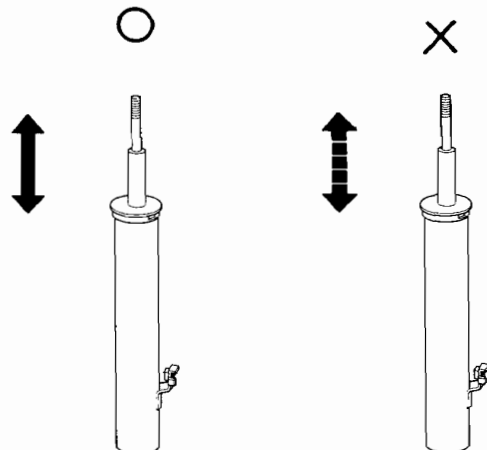
5. Remove the self locking nut and then remove the damper from the frame by compressing the shock absorber spring.

Disassembly/Inspection

1. Remove the self locking nut while holding the damper rod.



2. Disassemble the damper assembly.
3. Slowly move the damper piston rod a full stroke and check for smooth operation.
4. Jerk the piston rod back and forth 5–10 mm (2–4 in.) to check for smooth operation.
5. Inspect for an oil leak cracks in the piston rod.
6. Listen for abnormal noises.

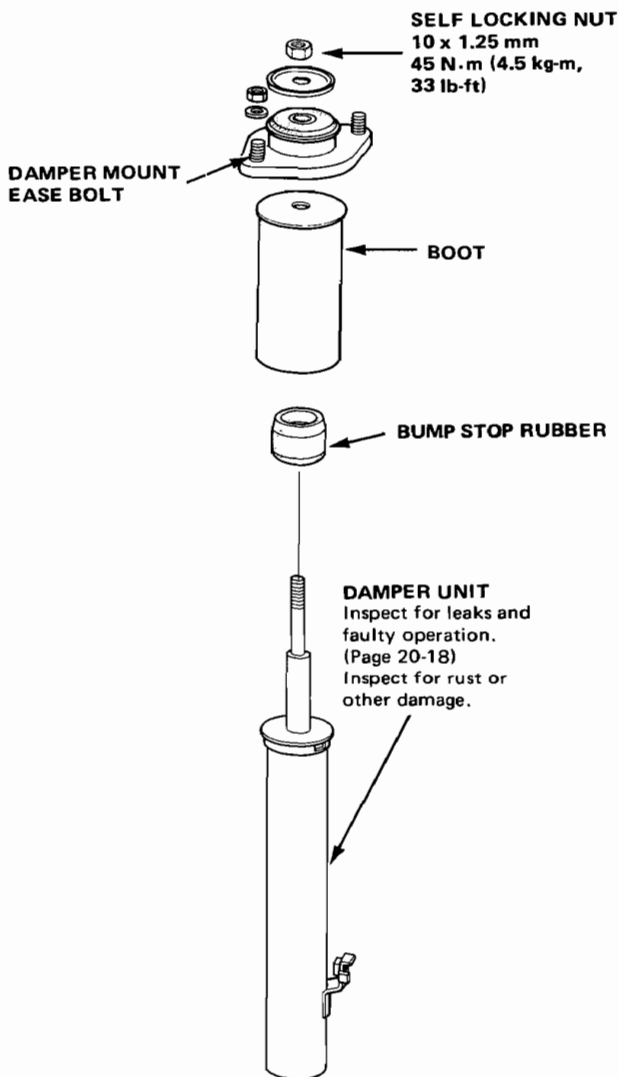


NOTE: The damper cannot be disassembled. If it does not operate smoothly, or if it makes any abnormal noises during operation, replace it.

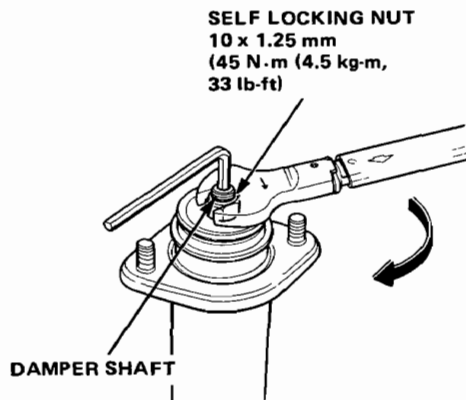


Assembly/Installation

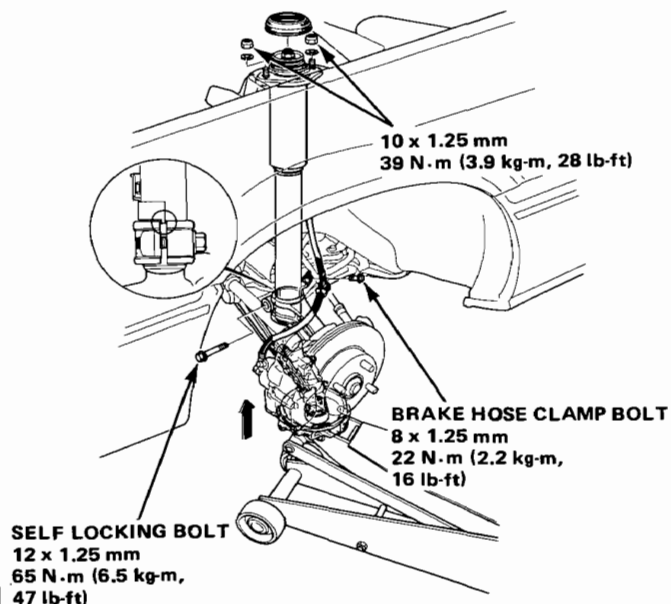
NOTE: Use a new self locking nut for reassembly



1. Install the damper assembly.



2. Install the self locking nut and tighten it while holding the damper shaft.
3. Compress the damper assembly and fit it into the frame.
4. Extend the damper, align the projection on the damper with the slot in the steering knuckle, then fit the damper into the knuckle and tighten the self locking bolt.

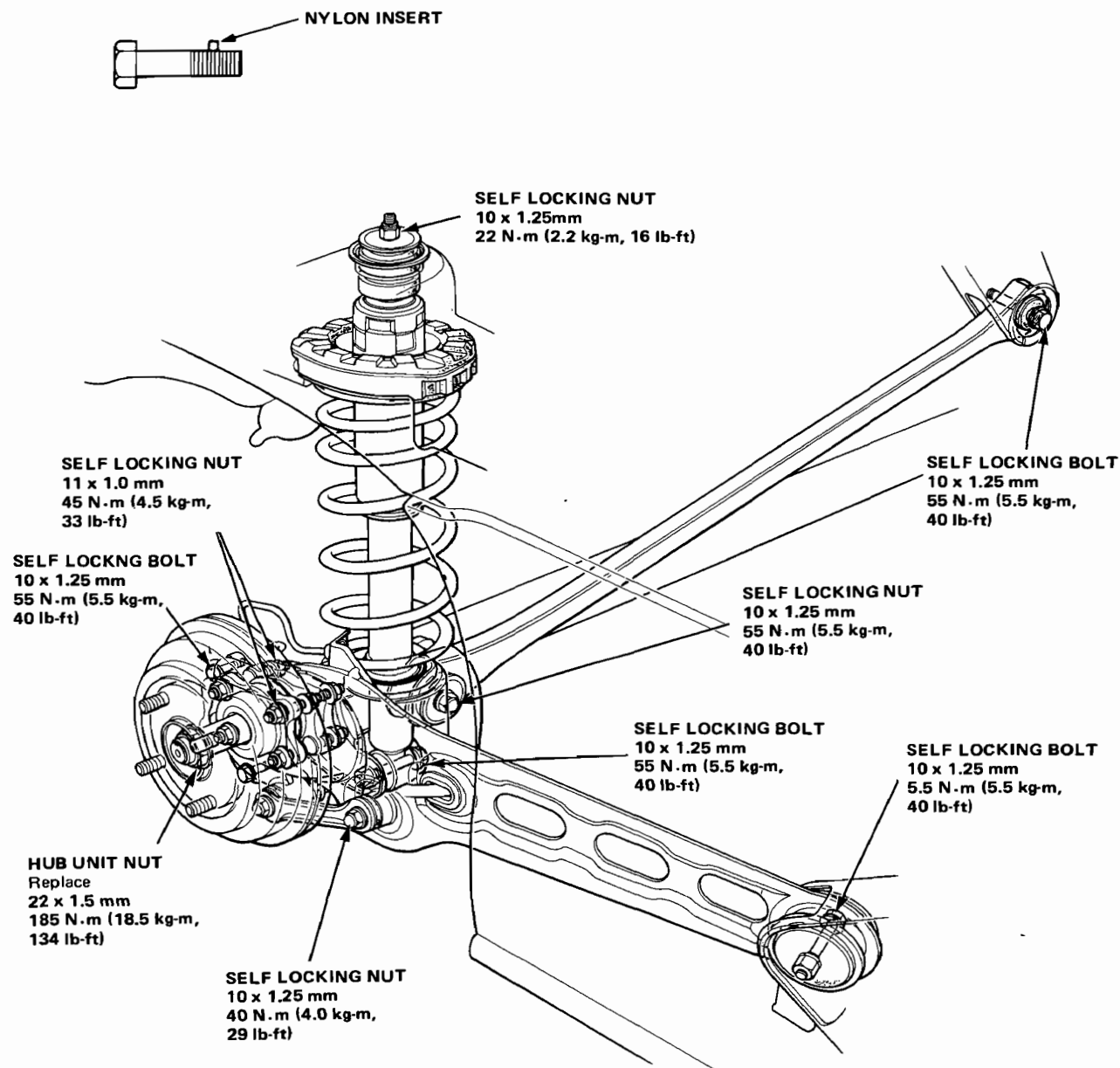


5. Install and tighten the brake hose clamp bolt.

Rear Suspension

Index

NOTE: Replace self-locking bolts if you can easily thread a nut past their nylon locking inserts.

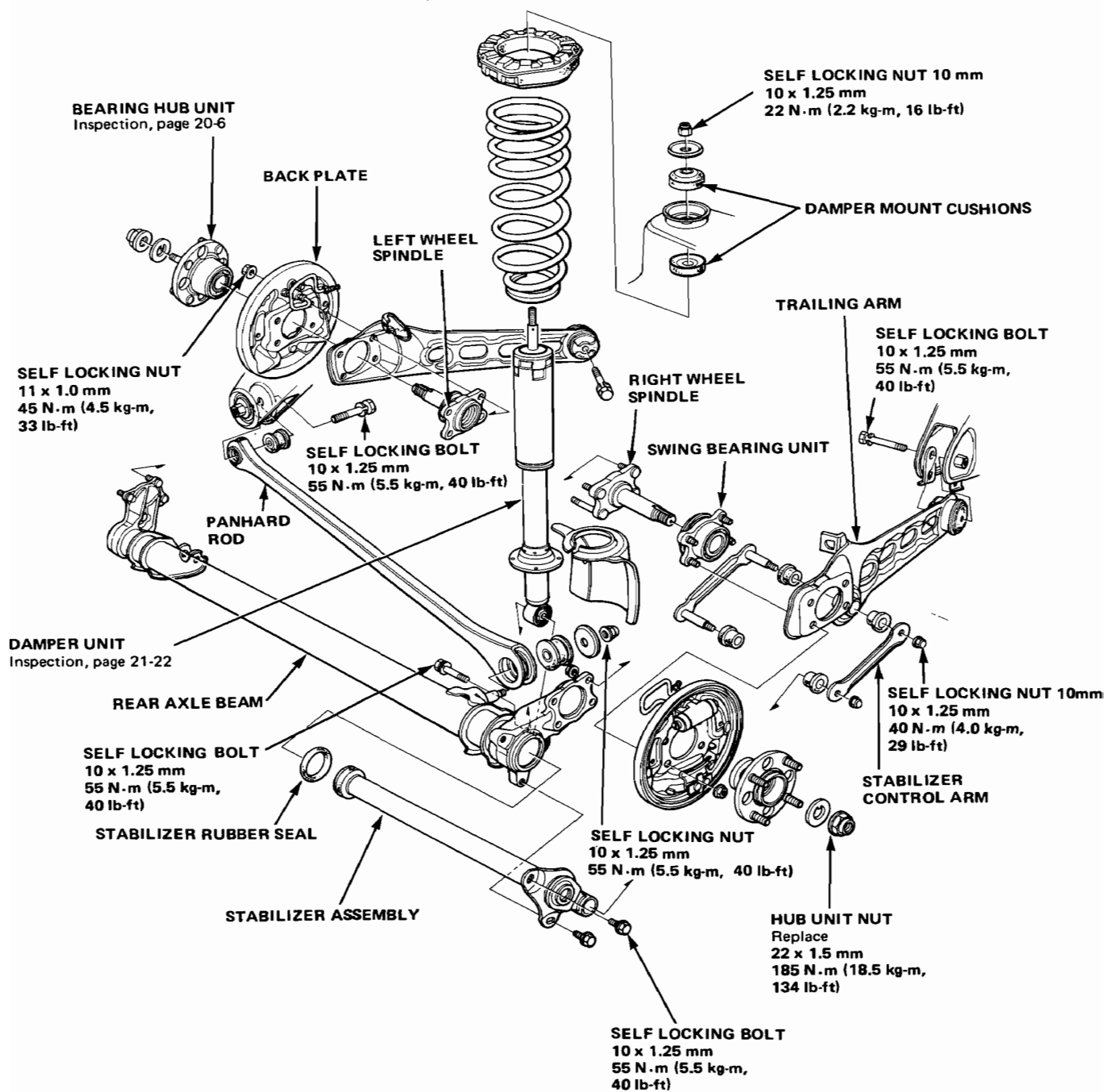




Disassembly/Inspection

NOTE: Inspect all bushings and rubber parts for wear and damage.

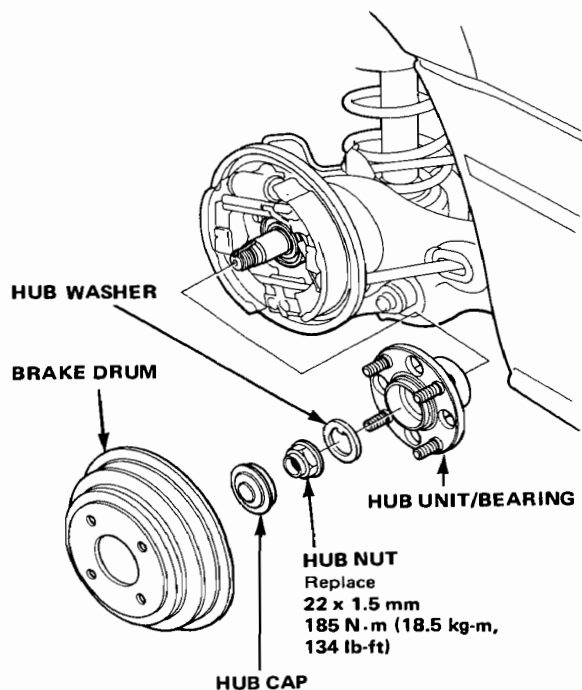
Replace all self locking nuts after disassembly.



Hub Unit

Removal/Installation

1. Raise the rear of the car and support it with safety stands (see page 1-7 for the proper locations for the safety stands).
2. Remove the rear wheels.
3. Remove the brake drum and inspect the wheel bearing end play. (Page 20-5)
4. Remove the hub unit cap and nut.

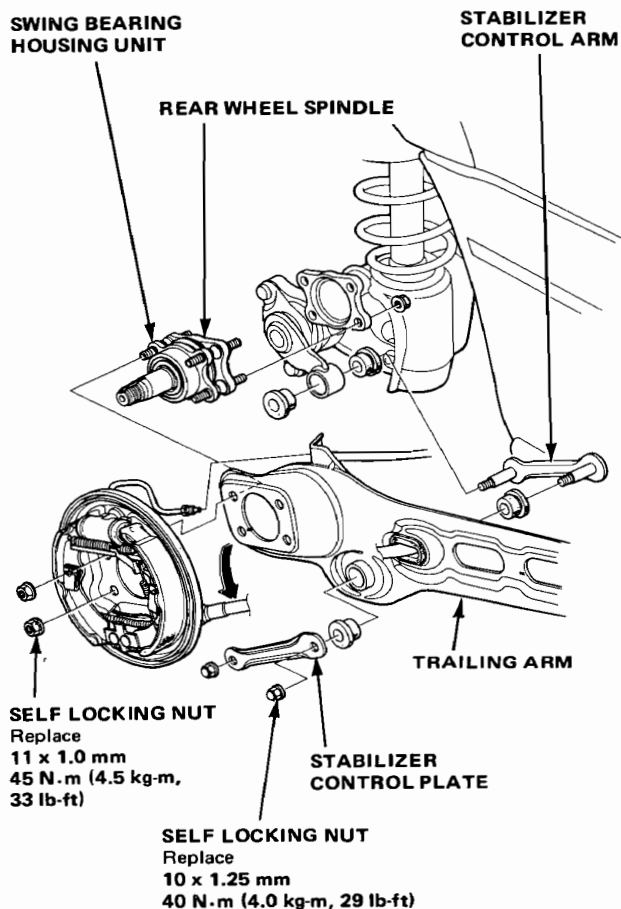


5. Remove the hub unit.
 6. Installation is the reverse order of removal.
- NOTE:** Stake the shoulder of the hub unit nut against the groove in the spindle after tightening.

Swing Bearing

Removal

1. Remove the right backing plate.
2. Remove the stabilizer control plate, then remove the trailing arm from the swing bearing housing unit.



3. Remove the rear wheel spindle from the axle beam.

Stabilizer Assembly

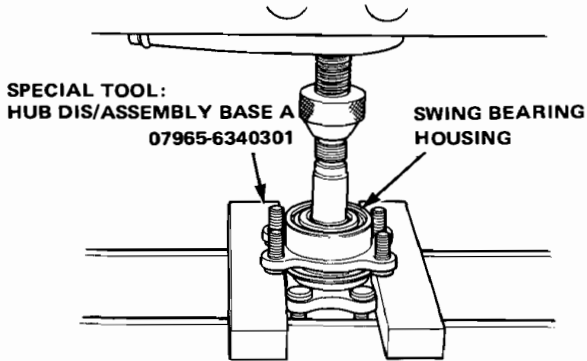


Replacement

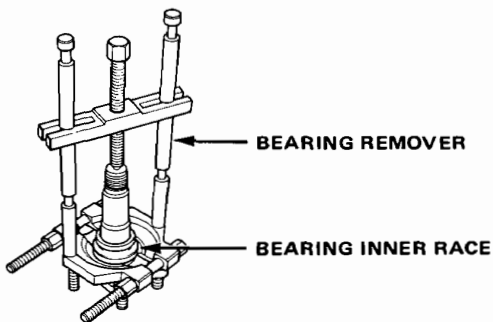
4. Separate the rear wheel spindle from the swing bearing housing with a hydraulic press.

SPECIAL TOOL:

HUB DIS/ASSEMBLY BASE A 07965-6340301

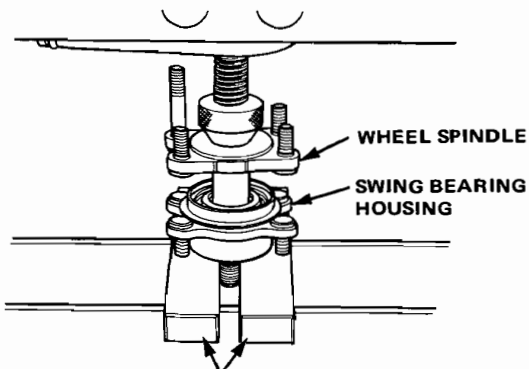


5. Remove the inner race with a bearing remover.



6. Install a new swing bearing when reassembling.
7. Install the wheel spindle into the swing bearing housing with the special tool and hydraulic press.

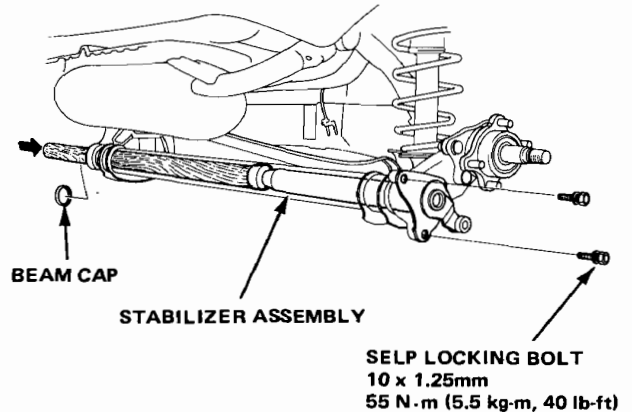
NOTE: Set the hub dis/assembly base securely at the inner race of the swing bearing housing.



SPECIAL TOOL;
HUB DIS/ASSEMBLY BASE A 07965-6340301

Removal/Installation

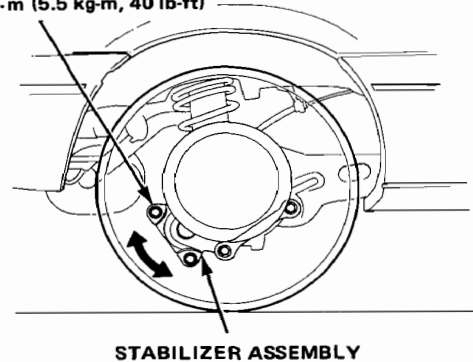
1. Raise the rear of the car and support with safety stands (see page 1-7 for the proper locations for the safety stands).
Remove the rear wheels.
2. Remove both right and left backing plates. (Page 21-23)
3. Remove the stabilizer control arm. (Page 20-21)
4. Remove the stabilizer assembly mounting bolt.



5. Remove the right beam cap.
6. Remove the stabilizer assembly by tapping it with a mallet.
7. Stabilizer reassembly is the reverse of removal.

NOTE: When fitting the stabilizer assembly into the axle beam, tighten the two self locking bolts last.

SELF LOCKING BOLT
10 x 1.25 mm
55 N·m (5.5 kg-m, 40 lb-ft)

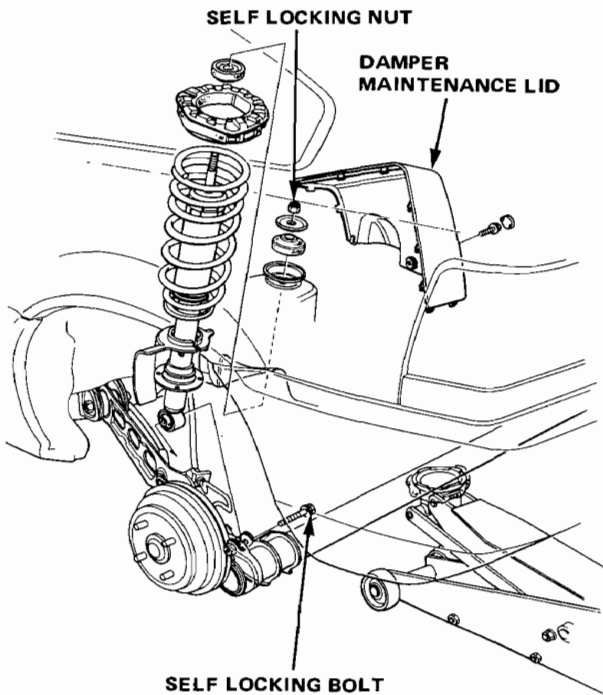


8. Tighten all bushings and rubber dampened parts after the car is back on the ground.

Rear Damper Assembly

Removal

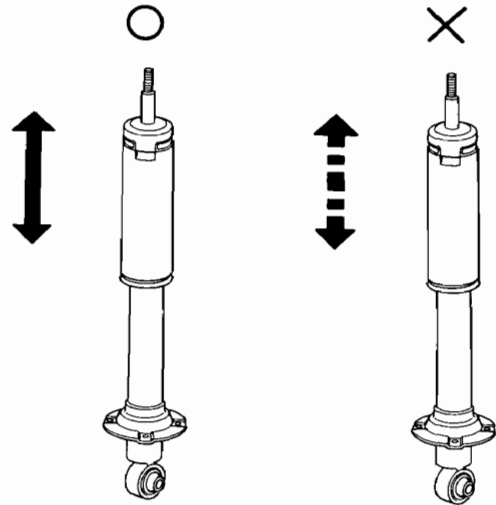
1. Raise the rear of the car and support it with safety stands (see page 1-7 for the proper locations for the safety stands).
2. Remove the rear wheels.
3. Place a jack under the rear axle beam.
4. Remove the damper maintenance lid and the self locking nut.



5. Remove the self locking bolt, damper assembly, spring and the spring cushion.

Inspection

1. Slowly move the damper piston rod a full stroke and check for smooth operation.
2. Jerk the piston rod back and forth 5–10 mm (2–4 in.) and check for smooth operation.
3. Inspect for an oil leak or cracks in the piston rod.
4. Listen for abnormal noises.

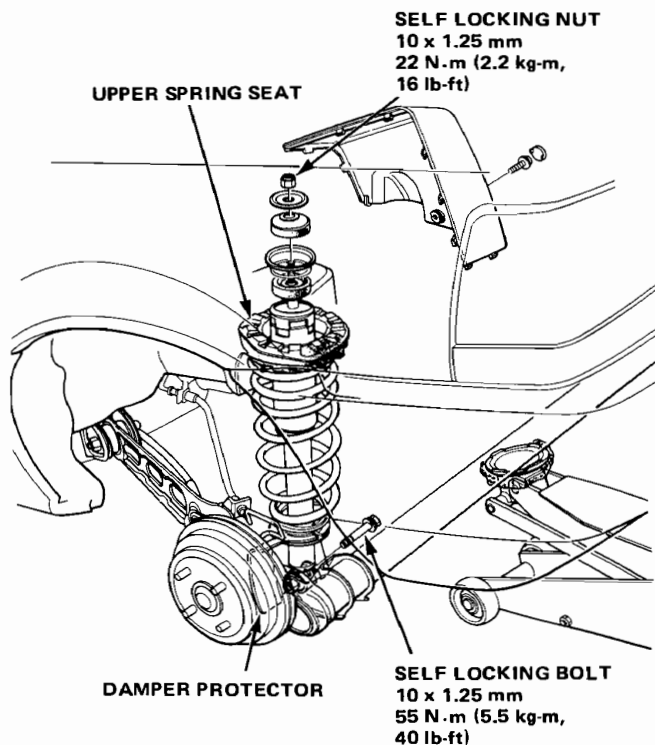


NOTE: The damper cannot be disassembled. If it does not operate smoothly, or if it makes any abnormal noises during operation, replace it.



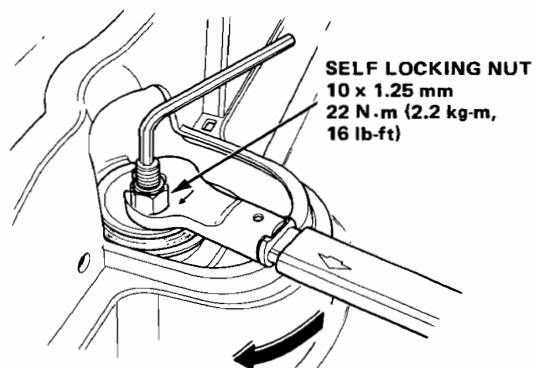
Installation

1. Fit the upper spring seat into the frame.
2. Install the damper protector on the damper unit, install the damper spring, and then temporarily tighten the damper at the rear axle beam.



3. Fit the inner damper mount cushion into the frame.
4. Jack up the axle beam so that the damper shaft fits into the hole in the frame.

5. Install the outer damper mount cushion and washer and then tighten the self locking nut.

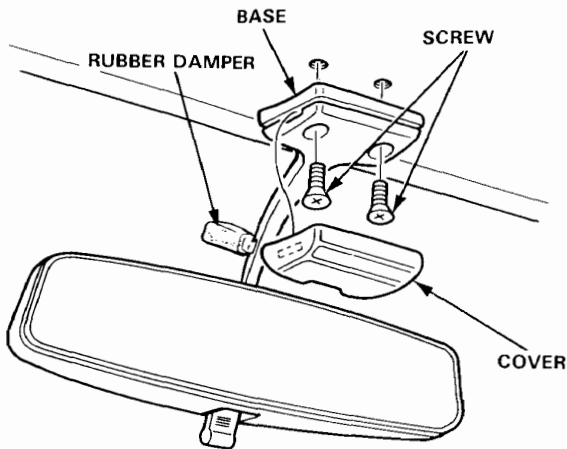


6. Install the damper maintenance lid.
 7. Tighten the damper on the rear axle beam.
- NOTE:** Tighten the rubber and bushing with the vehicle placed on the ground.

Rear View Mirror

Replacement

1. Remove the rubber damper.
2. Pry the cover off using the end of a slot-head screw-driver.



3. Remove both mounting screws from the mirror base, then remove the mirror assembly.
4. Remove the base from the rear view mirror.

